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A Note to Students

Known by many as simply "Turabian" in honor of its original author, *A Manual for Writers of Research Papers, Theses, & Dissertations* ia the authoritative student resource on "Chicago style."

This book has helped generations of students successfully research, write, and submit papers in virtually all academic disciplines.

Its guidelines for source citations and style have been condensed and adapted for student writers from another, more comprehensive reference work, *The Chicago Manual of Style*.

- Part 1 covers every step of the research and writing process.

  It provides practical advice to help you formulate the right questions, read critically, and build arguments.

  It also shows you how to draft and revise your papers to strengthen both your arguments and your writing.

- Part 2 offers a comprehensive guide to the 2 methods of Chicago-style source citation, beginning with helpful information on general citation practices in Chap. 15.

  In the humanities and most social sciences, you will likely use the notes-bibliography style detailed in Chaps. 16–17; in the natural and physical sciences and some social sciences, you will more likely use the author-date style described in Chaps. 18–19.

- Part 3 covers Chicago's recommended editorial style, which will help you bring consistency to your writing in matters e.g. punctuation, capitalization, and abbreviations; this section also includes guidance on incorporating quotations into your writing and on properly presenting tables and figures.

- The appendix presents formatting and submission requirements for theses and dissertations that many academic institutions use as a model, but be sure to follow any local guidelines provided by your institution.

Preface

students writing research papers, theses, and dissertations in today's colleges and universities inhabit in world filled with digital technologies that were unimagined in 1937 - the year dissertation secretary Kate L. Turabian 1st assembled a booklet of guidelines for student writers at the University of Chicago.

The availability of word-processing software and new digital sources has changed the way students conduct research and write up the results.

But these technologies have not altered the basic task of the student writer: doing well-designed research and presenting it clearly and accurately while following accepted academic standards for citation, style, and format.

Turabian's 1937 booklet reflected guidelines found in an already classic resource for writers and editors published by the University of Chicago Press that would ultimately be known as *The Chicago Manual of Style* (*CMOS*).

The Press began distributing Turabian's booklet in 1947 and 1st published the work in book form in 1955, under the title *A Manual for Writers of Term Papers, Theses, and Dissertations*.

Over time, Turabian's book has become a standard reference for students of all levels at universities and colleges across the country.

Turabian died in 1987 at age 94, a few months after publication of the book's 5th edition.

Beginning with that edition, members of the Press editorial staff have carried out the revisions to the chapters on source citation, style, and paper format.

For the 7th edition (2007), Wayne C. Booth, Gregory G. Colomb, and Joseph M. Williams expanded the focus of the book by adding extensive new material adapted from their book *The Craft of Research*, also published by the University of Chicago Press and now in its 4th edition (2016).

Among the new topics covered in their chapters were the nature of research, finding and engaging sources, taking notes, developing an argument, drafting and revising, and presenting evidence in tables and figures.

Following the deaths of this remarkable trio of authors, whose collective voice will always animate this work, Joseph Bizup and William T. FitzGerald have with this edition assumed the mantle of revising their chapters for a new generation of students, as they had previously for *The Craft of Research*.

Part 1, now aligned with the most recent edition of that book, incorporates updated advice for writers and responds to recent developments in information literacy, including the use of digital materials.

Part 2 offers comprehensive guide to the 2 Chicago styles of source citation - the notes-bibliography format used widely in the humanities and most social sciences and the author-date format favored in many of the sciences and some social sciences.

Thoroughly updated guidance related to online citation practices have been supplemented throughout by new examples featuring the types of sources students are most likely to consult.

Part 3 addresses matters of spelling, punctuation, abbreviation, and treatment of numbers, names, special terms, and titles of works.

The final 2 chapters in this section treat the mechanics of using quotations and graphics (tables and figures), topics that are discussed from a rhetorical perspective in Part 1.

Both Parts 2 and 3 have been updated for this edition in accordance with the 17th edition (2017) of *The Chicago Manual of Style*.

The recommendations in this manual in some instances diverge from *CMOS* in small ways, to better suit the requirements of academic papers as opposed to published works.

The appendix presents guidelines for paper format and submission that have become the primary authority for dissertation offices throughout the United States.

These guidelines have been updated to reflect the nearly universal electronic submission of papers and to feature new examples from recently published dissertations.

This appendix is intended primarily for students writing PhD dissertations and master's and undergraduate theses, but the sections on format requirements and electronic file preparation will also aid those writing class papers.

An extensive bibliography, organized by subject area and fully updated, lists sources for research and style issues specific to various disciplines.

The guidelines in this manual offer practical solutions to a wide range of issues encountered by student writers, but they may be supplemented - or even overruled - by the conventions of specific disciplines or the preferences of particular

institutions, departments, or instructors.

All of the chapters on style and format remind students to review the requirements of their university, department, or instructor, which take precedence over the guidelines presented in this book.

Updating a book that has been used by millions of students over 80 years is no small task, and many people participated in preparing this 9th edition.

The University of Chicago Press Editorial Staff

Part I

Research & Writing - Wayne C. Booth, Gregory G. Colomb, Joseph M. Williams, Joseph Bizup, and William T. FitzGerald

Overview of Part I

The authors know how daunting it can feel to start a substantial research project, whether it's a doctoral dissertation, a master's or senior thesis, or just a long class paper.

But you can handle any project if you break it into its parts, then work on them 1 step at a time.

Part I of this book shows you how.

The authors 1st discuss the aims of research and what readers will expect of any research paper (a term the authors use broadly to refer to all varieties of *research-based writing*).

The authors then focus on how to find a research question and problem whose answer is worth your time and your readers' attention; how to find and use information from sources to back up your answer; and then how to plan, draft, and revise your paper so your readers will see that your answer is based on sound reasoning and reliable evidence.

Several themes run through this part.

- You can't plunge into a project blindly; you must plan it, then keep the whole process in mind as you take each step.

  So think big, but break the process down into small goals that you can meet one at a time.

- Your best research will begin with a question that *you* want to answer.

  But you must then imagine readers asking questions of their own: *So what if you don't answer it? Why should I care?*

- From the outset, you should try to write every day, not just to record the content of your sources but to clarify what you think of them.

  You should also write down your own developing ideas to get them out of the cozy warmth of your head and into the cold light of day, where you can see if they still make sense.

  You probably won't use much of this writing in your final draft, but it is essential preparation for it.

- No matter how carefully you do your research, readers will judge it by how well you present it, so you must know what they will look for in clearly written paper that earns their respect.

If you're an advanced researcher, skim Chaps. 1–4.

You will see there much that's familiar; but if you're also teaching, it may help you explain what you know to your students more effectively.

Many experienced researchers tell us that Chaps. 5–12 have helped them not only to explain to others how to conduct and report research, but also to draft and revise their own writing more quickly and effectively.

If you're just starting your career in research, you'll find every chapter of Part 1 useful.

Skim it all for an overview of the process; then as you work through your project, reread chapters relevant to your immediate task.

You may feel that the steps described here are too many to remember, but you can manage them if you take them one at a time, as you do do more research they'll become habits of mind.

Don't think, however, that you must follow these steps in exactly the other the authors present them.

Researchers regularly think ahead to future steps as they work through earlier ones and revisit earlier steps as they deal with a later one.

(That explains why we so often refer you ahead to anticipate a later stage in the process and back to revisit an earlier one.)

And even the most systematic researcher has unexpected insights that send her off in a new direction.

Work from a plan, but be ready to depart from it, even to discard it for a new one.

If you're a very new researcher, you may also think that some matters the authors discuss are beyond your immediate needs.

The authors know that a 10-page class paper differs from a dissertation.

But both require a kind of thinking that even the newest researcher can start practicing.

You begin your journey toward full competence when you not only know what lies ahead but also can start practicing the skills that experienced researchers began to learn when they were where you are now.

No book can prepare you for every aspect of every research project.

And this one won't help you with the specific methodologies in fields e.g. psychology, economics, and philosophy, much less physics, chemistry, and biology.

Nor does it tell you how to adapt what you learn about academic research to business or professional settings.

But it does provide an overview of the processes and habits of mind that underlie all research, wherever it's done, and of the plans you must make to assemble a paper, draft it, and revise it.

With that knowledge and with help from your teachers, you'll come to feel in control of your projects, not intimated by them, and eventually you'll learn to manage even the most complex projects on your own, in both the academic and the professional worlds.

The 1st step in learning the skills of sound research is to understand how experienced researchers think about its aims.

Chapter 1

What Research Is & How Researchers Think about It

Whenever we read about a scientific breakthrough or a crisis in world affairs, we benefit from the research of others, who likewise benefited from the research of countless others before them.

When we walk into a library, we are surrounded by more than 25 centuries of research.

When we go on the internet, we can read the work of millions of researchers who have posed questions beyond number, gathered untold amounts of information from the research of others to answer them, and then shared their answers with the rest of us.

We can carry on their work by asking and, we hope, answering new questions in turn.

Governments spend billions on research, businesses even more.

Research goes on in laboratories and libraries, in jungles and ocean depths, in caves and in outer space, in offices and, in the information age, even in our own homes.

Research is in fact the world's biggest industry.

So what, exactly, is it?

1.1 What Research Is

You already have a basis understanding of research: answering a question by obtaining information.

In this case, research can be as simple as choosing a new phone or as complex as discovering the origin of life.

In this book the authors use *research* in a specific way to mean a process of systematic inquiry to answer a question that not only the researcher but also others want to solve.

Research thus includes the steps involved in presenting or reporting it.

To be a true researcher, as we are using the term, you must share your findings and conclusions with others.

If you are new to research, you may think that your paper will add little to the world's knowledge.

But done well, it will add a lot to *your* knowledge and to your ability to communicate that knowledge.

As you learn to do your own research, you also learn to use and judge that of others.

In every profession, researchers must read and evaluate the work of others before they make a decision.

This is a job you will do better after you have learned how others judge yours.

This book focuses on research in the academic world, but every day we read or hear about research that affects our lives.

Often we get news of research secondhand, and it can be difficult to know what reasoning and evidence support a claim.

But research doesn't ask for our blind trust or that we accept something on the basis of authority.

It invites readers to think critically about evidence and reasoning.

That is how *research-based writing* differs from other kinds of persuasive writing: it must rest on shared facts that readers accept as truths independent of your feelings and beliefs.

Your readers must be able to follow your reasoning from evidence they accept to the claim you draw from it.

Your success as a researcher thus depends not just on how well you gather and analyze data but also on how clearly you report your reasoning so that your readers can test and judge it before making your claims part of their knowledge and understanding.

1.2 How Researchers Think about Their Aims

All researchers collect information, what we're calling *data*.

But researchers do not merely gather facts on a topic - *stories about the Battle of the Alamo*, e.g.

They look for specific data to test and support an answer to a question that their topic inspired them to ask, e.g. *Why has the Alamo story become a national legend?*

In doing so, they also imagine a community of readers who they believe will share their interest and help them test and support an answer to that question.

Experienced researchers, however, know that they must do more than convince us that their answer is sound.

They must also show us why their question was worth asking, how its answer helps us understand some bigger issue in a new way.

If we can figure out why the Alamo story has become a national legend, we might then answer a larger question: how have regional myths shaped the American character?

You can judge how closely your thinking tracks that of an experienced researcher by describing your project in a sentence like this:

1. **Topic**: I am working on X (*stories about the Battle of the Alamo*)
2. **Question**: because I want to find out Y (*why its story became a national legend*)
3. **Significance**: so that I can help others understand Z (*how such regional myths have shaped the American character*).

That sentence is worth a close look, because it describes not just the progress of your research but your personal growth as a researcher.

- **Topic**: "I am working on X ...": Those new to research often begin with a simple topic like *the Battle of the Alamo*.

  But too often they stop there, with nothing but a broad topic to guide their work.

  Beginning this way, they may pile up dozens or hundreds of notes but then can't decide what data to keep or discard.

  When it comes time to write, their papers become "data dumps" that leave readers wondering what all those data add up to.

- **Question**: "... because I want to find out Y ...": More experienced researchers begin not just with a topic but with a *research question*, e.g. *Why has the story of the Alamo become a national legend?*

  They know that readers will think their data add up to something only when they serve as evidence to support an answer.

  Indeed, only with a question can a researcher know what information to look for and, once obtained, what to keep - and not just data that support a particular answer but also data that test or discredit it.

  With sufficient evidence to support an answer, a researcher can respond to data that seem to contradict it.

  In writing a paper, the researcher tests that answer and invites others to test it too.

- **Significance**: "... so that I can help others understand Z": The best researchers understand that readers want to know not only that an answer is sound but also why the question is worth asking: *So what? Why should I care why the Alamo story has become a national legend?*

  Think of it this way: what will be lost if you *don't* answer your question?

  Your answer might be *Nothing. I just want to know.*

  Good enough to start but not to finish, because eventually your readers will want an answer beyond *Just curious.*

Answering *So what?* is tough for all researchers, beginning and experienced alike, because when you only have a question stemming from a topic of personal interest, it's hard to predict whether others will find its answer significant.

Some researchers therefore work backwards: they begin not by following their own curiosity but by crafting questions with implications for bigger ones that others in their field already care about.

But many researchers, including us, find that they cannot address that 3rd step until they finish a 1st draft.

So it's fine to *begin* your research without being able to answer *So what?*, and if you are a student, your teacher may even let you skip that last step.

But if you are doing advanced research, you *must* take it, because your answer to *So what?* is what makes your research matter to others.

In short, not all questions are equally good.

We might ask how many cats slept in the Alamo the night before the battle, but so what if we find out?

It is hard to see how an answer would help us think about any larger issue worth understanding, so it's a question that's probably not worth asking (though as we'll see, we could be wrong about that).

How good a question is depends on its significance to some community of readers.

Exactly what community depends on your field but also on how you frame your research.

You can try to expand your potential readership by connecting Z to even broader questions: *And if we can understand what has shaped the American character, we might understand better who Americans think they are. And when we know that, we might better understand why others in the world judge them as they do.*

Now perhaps political scientists will be as interested in this research as historians.

On the other hand, if you try to widen your audience too much, you risk losing it altogether.

Sometimes it's better to address a smaller community of specialists.

The authors can't tell you the right choice, but they can tell you 2 wrong ones: trying to interest everyone (some people just won't care no matter how you frame your research) or not trying to interest anyone at all.

1.3 Conversing with Your Readers

When you can explain the *significance* of your research, you enter into a kind of conversation with your research community.

Some people, when they think of research, imagine a lone scholar or scientist in a hushed library or lab.

But no places are more crowded with the presence of others than these.

When you read a book or an article or a report, you silently converse with its authors - and through them with everyone else th

In fact, every time you go to a written source for information, you join a conversation between writers and readers that began millennia ago.

And when you report your own research, you add your voice and hope that other voices will respond to you, so that you can in turn respond to them.

And so it goes.

Experienced researchers understand that they are participating in such conversations and that genuine research must matter not only to the researcher but also to others.

That is why the authors' formula - *I am working on X to find out Y so that others can better understand Z* - is so powerful: because it makes informing others the end of research.

But these silent conversations differ from the face-to-face conversations we have every day.

We can judge how well everyday conversations are going as we have them, and we can adjust our statements and behavior to repair mistakes and misunderstandings as they occur.

But in writing we don't have that opportunity: readers have to *imagine* writers in conversation with 1 another, as well as with themselves, and writers have to imagine their readers and their relationship to them.

In other words, writers have to offer readers a social contract: *I'll play my part if you play yours.*

Doing this is 1 of the toughest tasks for beginning researchers: get that relationship wrong and your readers will think you are naive or, worse, won't read your work at all.

Too many beginning researchers offer their readers a relationship that caricatures a bad classroom: *Teacher, I know less than you. So my role is to show you how many facts I can dig up. Yours is to say whether I've found enough to give me a good grade.*

Do that and you turn your project into a pointless drill, casting yourself in a role exactly opposite to that of a true researcher.

In true research, you must switch the roles of student and teacher.

You must imagine a relationship that goes beyond *Here are some facts I've dug up about 14th-century Tibetan weaving. Are they enough of the right ones?*

There are 3 better reasons to share what you've found.

You could say to your reader, *Here is some information that you may find interesting.*

This offer assumes, of course, that your reader wants to know.

You could also say not just *Here is something that should interest you* but *Here is something that will help you remedy a situation that troubles you.*

People do this kind of research every day in business, government, and the professions when they try to figure out how to address problems ranging from insomnia to falling profits to climate change.

In Chap. 2 the authors call such situations and their consequences *practical problems.*

When academic researchers address such practical problems, we say they are doing *applied* research.

Most commonly, though, academic researchers do *pure* research that addresses what we call *conceptual problems* - i.e., not troubling situations in the world but the limitations of our understanding of it (again see Chap. 2).

In this case, you say to your readers, *Here is something that will help you better understand something you care about.*

When you make this last sort of appeal, you imagine your readers as a community of receptive but also skeptical colleagues who are open to learning from you and even changing their minds - if you can make the case.

We now understand the *goal of research, at least in its pure form: it is not to have the last word but to keep the conversation going.*

The best question are those whose answers raise several more.
When that happens, everyone in the research community benefits.

Chapter 2

Defining a Project: *Topic, Question, Problem, Working Hypothesis*

A research project begins well before you search the internet or head for the library and continues long after you have collected all the data you think you need.

Every project involves countless specific tasks, so it is easy to get overwhelmed.

But in all research projects, you have just 5 general aims:

1. Ask a question worth answering.
2. Find an answer that you can support with good reasons.
3. Find good data that you can use as reliable evidence to support your reasons.
4. Draft an argument that makes a good case for your answer.
5. Revise that draft until readers will think you met the 1st 4 goals.

You might even post those 5 goals in your workspace.

Research projects would be much easier if we could march straight through these steps.

But you will discover (if you have not already) that the research process is not so straightforward.

Each task overlaps with others, and frequently you must go back to an earlier one.

The truth is, research is messy and unpredictable.

But that's also what makes it exciting and ultimately rewarding.

2.1 Find a Question in Your Topic

Researchers begin projects in different ways.

Many experienced researchers begin with a question that others in their field want to answer: *What caused the extinction of most large North American mammals?*

Others begin with just basic curiosity, a vague intellectual itch that they have to scratch.

They might not know what puzzles them about a topic, but they're willing to spend time to find out whether that topic can yield a question worth answering.

They realize, moreover, that the best research question is not one whose answer they want to know just for its own sake; it is one that helps them and others understand some larger issue.

E.g., if we knew why North American sloths disappeared, we might be able to answer a larger question that puzzles many historical anthropologists: *Did early Native Americans live in harmony with nature, as some believe, or did they hunt its largest creatures to extinction? And if we knew that, then we might also understand* ... (*So what?* again. See **1.2**)

Then there are those questions that just pop into a researcher's mind with no hint of where they'll lead, sometimes about matters so seemingly trivial that only the researcher thinks they're worth answering: *Why does a coffee spill dry up in the form of a ring?*

Such a question might lead nowhere, but you can't know that until you see its answer.

In fact, the scientist puzzled by coffee rings made discoveries about the behavior of fluids that others in this field thought important - and that paint manufacturers found valuable.

If you cultivate the ability to see what's odd in the commonplace, you'll never lack for research projects as either a student or a professional.

If you already have a focused topic, you might skip to **2.1.3** and begin asking questions about it.
If you already have some questions, skip to **2.1.4** to test them using the criteria listed there.
Otherwise, here's a plan to help you search for a topic.

2.1.1 Search Your Interests

Beginning researchers often find it hard to pick a topic or believe they lack the expertise to research a topic they have.

But a research topic is an interest stated specifically enough for you to imagine *becoming* a local expert on it.

That doesn't mean you *already* know a lot about it or that you'll know more about it than others, including a teacher or advisor.

You just want to know more about it than you do now.

If you can work on any topic, the authors offer only a cliché: start with what interests you.
Ask these questions:

- What special interests do you have - chess, old comic books, scouting?

  The less common, the better.

  Choose one and investigate something about it that you don't know.

- Where would you like to travel?

  Find out all you can about your destination.

  What particular aspect surprises you or makes you want to learn more?

- Can you find an online discussion list or social media page focused on issues that interest you?

- Visit a museum or a "virtual museum" on the internet with exhibitions that appeal to you.

  What catches your interest that you would like to know more about?

- Have you taken positions on issues in your fields or in debates with others but found that you couldn't back up your views with good reasons and evidence?

- What issues in your field do people outside your field misunderstand?

- What topic is your instructor or advisor working on?

  Would she like you to explore a part of it?

  Don't be too shy to ask.

- Does your library have rich resources in some field?

  Ask your instructor or a librarian.

- What intrigues you in your reading?

  What connections do you see among different things you are reading?

- What other courses will you take in your field or out of it?

  Find a textbook and skim it for study questions.

- If you have a job in mind, what kind of writing might help you get it?

  Employers often ask for samples of an applicant's work.

Once you have a list of possible topics, choose 1 or 2 that interest you most and explore their research potential.

Sometimes beginning researchers choose a topic because they already know what they want to say about it, even before they've done any research.

That's a mistake: the best topics provoke good questions; the worst come with ready-made answers.

To gauge a topic's potential, do these things:

- In the library, look up your topic in a general guide e.g. *CQ Researcher* and skim the subheadings.

  In an online database e.g. Academic Search Premier, you can explore your topic through subject terms.

  If you have a narrower focus, you can do the same with specialized guides e.g. *Women's Studies International*.

  At most libraries today, such guides are found online.

- On the internet, google your topic, but don't surf indiscriminately.

  Look 1st for websites that are roughly like the sources you would find in a library, e.g. online encyclopedias.

  Read the entries on your general topic, an then copy their lists of references for a closer look.

  Few experienced researchers trust Wikipedia as a reliable source to cite as evidence, but most would *use the site to find ideas and more specific sources.*

- Finally, think ahead: you may be in for a long relationship with your topic, so be sure it interests you enough to get you through the inevitable rocky stretches.

2.1.2 Make Your Topic Manageable

If you pick a topic that sounds like an encyclopedia entry - *bridges, birds, masks* - you'll find so many sources that you could spend a lifetime reading them.

You must carve out of your topic a manageable piece.

Before you start researching, limit your topic to reflect a special interest in it: *What is it about, say, masks that made you choose them? What particular aspect of them interests or puzzles you?*

Think about your topic in a context that you know something about, and then add words and phrases to reflect that knowledge:

Example 2.1.1. *masks in religious ceremonies*
masks as symbols in Hopi religious ceremonies
mudhead masks as symbols of sky spirits in Hopi fertility ceremonies

You might not be able to focus your topic until after you start reading about it.

That takes time, so start early (you can do much of this preliminary work online):

- Begin with an overview of your topic in a general encyclopedia (in the bibliography, see items in category 2 in the general sources); then read about it in a specialized one (see items in category 2 in your field).

- Skim a survey of your topic (encyclopedia entries usually cite a few).

- Skim subheads under your topic in an annual bibliography in your field (in the bibliography, see items in category 4 in your field).

  That will also give you a start on a reading list.

- Search the internet for the topic (but evaluate the reliability of what you find; see **3.3.2**).

Especially useful are topics that spark debate: *Fisher claims that Halloween masks reveal children's archetypal fears, but do they?*

Even if you can't resolve the debate, you can learn how such debates are conducted (for more on this, see **3.1.2**).

2.1.3 Question Your Topic

Once they have a focused topic, many new researchers start plowing through all the sources they can find, taking notes on everything they read.

Then they dump it all into a report with little sense of purpose or direction.

Experienced researchers, however, document information not for its own sake but to support an answer to a question they (and they hope their readers) think worth asking.

So the best way to begin working on a focused topic is to pose questions that direct you to just the information you need to answer them.

Do this not just once, early on, but throughout your project.

Ask questions as you read, especially *how* and *why* (see also **4.1.1**–**4.1.2**).

Try the following kinds of questions (the categories are loose and overlap, so don't worry about keeping them distinct).

1. Ask how the topic fits into a larger context (historical, social, cultural, geographic, functional, economic, and so on):

   - How does your topic fit into a larger story?

Example 2.1.2. *What cam before masks?*
How did masks come into being?
Why?
What changes have they caused in other parts of their social or geographic setting?
How and why did that happen?
Why have masks become a part of Halloween?
How and why have masks helped make Halloween the biggest American holiday after Christmas?

- How is your topic a functioning part of a larger system?

  Example 2.1.3. *How do masks reflect the values of specific societies and cultures?*
  What roles do masks play in Hopi dances?
  In scary movies?
  In masquerade parties?
  For what purposes are masks used other than disguise?
  How has the booming market for kachina masks influenced traditional designs?

- How does your topic compare to and contrast with other topics like it?

  Example 2.1.4. *How do masks in Native American ceremonies differ from those in Africa?*
  What do Halloween masks have to do with Mardi Gras masks?
  How are masks and cosmetic surgery alike?

2. Ask questions about the nature of the thing itself, as an independent entity:

   - How has your topic changed through time? Why? What is its future?

     Example 2.1.5. *How have Halloween masks changed?*
     Why?
     How have Native American masks changed?
     Why?

   - How do the parts of your topic fit together as a system?

     Example 2.1.6. *What parts of a mask are most significant in Hopi ceremonies?*
     Why?
     Why do some masks cover only the eyes?
     Why do so few masks cover just the bottom half of the face?

   - How many different categories of your topic are there?

     Example 2.1.7. *What are the different kinds of Halloween masks?*
     What are the different qualities of masks?
     What are the different functions of Halloween masks?

3. Turn positive questions into a negative ones:

   Example 2.1.8. *Why have masks* not *become a part of Christmas?*
   How do Native American masks not *differ from those in Africa?*
   What parts of masks are typically not *significant in religious ceremonies?*

4. Ask speculative questions:

   Example 2.1.9. *Why are masks common in African religions but not in Western ones?*
   Why are children more comfortable wearing Halloween masks than are most adults?
   Why don't hunters in camouflage wear masks?

5. Ask *What is?* questions: how would things be different if your topic never existed, disappeared, or were put into a new context?

Example 2.1.10. *What if no one ever wore masks except for safety reasons?*

What if everyone wore masks in public?

What if movies and TV were like Greek plays and all the actors wore masks?

What if it were customary to wear masks on blind dates?

In marriage ceremonies?

At funerals?

6. Ask questions that reflect disagreements with a source: if a source makes a claim you think is only weakly supported or even wrong, make that disagreement a question (see also **4.1.2.**).

   Example 2.1.11. *Martinez claims that carnival masks uniquely allow wearers to escape social norms.*

   But I think religious masks also allow wearers to escape from the material realm to the spiritual.

   Is there a larger pattern of all masks creating a sense of alternative forms of social or spiritual life?

7. Ask question that build on agreement: if a source offers a claim you think is persuasive, ask questions that extend its reach (see also **4.1.1**).

   Example 2.1.12. *Elias shows that masked balls became popular in 18th-century London in response to anxiety about social mobility.*

   Is the same anxiety responsible for similar developments in other European capitals?

   You can also ask a question that supports the same claim with additional evidence.

   Example 2.1.13. *Elias supports his claim about masked balls entirely with published sources.*

   It is also supported by evidence from unpublished sources e.g. letters and diaries?

8. Ask questions analogous to those that others have asked about similar topics.

   Example 2.1.14. *Smith analyzed the Battle of Gettysburg from an economic point of view.*

   What would an economic analysis of the Battle of the Alamo turn up?

9. Look for questions that other researchers pose but don't answer.

   Many journal articles end with a paragraph or 2 about open questions, ideas for more research, and so on.

   You might not be able to do all the research they suggest, but you might carve out a piece of it.

10. Find a professional discussion forum on your topic, then "lurk," just reading the exchanges to understand the kinds of questions being asked.

    If you can;t find one using a search engine, ask a teacher or visit websites of professional organizations in your field.

    Look for questions that spark your interest.

    If questions from students are welcomed, you can even post one yourself, so long as it is very specific and narrowly focused.

2.1.4 Evaluate Your Questions

After asking all the questions you can think of, evaluate them.

Not all questions are equally good.
Look for questions whose answers might make you (and your readers) think about your topic in a new way.
Avoid questions like these:

- Their answers are settled fact that you could just look up.

  Example 2.1.15. *What was Audre Lorde's 1st published poem?*

  Question that ask *how* and *why* call for interpretations, not just the discovery of facts.

  That's why they invite deeper thinking than questions beginning *who, what, when*, or *where*, and deeper thinking leads to more interesting answers.

- Their answers can't be plausibly disproved.

  Example 2.1.16. *How important are masks in Inuit culture?*

  The answer is obvious: *Very.*

  If you can't imagine disproving a claim, then proving it is pointless.

  (On the other hand, world-class reputations have been won by those who questions a claim that seemed self-evidently true - e.g., that the sun circled the earth - and dared to disprove it.)

- Their answers would be merely speculative.

  Example 2.1.17. *Would church services be as well attended if the congregation all wore mask?*

  If you can't imagine finding data that would settle the question, it's not a question you can answer.

- Their answers are dead ends.

  Example 2.1.18. *How many black cats slept in the Alamo the night before the battle?*

  It's hard to see how an answer would help us think about any larger issue worth understanding better, so the question is probably not worth asking.

- Their answers require different capacities from the ones you have.

  Example 2.1.19. *How do Japanese translations of* The Great Gatsby *treat early 20th-century America?*

  If you can't read Japanese, this question is not for you to answer.

- The answers require more or different resources - materials, technology, money, especially time - than you have.

  Example 2.1.20. *How is childhood represented in the Victorian novel?*

  Can you read enough of them in the time you have to arrive at a reasonable answer?

Don't reject a question because you think someone must already have asked it.

Until you know, pursue its answer as if you asked 1st.

Even if someone has answered it, you might come up with a better answer or at least one with a new slant.

In fact, in the humanities and social sciences the best questions usually have more than 1 good answer.

You can also organize your project around comparing and contrasting competing answers and supporting the best one (see **6.2.5**).

The point is to find a question that *you* want to answer.

Too many students, both graduate and undergraduate, think that the aim of education is to learn settled answers to someone else's questions.

It's not.

It is to find your own answers to your own questions.

To do that, you must learn to wonder about things, to let them puzzle you - particularly things that seem commonplace.

2.2 Understanding Research Problems

In Chap. 1 the authors gave you a formula that expresses how experienced researchers think about their work:

1. **Topic**: I am working on X (*stories about the Battle of the Alamo*)
2. **Question**: because I want to find out Y (*why its story became a national legend*)
3. **Significance**: so that I can help others understand Z (*how such regional myths have shaped the American character*).

When you can state that significance from the point of view of your readers, you have more than a question: you have posed a *research problem* that they recognize needs a solution.

Among researchers, the term *problem* has a special meaning that sometimes confuses beginners.

In our everyday world, a problem is something we try to avoid.

But in academic research, a problem is something we seek out, even invent.

Indeed, without a problem to work on, a research is out of work.

Experienced researchers often talk about their problems in shorthand.

When asked what they are working on, they often answer with what sounds like a general topic: *adult measles, mating calls of Wyoming elk.*

As a result, beginners may think that having a topic to read about is the same thing as having a problem to solve.

But without a specific question to answer and a reason to find that answer significant, researchers have no way of knowing when they have enough.

So they can be tempted to throw in everything just to be safe.

To avoid the judgment that your paper is just a *data dump*, you need a problem, one that focuses on finding just those data that will help you solve it.

To find one, you need to know how problems work.

2.2.1 Understanding Practical and Conceptual Problems

There are 2 kinds of research problems: *practical* and *conceptual*.

Each of them has a 2-part structure:

- a situation or *condition*, and
- undesirable *costs* or *consequences* caused by that condition.

Your research question is about your problem's condition; its significance follows from your problem's cost or consequence.

What differentiates practical and conceptual problems is the nature of those conditions and costs/consequences.

The condition of a practical problem can be any state of affairs in the world that troubles you or, better, your readers: a traffic jam, foreign competition, a disease we can't effectively treat.

The cost of a practical problem is always some tangible effect we don't like: inconvenience, expense, pain, even death.

Practical problems are often a matter of perspective: if my company's products are outselling yours, that's problem for you but not for me.

The condition of a conceptual problem is always some version of not knowing or understanding something.

A conceptual problem does not have a tangible cost but a consequence.

This consequence is a particular kind of ignorance: a lack of understanding that keeps us from understanding something else that is even more significant.

Put another way, because we haven't answered 1 question, we can't answer another that is more important.

In short, practical problems concern what we should *do*; conceptual problems concern what we should *think*.

Practical problems are most common in the professional world; conceptual problems are most common in academe.

2.2.2 Distinguishing Pure & Applied Research

We call research *pure* when it addresses a conceptual problem that does not have any direct practical consequences, when it only improves the understanding of a community of researchers.

We call research *applied* when it addresses a conceptual problem that does have practical consequences.

You can tell whether research is pure or applied by considering the significance of your project: *is it about understanding or doing?*

1. **Topic**: I am studying how readings from the Hubble telescope differ from readings for the same stars measured by earthbound telescopes
2. **Question**: because I want to find out how much the atmosphere distorts measurements of electromagnetic radiation
3. **Practical Significance**: so that astronomers can use data from earthbound telescopes to *measure* more accurately the density of electromagnetic radiation.

Applied research is common in academic fields e.g. business, engineering, and medicine and in companies and government agencies that do research to understand what must be known before they can solve a problem.

Some new researchers may be uneasy with pure research because the consequence of a conceptual problem - not knowing something - seems so abstract.

Since they are not yet part of a community that cares deeply about understanding its part of the world, they feel that their findings aren't good for much.

So they try to show the importance of their conceptual answer by cobbling onto it an implausible practical use:

1. **Topic**: I am studying differences among 19th-century versions of the Alamo story
2. **Question**: because I want to find out how politicians used stories of such events to shape public opinion
3. **Potential Practical Significance**: in order to protect ourselves from unscrupulous politicians.

Most readers will find the link between this research question and its asserted significance a stretch.

But for researchers in American history, the question does not need to have practical significance.

As the term *pure* suggests, many researchers value the pursuit of knowledge "for its own sake" as a reflection of humanity's highest calling to know more.

So if you are doing academic research, resist the urge to turn a conceptual problem into a practical one - unless you've specifically been asked to do so.

You are unlikely to solve any genuine practical problem in a course project.

And in any case, most academic researchers see their mission not as fixing the problems of the world but as understanding them better (which may or may not lead to fixing them).

2.3 Propose a Working Hypothesis

Before you get deep into your project, try 1 more step.

It is one that some beginners resist but that experienced researchers usually attempt.

Once you have a question, imagine some plausible answers, no matter how sketchy or speculative.

At this stage, don't worry whether they're right.

That comes later.

E.g., suppose you ask, *Why do some religions use masks in ceremonies while others don't?*

You might speculate:

Example 2.3.1. *Maybe cultures with many spirits need masks to distinguish them.*

Maybe masks are common in cultures that mix religion and medicine.

Maybe religions originating the Middle East were influenced by the Jewish prohibition against idolatry.

Even a general answer can suggest something worth studying:

Example 2.3.2. *Maybe it has to do with the role of masks in nonreligious areas of a culture.*

Try to imagine at least 1 plausible answer, no matter how tentative or speculative.

If 1 answer seems promising, call it your *working hypothesis* and use it to guide your research.

You can, of course, start gathering data with no more than a question to guide you, but even a tentative working hypothesis will help you think about the kind of data you'll need as evidence to support it: numbers? quotations? observations? images? historical facts?

In fact, until you have a hypothesis, you can't know whether any data you collect are relevant to your project.

If you can't imagine any working hypotheses, reconsider your question.

You may even decide to start over with a new topic.

That costs time in the short run, but it may save you from a failed project.

Under no circumstances should you put off thinking about a hypothesis until you begin drafting your paper or, worse, until you've almost finished it.

You might not settle on the *best* answer to your question until you're well into writing your paper, for writing is an act of discovery.

But you can't wait until that last draft to start thinking about *some* answer.

2.3.1 Beware the Risks in a Working Hypothesis

Don't settle on a final answer too soon: working hypotheses are meant to change.

But many new researchers and some experienced ones are afraid to consider *nay* working hypothesis early in their project, even one they hold lightly, because they fear it might bias their thinking.

There is some risk of that, but a working hypothesis need not close your mind to a better one.

Even the most objective scientist devises an experience to test for just a few predicted outcomes, often just one.

A working hypothesis is a risk only if it blinds you to a better one or if you can't give it up when the evidence says you should.

So as in all relationships, don't fall too hard for your 1st hypothesis: the more you like it, the less easily you'll see its flaws.

Despite that risk, it's better to start with a flawed hypothesis than with none at all.

2.3.2 If You Can't Find an Answer, Argue for Your Question

We have focused so much on questions that you might think your project fails if you can't answer yours.

Not so.

Much important research explains why a question no one has yet asked should be asked: *Do turtles dream? Why is yawning contagious?*

Papers addressing such questions don't argue for answers; they explain why the question is important and what a good answer might look like.

Or perhaps you find that someone has answered your question, but incompletely or even - if you're lucky - incorrectly.

If you can't find the right answer, you still help your research community by showing that a widely accepted one is wrong.

You can even organize your paper around a working hypothesis you abandon.

If after lots of research, you can't confirm it, you can explain why that answer seemed reasonable at the time but turned out to be wrong and so isn't worth the time of other researchers.

That in itself can be a valuable contribution to the conversation on your topic.

(See 10.1.1–10.1.2 for how to use an apparently good idea that turns out to be wrong.)

Only by asking question after question will you develop the critical imagination you need to excel at research.

Experienced researchers know there are few, if any, final answers, because there are no final questions.

They know that it's as important to ask a new question as it is to answer an old one, and that 1 day their new question will become old and yield to a newer researcher's still newer one.

That's how the conversations of research communities progress.

2.4 Build a Storyboard to Plan & Guide Your Work

For a short paper, you might not need a detailed plan - a sketch of an outline might do.

But for a long project, you'll usually need more, especially for one as long as a thesis or dissertation.

The 1st plan that comes to mind is usually an outline, with its *I*s and *II*s and *A*s and *B*s and so on (See **23.4.2.2**).

If you prefer an outline, use one, especially if your project is relatively short.

The problem is that an outline can force you to specify too much too soon and so lock up a final form before you've done your best thinking.

To avoid that risk, any researchers, including those outside the academic world, use a *storyboard* to plan and guide their work.

A storyboard is like an outline spread over several pages, with lots of spae for adding data and ideas as you go.

But it is more flexible than an outline because storyboard pages can be moved around, allowing you to try out new ways of organizing your ideas.

You can spread its pages across a wall, group related pages, and put minor sections below major ones to create a "picture" of your project that shows you at a glance the design of the whole and your progress through it.

For this reason, a storyboard is useful at every stage of your project.

It can help you plan your research, develop your argument, organize your paper, write a 1st draft, and test a final one.

Someday you may have the time to amble through sources, reading just what interests you.

Such random browsing has opened up many important lines of research.

But if your paper is due in a month or so, or even sooner, you need a plan.

A storyboard is a simple and reliable device to help you create one.

2.4.1 State Your Question & Working Hypotheses

To start a storyboard, state at the top of its 1st page your question and working hypothesis as precisely as you can.

Then add plausible alternatives and new hypotheses as you think of them, and cross off those you prove wrong.

But save them, because you might be able to use 1 of them in your introduction (see **10.1.1**).

2.4.2 State Your Reasons

The authors say more about the structure of arguments in Chap. 5.

For now, the commonsense understanding of an argument as a claim supported by reasons and evidence is enough.

Put at the top of separate pages each reason that might support your best hypothesis, even if you have only 1 or 2.

Imagine explaining your project to a friend.

You say, *I want to show that Alamo stories helped develop a unique Texan identity*, and your friend asks, *Why do you think so?*

Your reasons are the general statements that you offer to support your answer: *Well, 1st, the stories distorted facts to emphasize what became central to Texan identity; 2nd, the stories were 1st used to show that Texas (and the Wild West) was a new kind of frontier; 3rd, . . .* and so on.

If you can't think of more than 1 or 2 reasons, put placeholders at the tops of pages: *Reason 3: Something about Alamo stories making Texans feel special.*

If you know only *how* you want a reason to support your answer, state that: *Reason 4: Something to show that Alamo stories were more than just myth.*

Each reason, of course, needs support, so for each reason, ask, *Why do I think that? What evidence will I need to prove it?*

That will help you focus your search for evidence (see **2.4.3** and **5.4.2**).

If you're new to your topic or early in your project, your reasons may be only educated guesses that you'll later change.

But a list of reasons, no matter how speculative, cannot only guide your research but also focus your thinking and help you anticipate the argument you will eventually make.

2.4.3 Sketch the Kind of Evidence You Should Look For

Every field prefers its own kinds of evidence - numbers, quotations, observations, historical facts, images, and so on.

So for each reason, sketch the kind of evidence that you think you'll need to support it.

Even imagine what the most convincing evidence would look like.

If you can't imagine the kind of evidence you'll need, leave that part of the page blank, then read secondary sources to find out the kind of evidence researchers in your field typically use (see **3.1.2**).

2.4.4 Look at the Whole

Lay the pages on a table or tape them on a wall.

Then step back and look at their order.

When you plan a 1st draft, you must put its parts in some order, so you might as well think about one now.

Can you see a logic in your storyboard?

Cause and effect?

Narrative time?

Relative importance?

Complexity?

Length?

(See **6.2.5** for more principles of order.)

Try out different orders.

This storyboard isn't your final plan; it's only a tool to guide your thinking and organize what you find.

When you fill a page, try drafting that section, because writing out your ideas can improve your thinking at every stage of your project.

2.5 Join or Organize a Writing Group

1 of the best ways to stay on track with your project is to join or organize a writing group.

In many fields, especially in the humanities and social sciences, scholars read, think, and write mostly alone.

But it doesn't have to be that way, at least not entirely.

Find someone other than your instructor or advisor to talk with about your progress, to review your drafts, even to pester you about how much you have written.

That person might be a generous friend or, better, another writer with whom you can trade feedback on ideas and drafts.

Better yet is a writing group: 4 or 5 people working on their own projects who meet regularly to discuss each other's work.

Early on, start each meeting with a summary of each person's project in that 3-part sentence: *I'm working on the topic X, because I want to find out Y, so that I (and you) can better understand Z.*

As your projects develop, start with an "elevator story," a short summary of your research that you might give someone in the elevator on the way to the meeting.

It should include that 3-part sentence, a working hypothesis, and the major reasons supporting it (see **13.4**).

In later stages, share outlines and drafts so that the members of the group can serve as surrogate readers.

If your group has a problem with your draft, so will your final readers.

Your group can even help you brainstorm when you bog down.

All of this support is valuable.

But for many writers, a writing group is valuable simply for the discipline it imposes.

It is easier to meet a schedule when you know you must report your progress to others.

Writing groups are standard practice for those preparing theses or dissertations.

But the rules may differ for class papers.

Some teachers worry that writing groups or writing partners might provide more help than they should.

So if you join a group, be clear with your teacher about what it will do.

If you don't she may decide the assistance you have received is inappropriate (see **7.10**).

Chapter 3

Finding Useful Sources

Once you have a research question and perhaps a working hypothesis, you can start looking for the data you'll need to test your hypothesis and develop your argument.

In this chapter the authors explain how to find those data in sources and in the next how to work with them.

But don't think of finding sources and reading them as separate steps.

Once you have a promising source, read it to find other sources.

And as you fill your storyboard with notes, you'll discover gaps and new questions that only more sources can fill.

So while we discuss finding and using sources as distinct steps, you'll more often do them repeatedly and simultaneously.

3.1 3 Kinds of Sources & Their Uses

Sources are conventionally categorized into 3 kinds: *primary, secondary*, and *tertiary* (think 1st-, 2nd-, and 3rdhand).

Their boundaries are fuzzy, but knowing these categories can help you conduct your research.

3.1.1 Consult Primary Sources for Evidence

Primary sources are "original" materials that provide you with the "raw" data or evidence you will use to develop, test, and ultimately justify your hypothesis or claim.

In history, primary sources are artifacts or documents that come directly from the period you are studying: letters, diaries, objects, maps, even clothing.

In literature or philosophy, your main primary sources are usually the texts you are analyzing, and your data are the words on the page.

In the arts, your primary source would be the works of art you are interpreting.

In social sciences, e.g. sociology or political science, census or survey data would also count as primary sources, as would data obtained through observation or experiment in many fields.

The primary sources for those collected data are reports of original research in scholarly journals or government and commercial databases.

3.1.2 Read Secondary Sources to Learn from Other Researchers

Secondary sources are books, articles, papers, or reports that are based on primary sources and intended for scholarly or professional audiences.

An article in a scholarly journal analyzing Alamo stories would be a secondary source for researchers working on those stories.

The body of secondary sources in a field is sometimes called that field's *literature.*

The best secondary sources are books from reputable university presses and articles or reports that have been *peer reviewed*, meaning that they were vetted by experts in the field before they were published (see **3.3.2**).

Secondary sources also include specialized encyclopedias and dictionaries that offer essays written by scholars in a field.

You use secondary sources for 3 purposes:

1. *To keep up with current research.*

   Researchers read secondary sources to keep up with the work of other researchers, to inform and refine their thinking, and to motivate their own work by adding to a published line of research.

2. *To find other points of view.*

   A research paper is not complete until the researcher acknowledges and responds to the views of others and to this readers' predictable questions and disagreements.

   You can find most of those other points of view in secondary sources.

   What alternatives to your ideas do they offer?

   What evidence do they cite that you must acknowledge?

   Some new researchers think they weaken their case if they mention any view opposing their own.

   The opposite is the truth.

   When you acknowledge competing views, you show readers that you not only know those views but can confidently respond to them.

   (For more on this, see **5.4.3**.)

3. *To find models for your own research and analysis.*

   You can use secondary sources to find out not just *what* others have written about your topic but *how* they have written about it, as models for the form and style of your own paper.

   Imagine a secondary source as a colleague talking to you about your topic.

   As your respond, you'd want to sound like someone who knows the field, and so you'd try to learn how she reasons, the language she uses, the kinds of evidence she offers, and the kinds she rarely or never uses.

   The "conversation" would be in writing, so you'd even imitate stylistic details e.g. whether she writes in long paragraphs or breaks up her pages with subheads and bullet points (common in the social sciences, less common in the humanities).

You can also use a secondary source as a model for your conceptual analysis.

If, e.g., you were analyzing Alamo stories, you might study how a source treats Custer's Last Stand.

Is its approach psychological, social, historical, political?

Its particular reasons or evidence will probably be irrelevant to your project, but you might support your answer with the same kinds of data and reasoning, perhaps even following the same organization.

So if you come across a source that's not exactly on your topic but treats one like it, skim it to see how that researcher thinks about his material and presents it.

(You don't have to cite that source if you use only its general logic, but you may cite it to give your own approach more authority.)

You can even borrow evidence from secondary sources to use in your own arguments, but you should do so only if you do not have access to the primary sources from which that evidence was originally taken.

If you're doing advanced work, check the accuracy of important quotations, facts, or numbers you draw from secondary sources.

Of course, if you were studying how the Alamo story has been analyzed, then secondary sources offering those analyzes would be your primary sources.

If you're new to a field, you may find secondary sources hard to read: they assume a lot of background knowledge, and many aren't clearly written (see **11.2**).

If you're working on a topic new to you, you might begin with an overview in a specialized encyclopedia or reliable tertiary source.

3.1.3 Read Tertiary Sources for Introductory Overviews

Tertiary sources are books and articles that synthesize secondary sources for general readers.

They include textbooks, encyclopedias (including Wikipedia), and dictionaries, as well as articles in publications for broad audiences, like *Time* and the *Atlantic*.

In the early stages of research, you can use tertiary sources to get a broad overview of your topic.

But if you are making a scholarly argument, you should rely on secondary sources because these make up the conversation in which you are seeking to participate.

If you cite tertiary sources in a scholarly argument, you will mark yourself as a novice or outsider, and many readers won't take you - or your argument - seriously.

3.2 Search for Sources Systematically

Knowing where to begin your search for sources can be overwhelming at 1st.

It is temping to simply enter terms into a familiar search engine (e.g., Google) and see what comes up.

The authors do this too, but they also know that there are more systematic and productive ways for discovering useful, credible sources.

Make the library the focus of your search strategies even as you draw on the expertise of others inside or outside the academic community and use the resources of the internet.

3.2.1 Ask a Librarian

The best advice the authors can offer is to draw on the research expertise of librarians.

Both general reference librarians and (in larger libraries) subject area specialists can help you refine your search parameters and direct you to the right tools for your specific research question.

They can help you use the catalog to locate materials held by your library or by other libraries (and obtainable through interlibrary loan).

These same librarians typically design research guides that identify reference works and online databases for specific fields.

If you're a new researcher, seize every opportunity to learn online search techniques in your field.

And don't by shy.

Librarians love to assist researchers of all levels and at all stages of the research process.

They can help you formulate your research question and plan, develop search terms, and inventory your results to ensure you haven't overlooked something of value.

The only embarrassing question is the one you failed to ask.

If you already have a research question, share it: *I'm looking for data on X because I want to find out . . .*

If you have a working hypothesis and reasons, share them too: *I'm looking for data to show Y [your reason] because I want to claim Z [your hypothesis].*

Rehearse your questions to avoid wasting your time and theirs.

3.2.2 Consult Reference Works

Researchers in all fields share common values and habits of thought, but every field has its own ways of doing things.

To learn about the ways of your field, browse the shelves in your library's reference room that hold guides to your field's particular research methods, databases, and special resources (in the bibliography, see items in category 3 in your field).

At least familiarize yourself with the following resources (in the bibliography, see category 4 for lists of sources in your field; many are also online):

- a bibliography of works published each year in your field, e.g. *Philosopher's Index* or *Education Index*
- summary bibliographies of works on a specific topic collected over several years (*Bibliographic Index* is a bibliography of bibliographies)
- annual literature reviews; look for a title in your field beginning with *Reviews in . . .*
- specialized reference works e.g. the *Encyclopedia of Philosophy* or the *Concise Oxford Dictionary of Literary Terms*, where you may find an overview of your topic and often a list of standard primary and secondary sources (in the bibliography, see items in categories 1–2 in your field).

3.2.3 Explore Online Databases

What sets libraries apart from the internet are subscriptions to indexes and databases.

After books, these are arguably a library's most valuable assets, since they give researchers access to materials they could not obtain otherwise.

Each library's subscription will differ, with major research libraries offering the most comprehensive access to specialized indexes and databases.

However, academic libraries and many public libraries offer powerful online tools that greatly extend their actual collections.

You will certainly want to make use of these general and specialized resources in your research.

At least become familiar with the major databases to which your library subscribes, e.g. Academic Search Premier, MLA International Bibliography, or PubMed.

Many academic databases either provide abstracts or direct you to articles that include abstracts.

Looking at these can help you decide if an article itself is worth reading carefully.

Some databases allow you to access full-text articles or books.

But beware: if you library does not subscribe to a particular journal included in a database, you might be asked to pay a fee to access a full-text article.

Before doing so, *always* speak with a librarian about other means of access.

3.2.4 Search Your Library Catalog

In your research, you will likely use your library's catalog in 2 ways: keyword searching and browsing.

When you have examined some sources to identify a list of *keywords* related to your topic, you are ready to use these terms to search the catalog.

In most libraries you must choose the category (books, articles, journals, etc.) you wish to use for your search.

If your sources include books, you can use the Library of Congress subject headings, found either on the back of their little page or on their "details" page in the online catalog, to search for related materials.

E.g., the online entry for this book includes these 2 topics:

Example 3.2.1. *1. Dissertations, Academic. 2. Academic writing.*

If you search an online catalog, you will find all the books on that subject in that library.

A book may be cross-listed under multiple subject headings.

In that case, look at titles listed under those headings as well.

You can also *browse* the catalog for books with similar *call numbers*.

Also search your online catalog using keywords from your question or working hypothesis - *Alamo, Texas independence, James Bowie*.

If you find too many titles, start with those published in the last 10 or 15 years by well-known university presses.

For a wide selection, search WorldCat if your library subscribes.

Otherwise, search the Library of Congress catalog at https://www.loc.gov.

It has links to large university catalogs.

Start early if you expect to get books on interlibrary loan.

If most sources on your topic are articles, locate a recent one in your library's online databases.

Its database entry will include a list of keywords.

Search for them to find more articles on your topic.

In most cases you can just click on them.

Use the keywords to search the library catalog as well.

Some databases also provide abstracts of journal articles that you can skim for search terms.

3.2.5 Search Guides to Periodical Literature

If you've done research before, you're probably familiar with annual guides e.g. *Readers' Guide to Periodical Literature*, which cites sources e.g. magazines and newspapers.

Most specialized fields also have yearly guides to secondary sources, e.g. *Art Abstracts, Historical Abstracts*, and *Abstracts in Anthropology* (in the bibliography, see items in category 4 in your field).

Most are available online or in other digital forms.

All those resources will direct you to more sources, but none of them can substitute for the kind of in-library search that turns up an unexpectedly useful source.

3.2.6 Prowl the Stacks

Doing research online is faster than on foot, but if you never go into the stacks of your library (assuming you have access), you may miss crucial sources that you'd find only there.

More important, you'll miss the benefits of serendipity - a chance encounter with a source that you find only in person.

If you can get to the stacks, find the self with books on your topic, then scan the titles on that shelf and the ones above, below, and on their side, especially for books with new bindings published by university presses.

Then turn around and skim titles behind you; you never know.

When you spot a promising title, skim its table of contents, then its index for keywords related to your question or its answer.

Then skim its bibliography for titles that look relevant to your project.

You can do all that faster with books on a shelf than you can online.

If the book looks promising, skim its preface or introduction.

If it still looks promising, set it aside for a closer look.

Even if it doesn't seem relevant, record its Library of Congress call number and bibliographic data (author, title, publisher, date of publication and so on; see part 2 of this manual for the details), and in a few words summarize what the book seems to be about.

A month later, you might realize that it's more useful than you thought.

You can check tables of contents for many journals online, but browsing in the journals area of a library can be more productive.

Find the journals that have promising articles.

Skim tables of contents for the prior 10 years.

Most volumes include a yearly summary table of contents.

Then take a quick look at the journals shelved nearby.

Skim their most recent tables of contents.

You will be surprised at how often you find a relevant article that you would have missed had you done your work entirely online.

If you are new to a field, you can get a rough impression of the academic quality of a journal by its look.

If it's on glossy paper with lots of illustrations, even advertisements, it might be more journalistic than scholarly.

Those are not infallible signs of unreliable scholarship, but they are worth considering.

3.2.7 Follow Bibliographic Trails

Most sources will give you trailheads for bibliographic searches.

When you find a book or article that seems useful, skim its bibliography or works cited.

Its index will list the authors cited most often.

Journal articles usually begin with a review of previous research, all cited.

Finally, look for recent PhD dissertations even marginally related to your topic.

Almost every dissertation reviews research in its 1st of 2nd chapter.

By following these bibliographic trails, you can navigate the most difficult research territory, because 1 source leads to another, and another...

3.2.8 Search the Internet

Your practical experience using the internet for everyday research might lead you to believe that it is comprehensive and reliable, but that would be a mistake.

Your library's catalog and databases provide access to information that you cannot get through a search engine.

When doing research online, maintain a healthy skepticism: most of what we retrieve using Google or some other search engine is perfectly reliable, but not everything is.

In contrast to your library's catalog and databases, the internet is essentially unmonitored.

There is no one to vouch for the credibility of materials posted to countless websites.

And finally, keep in mind that companies offering free search engines make their money by acquiring data about you through online behavior and by selling advertising, and that webmasters routinely modify their sites to make them appear higher in search results.

But if you keep these limitations in mind, the internet can be an enormously valuable resource.

In particular, you can do useful preliminary work with a scholarly search engine e.g. Google Scholar or with a tertiary source e.g. Wikipedia (which you should never cite as a reference).

Of course, the internet can also be a primary source if you are researching the internet itself, including social media.

3.2.9 Look for Someone Who Knows Something about Your Topic

You might also ask around or search online to find someone who knows something about the topic and standard reference works on it: advanced students, faculty, even people outside the academic community.

You don't always find someone, but you might get lucky.

3.3 Evaluate Sources for Relevance & Reliability

You will probably find more sources than you can use, so you must evaluate their usefulness by skimming quickly for 2 criteria: relevance and reliability.

3.3.1 Evaluate Sources for Relevance

Once you decide a book might be relevant, do this:

- Skim its index for your keywords, then skim the pages on which those words occur.
- Skim the 1st and last paragraphs in chapters that use a lot of your keywords.
- Skim the book's introduction, especially its last page, where authors and editors often outline their text.
- Skim its last chapter or conclusion, especially the 1st and last several pages.
- Skim prologues, summary chapters, and so on.
- Check the bibliography for titles relevant to your topic.

Be sure that you're looking at a book's most recent edition.

Researchers change their views over time, refining them, even rejecting earlier ones.

If your source is an e-book, still follow these steps, but you can also search the whole text for your keywords.

If you are an advanced researcher, read book reviews of promising sources (see Sect. 4 of the bibliography of resources in your field).

If your source is an journal article, do this:

- Read the abstract, if it has one.
- Skim the introduction and conclusion; if they are not marked off by headings, skim the 1st 6 or 7 paragraphs and the last 4 or 5.
- Skim for section headings, and read the 1st and last paragraphs of those sections.
- Check the bibliography for titles relevant to your topic.

If your source is online, do this:

- If it looks like a printed article, follow the steps for journal article, and also search for your keywords.
- Skim sections labeled "Introduction," "Overview," "Summary," or the like.

  If there are none, look for a link labeled "About the Site" or something similar.
- If the site has a link labeled "Site map" or "Index," check it for your keywords and skim the referenced pages.
- If the site has a search function, type in your keywords.

3.3.2 Evaluate Sources for Reliability

You can't judge a source until you read it, but there are signs of its reliability:

1. **Is the source published by a reputable press?**

   Most books and journals published by university presses, whether in print or in electronic editions, are reliable, especially if you recognize the name of the university.

   You can also trust some commercial presses in some fields, e.g. Norton in literature, Ablex in sciences, or West in the law.

   Be skeptical of a commercial book that makes sensational claims, even if its author has a PhD after his name.

   Be especially careful on contested social issues e.g. gun control or climate change.

   Many books and articles are published by individuals or organizations driven more by ideology than by evidence.

2. **Is the book or article peer reviewed?**

   Most reputable presses and journals ask experts to review a book or article before it is published; this is called *peer review*.

   Essay collections published by university presses are often but not always peer reviewed; someones they are reviewed only the named editor or editors.

   Few commercial magazines use peer review, and fewer still check an author's facts.

   If a book or article hasn't been peer reviewed, use it cautiously.

3. **Is the author a reputable scholar?**

   This is a hard question to answer if you are new to a field.

   Most established scholars are reliable, but use good judgment: even reputable scholars can have axes to grind, especially if their research is supported by a special interest group.

4. **If the source is available only online, is it sponsored by a reputable organization?**

   A website is only as reputable as its sponsor.

   You can usually trust one that is sponsored and maintained by a reputable organization.

   Some sites run by individuals are reliable; most are not.

   Do a web search for the name of the sponsor.

5. **Is the source current?**

   You must use up-to-date sources, but what counts as current depends on the field.

   In computer science, a journal can be out of date in months; in the social sciences, 10 years pushes the limit.

   Publications have a longer shelf life in the humanities: literary or art criticism, e.g., can remain relevant for decades, even centuries.

6. **If the source is a book, dose it have notes and a bibliography?**

   If not, by suspicious, because you have no way to follow up on anything that the source claims.

7. **If the source is a website, does it include bibliographic data?**

   You cannot judge the reliability of a site that does not indicate who sponsors and maintains it, who wrote what's posted there, and when it was posted and last updated.

8. **If the source is a website, does it approach its topic judiciously?**

   Your readers are unlikely to trust a site that engages in heated advocacy, attacks other researchers, makes wild claims, uses abusive language, or makes errors of spelling, punctuation, and grammar.

   The following criteria are especially important for advanced researchers.

9. **If the source is a book, has it been well reviewed?**

   Many fields have indexes to published reviews that tell you how others evaluate a source.

10. **Has the source been frequently cited by others?**

    You can roughly estimate how influential a source is by how often others cite it.

    To determine that, consult a citation index (in the bibliography, see Sect. 4 in your field).

    If you find that a source has been cited repeatedly by other scholars, you can infer that experts in the field regard it as reliable and significant.

    Such sources are said to have an "high impact factor."

Those signs don't guarantee that a source is reliable, but they should give you reasonable confidence in it.

If you can't find reliable sources, acknowledge the limits of the ones you have.

Of course, you may find an exciting research problem when you discover that a source thought to be reliable is not.

3.4 Look beyond the Usual Kinds of References

For a class paper, you'll probably use sources typical in your field.

But if you are doing an advanced project e.g. an MA thesis or PhD dissertation, search beyond them.

If, e.g., you were doing a project on the economic effects of agricultural changes in late 16th-century England, you might read some Elizabethan plays, look at woodcuts of agricultural life, find commentary by religious figures on rural social behavior.

Conversely, if you were working on visual representations of daily life in London, you might work up the economic history of the period and place.

When you look beyond the standard kinds of references relevant to your question, you enrich not only your analysis but also your range of intellectual reference and your ability to synthesize diverse kinds of data, a crucial competence of an inquiring mind.

Don't ignore a work on your topic that is not mentioned in the bibliographies of your most relevant sources - you'll get credit for originality if you turn up good sources others have ignored.

3.5 Record Your Sources Fully, Accurately, & Appropriately

Once you decide a source is worth reading, record all of its bibliographic information.

Your 1st obligation as a researcher is to cite your sources accurately and fully so that your readers can find them.

3.5.1 Determine Your Citation Style

Most fields require a specific citation style.

The 2 most common ones are described in detail in part 2:

- *notes-bibliography style* (or simply *notes style*), used widely in the humanities and in some social sciences (see Chaps. 16–17)
- *author-date style*, used in most social sciences and in the natural sciences (see Chaps. 18–19).

If you are uncertain which style to use, consult your instructor or advisor.

Before you start compiling your list of sources, read the general introduction to citations in Chap. 15 and then, depending on the citation style you are required to use, read the introduction to notes style (Chap. 16) or author-date style (Chap. 18).

3.5.2 Record Complete Bibliographical Information

To save time and avoid errors, record all the citation information you will need when you 1st find a source.

Most of this information appears on the title page of a book or at the head of a journal article.

The specific information you need depends on the type of source, bur for each source, answer the following:

- Who wrote or assembled the source?
  - author(s)
  - editor(s) (if any)
  - translator(s) (if any)
- What data identify the source?
  - title (including subtitle)
  - page numbers (if the source appears in a larger work)
  - volume number
  - issue number
  - edition number
- Who published the source and in what context?
  - publisher's name
  - place of publication
  - name of the journal, collection, or other work in which the source appears
- When was the source published?
  - year of publication
  - season, month, or specific day (and, in some cases, time)
  - for online sources, the date you accessed the material (whether or not you include this date in your citation)
- Where can the source be found?
  - for online sources, a URL or the name of the database
  - for physical items in a 1-of-a-kind collection, the place that houses the collection

For your own use, you might record Library of Congress call numbers.

You won't include them in bibliographic citations, but you may find them helpful if you must consult the source again.

If you access a printed text online, record the URL and any other data for the online version in addition to the full bibliographic data for the original source.

If you scan or photocopy a passage from a book, also scan or photocopy its title page and the bibliographic information on the reverse side.

Then add the library call number if you know it.

At some point, you'll need to format this bibliographic information into your required citation style, so you should record your sources in that style now.

(You can find templates and examples for notes style in Fig. 16.1 and Chap. 17; for author-date style, refer to Fig. 18.1 and Chap. 19.)

For many types of sources, you can copy and paste citation data online if you know where to look.

And there are a number of programs that will help you collect and organize your citations and automatically insert them in your paper in the proper format.

These are useful aids, but they are not perfect, so plan on double-checking your citations not only as you acquire them but also later, after they've been inserted in your paper.

Chapter 4

Engaging Your Sources

Once you have a research problem, use it to guide your research for evidence, models, and arguments to respond to.

But if you don't yet have 1 firmly in hand, you won't know which data, models, or arguments will prove relevant.

So read sources not randomly but deliberately to find a problem.

Look for claims that seem puzzling, inaccurate, or simplistic - anything you can disagree with.

You're more likely to find a research problem when you disagree with a source, but you can also find one in sources you agree with.

Experienced researchers don't read passively; they engage their sources actively, entering into conversation with them.

One you find a source worth a close look, don't read it mechanically, just mining it for data to record.

Note-taking is not clerical work.

When you talk notes on a source thoughtfully, you engage not just its words and ideas but also its implications, consequences, shortcomings, and new possibilities.

4.1 Read Generously to Understand, Then Critically to Engage

If you can, read promising sources twice.

1st, read generously.

Pay attention to what sparks your interest.

Reread passages that puzzle or confuse you.

Don't look for disagreements right away, but read in ways that help the source make sense.

Disagree too soon and you will misunderstand your source or exaggerate a weakness if it presents an argument that challenges yours.

Then, if a source seems important or challenges your own position, read it a 2nd time slowly and more critically.

When you read a passage, think not only about what it says but about how you would respond.

Record those responses in your notes or - if you own the source or a copy of it - in the margins of the source.

Test your understanding by summarizing; if you can't sum up a passage in your mind, you don't understand it well enough to disagree.

You probably won't be able to engage your sources fully until you've done a bit of reading and have developed your own ideas further.

But from the outset, be on alert for ways to engage your sources, actively and creatively.

At some point, the earlier the better, you must look for ways to go *beyond* your sources, too, even when you agree with them.

4.1.1 Look for Creative Agreement

It's a happy moment when a source confirms our views.

But if we just passively agree, we don't develop our own ideas.

So if you believe what a source claims, try to extend that claim: What new cases might it cover?

What new insights can it provide?

Is there confirming evidence the source hasn't considered?

Here are some ways to find a problem through creative agreement.

1. **Offer additional support.** You can offer new evidence to support a source's claim.

Example 4.1.1. *Smith uses anecdotal evidence to show that the Alamo story had mythic status beyond Texas, but editorials in big-city newspapers off better evidence.*

- Source supports _____ with old evidence, but maybe you can offer new evidence.
- Source supports _____ with weak evidence, but maybe you can offer stronger evidence.

2. **Confirm unsupported claims.** You can prove something that a source has only assumed or speculated.

   Example 4.1.2. *Smith recommends visualization to improve sports performance, but a study of the mental activities of athletes shows why that is good advice.*

   - Source speculates _____ might be true, but you can offer evidence to show that it is.
   - Source assumes _____ is true, but maybe you can prove it.

3. **Apply a claim more widely.** You can extend a position to new areas.

   Example 4.1.3. *Smith argues that medical students learn physiological processes better when they are explained with many metaphors rather than with just 1.*

   The same appears to be true for engineers learning physical processes.

   - Source correctly applies _____ to 1 situation, but maybe it can apply to new ones.
   - Source claims that _____ is true in a specific situation, but maybe it's true in general.

4.1.2 Look for Creative Disagreement

If you read actively, you'll inevitably find yourself disagreeing with your sources.

Don't brush those disagreement aside, because they often point to new research problems.

So instead of just noting that you disagree with another writer's views, use that disagreement to encourage your own productive thinking.

Look for these types (the list is not exhaustive, and some overlap).

1. **Contradictions of kind.** A source says something is 1 kind of thing, but maybe it's another kind.

   Example 4.1.4. *Smith says that certain religious groups are considered "cults" because of their strange beliefs, but those beliefs are no different in kind from standard religions.*

   - Source claims that _____ is a kind of _____, but it's not.
   - Source claims that _____ always has _____ as 1 of its features or qualities, but it doesn't.
   - Source claims that _____ is normal/good/significant/useful/moral/interesting, but it's not.

   You can reverse those claims and the ones that follow to state the opposite:

   - Though a source says _____ is *not* a kind of _____, you can show that it is.

2. **Part-whole contradictions.** You can show that a source mistakes how the parts of something are related.

   Example 4.1.5. *Smith has argued that coding is irrelevant to a liberal education, but in fact, it is essential.*

   - Source claims that _____ is a part of _____, but it's not.
   - Source claims that 1 part of _____ relates to another in a certain way, but it doesn't.
   - Source claims that every _____ has _____ as 1 of its parts, but it doesn't.

3. **Developmental or historical contradictions.** You can show that source mistakes the origin and development of a topic.

   Example 4.1.6. *Smith argues that the world population will rise, but it won't.*

   - Source claims that _____ is changing, but it's not.
   - Source claims that _____ originated in _____, but it didn't.
   - Source claims that _____ develops in a certain way, but it doesn't.

4. **External cause-effect relations.** You can show that a source mistakes a causal relationship.

   Example 4.1.7. *Smith claims that legalizing marijuana will increase its use among teenagers, but evidence shows that it doesn't.*

   - Source claims that _____ causes _____, but it doesn't.
   - Source claims that _____ causes _____, but it doesn't; they are both caused by _____.
   - Source claims that _____ is sufficient to cause _____, but it's not.
   - Source claims that _____ causes only _____, but it also causes _____.

5. **Contradictions of perspective.** Most contradictions don't change a conceptual framework, but when you can contradict a "standard" view of things, you urge others to think in a new way.

   Example 4.1.8. *Smith assumes that advertising has only an economic function, but it also serves as a laboratory for new arts forms.*

   - Source discusses _____ from the point of view of _____, but a new context or point of view reveals a new truth. (The new or old context can be social, political, philosophical, historical, economic, ethical, gender specific, etc.)
   - Source analyzes _____ using theory/value system _____, but you can analyze it from a new point of view and see it in a new way.

As the authors said, you probably won't be able to engage sources in these ways until after you've read enough to form some views of your own.

But if you keep these ways of thinking in mind as you begin to read, you'll engage your sources sooner and more productively.

Of course, once you discover that you can productively agree or disagree with a source, you should ask *So what?*

So what if you can show that while Smith claims that easterners did not embrace the story of the Alamo enthusiastically, in fact many did?

4.2 Take Notes Systematically

Once you find and record a source you think you can use, you must read it purposefully and carefully.

Then take notes in a way that will help you not only to remember and use what you have read but also to further your own thinking.

Like the other steps in a research project, note-taking goes better with a plan.

4.2.1 Taking Notes on Paper

Years ago, the standard way to take notes on sources was to create a file of index cards (see Fig. 4.1).

Fig. 4.1. Example of a note card.

At the top left is the author, short title, and page number.

At the top right are keywords that let the researcher sort and re-sort notes into different categories and orders.

The body of the card summarizes the source, records a direct quotation, and includes a comment or thought about further research that is clearly distinguished from the quotation.

At the bottom left is the all number for the source.

A card like this seems old-fashioned, but it provides a template for efficient note-taking:

- Record complete bibliographic information for each source so that you can cite it properly and find it again easily.
- Separate notes on different topics, even if they come from the same source.
- Make sure your notes are accurate, because you need to be able to rely on them later.

  (If you want to quote more than a few lines, copy or save the passage or the whole document.)
- Clearly distinguish
  1. what you quote from a source,
  2. what you paraphrase or summarize from a source, and
  3. your own thoughts.

If you are writing on paper, use headings or brackets or distinct colors to differentiate these 3 different kinds of notes.

You might also create a section specially dedicated to *your own* responses, agreements, disagreements, speculations, and so on (see the italicized section in Fig. 4.1).

That will encourage you to do more than simply record the content of what you read.

4.2.2 Taking Notes Electronically

When you take notes electronically, you have some points:

- You can use a word processor.

  Create a separate file (or at least a separate page) for each source, and be sure to unambiguously distinguish your words from those of your source.

  Though word processors are easy to use, they also limit your ability to index, organize, sort, and search your notes.

  For long or complex projects, you might consider other options.

- You can use a dedicated note-taking application to create and organize your notes.

  Such applications can help you to index, sort, and access your notes.

  But since they sometimes use proprietary formats, they can make it difficult to share your notes or use them with other programs.

- You can use a full-featured citation management program.

  In addition to allowing you to make your own notes, these programs can often pull information directly from online library catalogs and databases, and they can format and update your bibliographies as you write.

  Some will even store full electronic copies of your sources, creating a personal library for your project.

  But like note-taking applications, these programs sometimes use proprietary formats.

In electronic notes, as in notes on paper, you must clearly distinguish your own words and ideas from those of your source.

- Record quotations from your source in a distinctive color or font so that you can recognize them at a glance, and enclose them in large quotation marks in case the file loses its formatting.

- Record paragraphs from your source in a 2nd color or front so that you can't possibly mistake them for your own ideas, and enclose them in curly brackets in case the file loses its formatting.

- Record your own thoughts in a 3rd color or font, and enclose them in square brackets.

  Put longer responses in a separate section so there is no chance you will mistake your own ideas for your source's, or vice versa.

4.2.3 Decide Whether to Summarize, Paraphrase, or Quote

If you can photocopy, scan, or otherwise reproduce your source, or you know you can access it online when you write, you can focus less on preserving its exact words than on your engagement with it.

Summarize the source, which will help you understand it, and note passages you might want to quote or paraphrase when you write.

Note also your response to the source.

Where did you find yourself agreeing or disagreeing with it?

If you can't preserve your source and you don't know whether you will be able to access it later, you have a tougher choice.

It takes too long to transcribe the exact words of every passage you might want to use,so when taking notes, you mus choose as you go whether to quote, paraphrase, or summarize.

Every choice depends on how you plan to use your source:

- Summarize when you need only the point of a passage, section, or even whole article or book.

  Summary is useful for context and for data or claims that are related - but not directly relevant - to your project.

  A summary of a source can never serve as good evidence (see **5.4.2**).

- Paraphrase when the specific words of a passage are less important than its meaning.

  Paraphrase doesn't mean changing just a word or 2.

  You must replace most of the words and phrasing of the original with your own (see **7.9.2**).

  As evidence, a paraphrase is never as good as a direct quotation.

- Record exact quotations for these purposes:

  - The quoted words are evidence that backs up your reasons.
    If, e.g., you claimed that different regions responded to the Battle of the Alamo differently, you would quote exact words from different newspapers.
    You would paraphrase them if you needed only their general sentiments.
  - The words are from an authority you plan to rely on or challenge.
  - The words are strikingly original or so compelling that the quotation can frame the rest of your discussion.
  - The words express a claim that you disagree with, and to be fair you want to state it exactly.

If you don't record important words now, you can't quote them later.

So copy or photocopy more passages than you think you'll need (for more on photocopying, see **4.3.1**).

Never abbreviate a quotation thinking you can accurately reconstruct it later.

You can't.

And if you misquote, you'll damage your credibility.

4.2.4 Guard against Inadvertent Plagiarism

For students and professionals alike, sloppy note-taking has caused grief ranging from ridicule for trivial errors to professional exile for inadvertent plagiarism.

To avoid that risk, commit to heart these 2 iron rules for recording information in notes:

- *Always* unambiguously identify words and ideas from a source so that weeks or months later you cannot possibly mistake them for your own.

  As recommended above, record quotations and paraphrases with quotation marks, as well as in a font or color that unambiguously distinguishes them from your own ideas.

- *Never* paraphrase a source so closely that a reader can match the phrasing and sense of your words with those in your source (see **7.9.2**).

In fact, rather than retyping quotations of more than a few lines, download or photocopy them.

Add to the top of the downloaded or photocopied pages the name of the source and keywords for sorting.

This is important: *never* assume that you can use what you find online without citing its source, even if it's free and publicly available.

You must acknowledge your use of *anything* you did not create yourself.

(For more on plagiarism, see **7.9**.)

4.3 Take Useful Notes

Readers will judge your work not just by the quality of your sources and how accurately you report them but also by how deeply you engage them.

To do that, you must take notes in a way that not only reflects but encourages a growing understanding of your project.

4.3.1 Use Note-Taking to Advance Your Thinking

Many inexperienced researchers think that note-taking is a matter of merely recording data.

Recording or photocopying can help you quote or paraphrase accurately, but if you don't *engage* your sources, you simply accumulate inert data.

To advance your thinking, on any pages you've copied, annotate key sentences and passages by highlighting or labeling them in the margin.

Mark ideas or data that you expect to use in your paper.

Summarize what you have highlighted, or sketch a response to it, or add notes in the margin that help you interpret your highlighting.

The more you write about a source now, the better you will understand it later.

4.3.2 Take Notes Relevant to Your Question & Working Hypothesis

For sources you think especially useful, record not just facts that you think you can use as evidence but also other information that helps you explain those facts and their relationship to your claim.

You can create a template to help with this (see **4.2.1**).

The 1st 3 items directly support or challenge your working hypothesis:

- reasons that support your hypothesis or suggest a new one
- evidence that supports your reasons
- views that undermine or even contradict your hypothesis (see **5.4.3**)

These next items might not support or challenge your hypothesis, but they may help you explain its context or simply make your paper more readable:

- historical background of your question and what authorities have said about it, particularly earlier research (see **6.2.2** an **10.1.1**)
- historical or contemporary context that explains the importance of your question
- important definitions and principles of analysis
- analogies, comparisons, and anecdotes that might not directly support your hypothesis but do explain or illustrate complicated issues or simply make your analysis more interesting
- strikingly original language relevant to your topic.

4.3.3 Get the Context Right

You can't record everything, but you have to record enough to ensure that you accurately capture the source's meaning.

As you use material from sources, record not just what they say but how they use the information.

Here are some ways to guard against misleading your reader.

1. **When you quote, paraphrase, or summarize, be sure to capture the context.**

   When you note an important conclusion, record the author's line of reasoning:

   Example 4.3.1. *Not: Bartolli (p. 123): The war was caused by Z.*

   But Bartolli: The war was caused by X, Y, and Z (p. 123). But the most important cause was Z (p. 123), for 2 reasons: reason 1 (pp. 124–26); reason 2 (p. 126).

   Sometimes you care only about a conclusion, but readers usually want to see how a conclusion emerges from the argument supporting it.

   So when you take notes, record not only conclusions but also the arguments that support them.

2. **When you record a claim, note its role in the original.** Is it a main points?

   A minor point?

   A qualification or concession?

   By noting these distinctions, you avoid this kind of mistake:

   Example 4.3.2. *Original by Jones: "Researchers recognize that lung cancer has a number of causes, including genetic predisposition and exposure to environmental factors e.g. asbestos, radon, and fine particulates. But no one who has studied the data doubts that lung cancer's leading cause is smoking."*

   Misleading report about Jones: Smoking is just 1 cause of lung cancer among many. Jones, e.g., claims that "lung cancer has a number of causes, including genetic predisposition and exposure to environmental factors e.g. asbestos, radon, and fine particulates."

   Jones did not make that point at all.

   He *conceded* a point to set up a point he wanted to make.

   Anyone who deliberately misrepresents an author in this way violates basic standards of truth.

   But you can make such a mistake inadvertently if you note only a source's words and not their role in an argument.

3. **Record the scope & confidence of a claim.** These are not the same:

   Example 4.3.3. *Chemicals in french fries cause cancer.*

   Chemicals in french fries may be a factor in causing cancer.

   Some chemicals in french fries correlate with a higher incidence of cancer.

4. **Don't mistake a summary of another writer's views for those of an author summarizing them.** Some writers do not clearly indicate when they summarize another's argument, so it's easy to quote them as saying what they set out to disprove rather than what they in fact believe.

5. **Note *why* sources agree & disagree.** 2 social scientists might claim that a social problem is caused by personal factors, not by environmental forces, but one might cite evidence from genetic inheritance while the other points to religious beliefs.

   How and why sources agree is as important as the fact that they do.

   In the same way, sources might disagree because they interpret the same evidence differently or take different approaches to the problem.

   It is risky to attach yourself to what any 1 researcher says about an issue.

   It is not "research" when you uncritically summarize another's work.

   If you rely on at least 2 sources, you'll usually find that they do not agree entirely, and that's where your own research can begin.

4.4 Review Your Progress

Regularly review your notes and storyboard to see where you are and where you have to go.

In a storyboard, full pages indicate reasons with support; empty pages indicate work to do.

Consider whether your working hypothesis is still plausible.

Do you have good reasons supporting it?

Good evidence to support those reasons?

Can you add new reasons or evidence?

4.4.1 Search Your Notes for an Answer

The authors have urged you to find a working hypothesis or at least a question to guide your research.

But some writers start with a question so vague that it evaporates as they pursue it.

If that happens to you, search your notes for a generalization that might be a candidate for a working hypothesis, then work backward to find the question it answers.

Look 1st for questions, disagreements, or puzzles in your sources and in your reaction to them (see **2.1.3** and **4.1**).

What surprises you might surprise others.

Try to state that surprise:

Example 4.4.1. *I expected the 1st mythic stories of the Alamo to originate in Texas, but they didn't.*
They originated in . . .

That working hypothesis suggests that the Alamo myth began as a national, not a regional, phenomenon - a modest but promising start.

If you can't find a working hypothesis in your notes, look for a pattern of ideas that might lead you to one.

If you gathered data with a vague question, you probably sorted them under predictable keywords.

For masks, the categories might be their origins (*African, Indian, Japanese,. . .*), uses (*drama, religion, carnival,. . .*), materials (*gold, feather, wood,. . .*), and so on.

E.g.,:

Example 4.4.2. *Egyptians - mummy masks of gold for nobility, wood for others*
Aztecs - masks from gold and jade buried only in the graves of the nobility
New Guinea tribes - masks for the dead from feathers from rare birds

Those facts could support a general statement e.g. *Mask-making cultures use the most valuable materials available to create religious masks, especially for the dead.*

Once you can generate 2 or 3 such statements, try to formulate a still larger generalization that might include them all:

Example 4.4.3. *Many cultures invest great material and human resources in creating masks that represent their deepest values. - generalization*

Egyptians, Aztecs, and Oceanic cultures all created religious masks out of the rarest and most valuable materials.

If you think that some readers might plausibly disagree with that generalization, you might be able to offer it as a claim that corrects their misunderstanding.

4.4.2 Invent Your Question

Now comes a tricky part.

It's like reverse engineering: you've found the answer to a question that you haven't yet asked, so you have to reason backward to invent the question that your new generalization answers.

In this case, it might be *What signs indicate the significance of masks in the societies of those who make and use them?*

As paradoxical as it may seem, experienced researchers often discover their question *after* they answer it, the problem they should have posed after they solve it.

4.4.3 Re-sort Your Notes

If none of that helps, try re-sorting your notes.

When you 1st selected keywords for your notes, you identified general concepts that could organize not just your evidence but your thinking.

If you chose keywords representing those concepts carefully, you can re-sort your notes in different ways to get a new slant on your material.

If your keywords no longer seem relevant, review your notes to create new ones and reshuffle again.

4.5 Manage Moments of Normal Anxiety

As you get deeper into your project, you may experience moments when everything seems to run together into a hopeless muddle.

That usually happens when you accumulate notes faster than you can sort them.

Such moments can be stressful, but they can also be a sign that you are on the verge of a new insight or discovery.

You can minimize the anxiety by taking every opportunity to organize and summarize what you have gathered by *writing as you go* and by returning to the central questions: *What question am I asking? What problem am I posing?*

Keep rehearsing that formula: *I am working on X to learn more about Y, so that my readers can better understand Z.*

Writing regularly about these questions does more than help you stay focused; it also helps you think.

You can turn to friends, classmates, teachers - anyone who will serve as a sympathetic but critical audience.

Explain how what you have learned bears on your question and helps you resolve your problem.

Ask them, *Does this make sense? Am I missing something important? What else would you like to know?*

You will profit from their reactions and even more from the mere act of explaining your ideas to nonspecialists.

Chapter 5

Constructing Your Argument

Most of us would rather read than write.

There is always another article to read, 1 more source to track down, just a bit more data to gather.

But well before you've done all the research you'd like to do, there comes a point when you must start thinking about the 1st draft of your paper.

You might be ready when your storyboard starts to fill up and you're satisfied with how it looks.

You will know you're ready when you think you can sketch a reasonable case to support your working hypothesis (see **2.3**).

If your storyboard is full and you still can't pull together a case strong enough to plan a draft, you may have to rethink your hypothesis, perhaps even your question.

But you can't be certain where you stand in that process until you at least try to plan that 1st draft.

In this chapter the authors explain how to build your argument; in the next, how to organize it.

As you gain experience, you'll learn to combine those 2 steps into 1.

5.1 What a Research Argument Is & Is Not

The word *argument* has negative associations these days because it evokes images of people shouting at one another.

In that kind of argument the goal is to win, to bludgeon or intimidate one's opponent into assent or silence.

But a research argument isn't like that.

As the authors suggested in Chap. 1, it is more like a conversation with a community of receptive but skeptical colleagues.

Such readers won't necessarily oppose your claims (although they might), but they also won't accept them until they see good reasons based on reliable evidence and until you respond to their questions and reservations.

When you make (not *have*) an argument in a face-to-face conversation, you cooperate with your listeners.

You state your reasons and evidence not as a lecturer would to a silent audience but as you would engage friends sitting around a table: you offer a claim and reasons to believe it; they probe for details, raise objections, or offer their points of view; you respond, perhaps with questions of your own; they ask more questions.

At its best, it's an amiable but thoughtful back-and-forth that develops and tests the best case that you and they can make *together*.

In writing, even when done collaboratively, that kind of cooperation is harder.

You must not only answer your imagined readers' questions but also *ask them on their behalf* - as often and as sharply as real readers will.

Your aim isn't to think up clever rhetorical strategies that will persuade readers to accept your claim regardless of how good it is.

It is to test your claim and especially its support, so that you offer your readers the best case you can make.

In a good research paper, readers hear traces of that imagined conversation.

As the authors have said, reasoning based on evidence isn't the only way to reach a sound conclusion, sometimes not even the best way.

We often make good decisions by relying on intuition, feeling, even spiritual inspiration.

But when we try to *explain* why we believe our claims are sound and why others should too, we have no way to *demonstrate* how we reached them, because we can't offer intuitions, feelings, or inspirations as evidence for readers to evaluate.

We can only say we had them and ask readers to take our claim on faith, a request that readers of research papers rarely grant.

When you make a research argument, therefore, you must lay out your reasons and evidence so that your readers can consider them; then you must imagine both their questions and your answers.

Doing all this is hard, but remembering how arguments work in everyday conversations will help you.

5.2 Build Your Argument around Answers to Readers' Questions

Consider the kind of conversation you have every day:

Example 5.2.1. *Abby: I hear you had a hard time last semester. How do you think this one will go?* [Abby poses a problem in the form of a question.]

Brett: Better, I hope. [Brett answers the question.]

Abby: Why so? [Abby asks for a reason to believe Brett's answer.]

Brett: I'm taking courses in my major. [Brett offers a reason.]

Abby: Like what? [Abby asks for evidence to back up Brett's reason.]

Brett: History of Art, Intro to Design. [Brett offers evidence to back up his reason.]

Abby: Why will taking courses in your major make a difference? [Abby doesn't see the relevance of Brett's reason to his claim that he will do better.]

Brett: When I take courses I'm interested in, I work harder. [Brett offers a general principle that relates his reason to his claim that he will do better.]

Abby: What about that math course you have to take? [Abby objects to Brett's reason.]

Brett: I know I had to drop it last time I took it, but I found a good tutor. [Brett acknowledges Abby's objection and responds to it.]

If you can imagine yourself in that conversation, you'll find nothing strange about assembling a research argument.

That's because the 5 elements of any arguments are just answers to the kinds of questions.

Abby asks Brett - and that you must ask yourself on your reader's behalf.

- **Claim**: What do you want me to believe?

  What is your point?

- **Reason**: Why do you say that?

  Why should I agree?

- **Evidence**: How do you know?

  Can you back it up?

- **Acknowledgment & response**: But what about ...?

- **Warrant**: How does that follow?

  Can you explain your reasoning?

Think of your research as the process of figuring out answers to those questions.

5.3 Turn Your Working Hypothesis into a Claim

The authors described the early stages of research as finding a question and imagining a tentative answer.

The authors called that answer your *working hypothesis*.

Now as we discuss building an argument to support that hypothesis, the authors change their terminology 1 last time.

When you think you can back up your hypothesis with good reasons and evidence, you'll present that hypothesis as your argument's *claim*.

A claim is an assertion (which could be 1 sentence or several) that demands support.

Your *main claim* is the assertion supported by your whole research argument.

Some call this assertion your *thesis*.

5.4 Assemble the Elements of Your Argument

At the core of your argument are 3 elements: your claim, your reasons for accepting it, and the evidence on which those reasons are based.

To that core you'll add 1 and perhaps 2 more elements: one responds to questions, objections, and alternative points of view; the other answers those who do not understand how your reasons are *relevant* to your claim.

5.4.1 State & Evaluate Your Claim

Start a new 1st page of your storyboard (or outline).

At the bottom, state your claim in a sentence or 2.

Be specific as you can, because the words in this claim will help you plan and execute your draft.

Avoid vague value words like *important, interesting*, and *significant.*

Compare the 2 following claims:

Example 5.4.1. *Masks play a significant role in many religious ceremonies.*

In cultures from pre-Columbian America to Africa and Asia, masks allow religious celebrants to bring deities to life so that worshipers experience them directly.

Now judge the *significance* of your claim (*So what?* again).

A significant claim doesn't make a reader think *I know that*, but rather *Really? How interesting. What makes you think so?* (Review **2.1.4**.)

These next 2 claims are too trivial to justify reading, much less writing, an argument to back them up:

Example 5.4.2. *This paper discusses teaching popular legends e.g. the Battle of the Alamo to elementary school students.* (So what if it does?)

Teaching United States history through popular legends e.g. the Battle of the Alamo is common in elementary education. (So what if it is?)

Of course, what your readers will count as interesting depends on what they know, and if you're early in your research career, that's something you can't predict.

If you're writing 1 of your 1st papers, assume that your most important reader is you.

It is enough if *you alone* think your answer is significant, if it makes you think, *Well, I didn't know that when I started.*

As you become more experienced and come to understand your particular research community better, you'll learn to frame your arguments in terms of your readers' interests (see Chap. 1).

If, however, you think you own claim is vague or trivial, you're not ready to assemble an argument because you have no reason to make one.

5.4.2 Support Your Claim with Reasons & Evidence

At the core of every research argument is the answer to your research argument is the answer to your research question, the solution to your problem - your main claim.

You have to back up that claim with 2 kinds of support: reasons and evidence.

A reason is a statement that leads readers to accept a claim.

The authors often join a reason to a claim with *because*:

Example 5.4.3. *Elementary schools should make teaching foreign languages a priority [claim] because we acquire languages best and most easily when we are young. [reason]*

You often need more than 1 reason to support a claim, and in a complex argument, your reasons will be claims themselves, requiring support with additional reasons in turn.

Evidence is the data on which you base your reasons.

It may seem obvious that you must back up a claim with reasons and evidence, but it's easy to confuse those 2 words because we often use them as if they mean the same thing:

Example 5.4.4. *You have to base your claim on good reasons.*

You have to base your claim on good evidence.

Reasons and evidence are not the same thing, and distinguishing them is crucial in making sound arguments.

Compare these 2 sentences:

Example 5.4.5. *On what evidence do you have your reasons?*

On what reasons do you base your evidence?

That 2nd sentence is odd: we don't base evidence on reasons; we base reasons on evidence.

We use our minds to *think up* reasons.

We have to *search* for evidence "out there" in the world and then make it available for everyone to see.

Reasons need the support of evidence; evidence should need to support beyond careful demonstration or a reference to a reliable source.

When assembling your evidence, be aware that what you think is a true fact and therefore hard evidence, your readers might not.

E.g., suppose a researcher offers this claim and reason:

Example 5.4.6. *Early Alamo stories reflected values already in the American character.* [claim] *The story almost instantly became a legend of American heroic sacrifice.* [reason]

To support that reason, she offers this "hard" evidence:

Example 5.4.7. *Soon after the battle, many newspapers used the story to celebrate our heroic national character.* [evidence]

The researcher treats this statement as fact.

But skeptical readers, the kind you should expect (even hope for), might ask *How soon is "soon"? How many is "many"? Which papers? In news stories or editorials? What exactly did they say? How many papers didn't mention it?*

Such readers will accept that statement as evidence only when they're satisfied their questions about it have been answered.

To be sure, readers may accept a claim based only on a reason, without any evidence at all, if that reason comes from a trusted authority or seems clearly - or *self-evidently* - true:

Example 5.4.8. *We are all created equal,* [reason] *so no one has a natural right to govern us.* [claim]

In research papers written for introductory courses, it is often sufficient to support reasons only by what authoritative sources say: *Wilson says X, Yang says Y, Schmidt says Z.*

But in advanced work, readers expect more: they want evidence drawn not just from secondary sources but from primary sources or your own observations, demonstrations, or experiments.

Review your storyboard: Can you support each reason with what your readers will think is evidence of the right kind, quantity, and quality?

Might your readers challenge what you offer as evidence?

If so, how?

Do you need to offer a better demonstration or a better source?

If so, you must produce more or better data or acknowledge the limits of what you have.

Your claim, reasons, and evidence make up the core of your argument, but it needs at least 1 more element, maybe 2.

5.4.3 Acknowledge & Respond to Anticipated Questions & Objections

Careful readers will be fair, but they will also question every part of your argument.

So you must anticipate as many of their questions as you can and then acknowledge and respond to the most important ones.

Doing this may be hard because you know your own argument too well and may believe in it too much to seriously challenge it.

Still, you must *imagine* your readers' questions and take their views into account.

That's how you establish a cooperative relationship with your readers.

Readers can challenge both the *intrinsic* and *extrinsic* soundness of your argument: they might point to problems *inside* your argument, usually with its evidence, or they might raise questions from *outside* your argument by noting alternatives or exceptions.

Try to imagine and respond to both sorts of challenges.

1. To address potential challenges to your argument's *intrinsic* soundness, imagine a reader making any of these criticisms, and then construct a subargument in response.

   Imagine a reader challenging the *nature* of your evidence:

   - "I want to see a different *sort* of evidence - hard numbers, not anecdotes."
     Or "... stories about real people, not cold statistics."

   Imagine a reader questioning its *quality*:

   - "It isn't accurate. The numbers don't add up."
   - "It isn't precise enough. What do you mean by 'many'?"
   - "It isn't current. There's newer research than this."
   - "It isn't representative. You didn't get data on all the groups."
   - "It isn't authoritative. Smith is no expert on this matter."

   Now imagine a reader questioning its quantity (usually the strongest objection of all):

   - "You need more evidence. A single data point/quotation/number/anecdote is not sufficient."

Most researchers have difficulty finding enough good evidence to make an airtight case, especially when working to a deadline.

Research is always a compromise between being thorough and being timely.

If you feel that your evidence is less than unassailable, you might want to admit its limitations candidly, before readers reject your argument because you overstated it.

Next, imagine these kinds of reservations about your reasons and how you would answer them:

- Your reasons are inconsistent or contradictory.
- They are too weak or too few to support your claim.
- They are irrelevant to your claim. (We discuss this matter in **5.4.4**.)

2. To address potential challenges yo your argument's *extrinsic* soundness, you have to *step back and view your argument from other perspectives.*

   Doing this is difficult, but you must try.

   It's important to get into the habit of asking yourself, *What could cast doubt on my claim?*

   Those who see the world differently from you are likely to define terms differently, reason differently, even offer evidence that you find irrelevant.

   If you and your readers see the world differently, you must acknowledge and respond to these issues as well.

   Do not treat differing points of view simply as objections.

   You will lose readers if you argue that your view is right and theirs is wrong.

   Instead, acknowledge the differences, then compare them so that readers can understand your argument on its own terms.

   They still might not agree, but you'll show them that you understand and respect their views; they are then more likely to try to understand and respect yours.

   If you're a new researcher, you'll find these questions hard to imagine because you might not know how your readers' views differ from your own.

   Even so, try to think of plausible questions and objections.

   But if you're writing a thesis of dissertation, you are responsible not just for supporting your own claim but also for knowing the positions of others in your research community and the issues they are likely to raise.

   Whatever your level of experience, practice imagining and responding to objections and alternative arguments.

   By doing so, you'll cultivate a habit of mind that your readers will respect and that may keep you from jumping to questionable conclusions.

   Add those acknowledgments and responses to your storyboard where you think readers will raise them.

5.4.4 Establish the Relevance of Your Reasons

The last element of an argument is called a *warrant*, and even experienced researchers find it hard to grasp.

You add a warrant to your argument when you think a reader might reject your claim not because a reason supporting it is factually wrong or is based on insufficient evidence, but because it seems *irrelevant* and so doesn't count as a reason at all.

E.g., imagine a researcher writes this claim.

Example 5.4.9. *The Alamo stories spread quickly* [claim] *because in 1836 the United States wasn't yet a confident player on the world stage.* [reason]

Imagine that she suspects that her readers will likely objects, *It's true that the Alamo stories spread quickly and that in 1836 the United States wasn't yet a confident player on the world stage. But I don't see how not being confident is relevant to the story's spreading quickly.*

The writer can't respond simply by offering more evidence that this country was not a confident player on the world stage or that the stories in fact spread quickly: her reader already accepts both as true.

Instead, she has to explain the *relevance* of that reason - *why* its truth supports the truth of her claim.

To do that, she needs a warrant.

5.4.4.1 How a warrant works in casual conversation

Suppose you make this little argument to a new friend from a faraway land:

Example 5.4.10. *It's* 5° *below zero,* [reason] *so you should wear a hat.* [claim]

To most of us, the reason seems obviously to support the claim and so needs no explanation of its relevance.
But suppose your friend asks this odd question:

Example 5.4.11. *So what if it is* 5° *below?*
Why does that mean I should wear a hat?

That question challenges not the *truth* of the reason (it is 5° below) but its *relevance* to the claim (you should wear a hat).
You might think it odd that anyone would ask that question, but you could answer with a general principle:

Example 5.4.12. *Well, when it's cold, people should dress warmly.*

That sentence is a warrant.
It states a general principle based on our experience in the world: when a certain general condition exists (*it's cold*), we're justified in saying that a certain general consequence regularly follows (*people should dress warmly*).
We think that the general warrant justifies our specific claim that our friend should wear a hat on the basis of our specific reason that it's 5° below, because we're reasoning according to this principle of logic: if a general condition and its consequence are true, then specific instances of it must also be true.
In more detail, it works like this (warning: what follows may sound like a lesson in Logic 101):

- In the warrant, the general condition is *it's cold.*

  It regularly leads us to draw a general consequence: *people should dress warmly.*

  We state that as a true and general principle: *When it's cold, people should dress warmly.*

- The specific reason, *it's* 5° *below*, is a valid instance of the general condition *it's cold.*

- The specific claim, *you should wear a hat*, is a valid instance of the general consequence, *people should dress warmly.*

- Since the general principle stated in the warrant is true and the reason and claim are valid instances of it, we're "warranted" to assert as true and valid the claim *wear a hat.*

But now suppose 6 months later you visit your friend and he says this:

Example 5.4.13. *It's above* 80° *tonight,* [reason] *so wear a long-sleeved shirt.* [claim]

That might baffle you: How could the reason (it's above 80°) be relevant to the claim (*wear a long-sleeved shirt*)?
You might imagine this general principle as a warrant:

Example 5.4.14. *When it's a warm night, people should dress warmly.*

But that isn't true.
And if you think the warrant isn't true, you'll deny that the reason supports the claim, because it's irrelevant to it.
But suppose your friend adds this:

Example 5.4.15. *Around here, when it's a warm night, you should protect your arms from insect bites.*

Now the argument would make sense, but only if you believe all this:

- The warrant is true (*when it's a warm night, you should protect your arms from insect bites*).

- The reason is true (*it's above* 80° *tonight*).

- The reason is a valid instance of the general condition (80° *is a valid instance of being warm*).

- The claim is a valid instance of the general consequence (*wearing a long-sleeved shirt is a valid instance or protecting your arms from insect bites*).

- No unstated limitations or exceptions apply (*a cold snap didn't kill all insects the night before, the person can't use insect repellent instead, and so on*).

If you believe all that, then you should accept the argument that when it's 80° at night, it's a good idea to wear a long-sleeved shirt, at least at that time and place.

We all know countless such principles, and we learn more every day.

If we didn't, we wouldn't make our way through our daily lives.

In fact, we express our folk wisdom in the form of warrants, but we call them proverbs: *When the cat's away, the mice will play.*

Out of sight, out of mind.

Cold hands, warm heart.

5.4.4.2 How a warrant works in an academic argument

Here is a more scholarly example, but it works in the same way:

Example 5.4.16. *Encyclopedias must not have been widely owned in early 19th-century America,* [claim] *because wills rarely mentioned them.* [reason]

Assume the reason is true: there is lots of evidence that encyclopedias were in fact rarely mentioned in early 19th-century wills.

Even so, a reader might wonder why that statement is *relevant* to the claim: *You may be right that most such wills didn't mention encyclopedias, but so what? I don't see how that is relevant to your claim that few people owned one.*

If a writer expects that question, she must anticipate it by offering a warrant, a general principle that shows how her reason is relevant to her claim.

That warrant might be stated like this:

Example 5.4.17. *When a valued object wasn't mentioned in early 19th-century wills, it usually wasn't part of the estate.* [warrant] *Wills at that time rarely mentioned encyclopedias,* [reason] *so few people must have owned one.* [claim]

We would accept the claim as sound *if and only if* we believe the following:

- The warrant is true.
- The reason is both true and a valid instance of the general condition of the warrant (encyclopedias were valued objects in the early 19th century).
- The claim is a valid instance of the general consequence of the warrant (not owning an encyclopedia is a valid instance of something valuable not being part of an estate).

And if the researcher feared that a reader might doubt any of those conditions, she would have to make an argument supporting it.

But that's not the end of the problem: is the warrant true *always and without exception?*

Readers might wonder whether in some parts of the country wills mentioned only land and buildings, or whether few people made wills in the 1st place.

If the writer thought that readers might wonder about such qualifications, she would have to make yet another argument showing that those exceptions don't apply.

Now you can see why we so rarely settle arguments about complex issues: even when we agree on the evidence, we can still disagree over how to reason about it.

5.4.4.3 Testing the relevance of a reason to a claim

The test the relevance of a reason to a claim, construct a warrant that bridges them.

1st, state the reason and claim, in that order.

Here's the original reason and claim from the beginning of this section:

Example 5.4.18. *In 1836, this country wasn't a confident player on the world stage,* [reason] *so the Alamo stories spread quickly.* [claim]

Now construct a general principle that includes that reason and claim.

Warrants come in all sorts of forms, but the most convenient is the *When - then* pattern.

This warrant "covers" the reason and claim.

Example 5.4.19. *When a country lacks confidence in its global stature, it quickly embraces stories of heroic military events.*

We can formally represent those relationship as in Fig. 5.1.

Fig. 5.1. Argument structure.

Example 5.4.20. *When this* ***General Condition*** *exists, this* ***General Consequence*** *follows.* [warrant]

When a country lacks confidence, [general condition] *it quickly embraces stories of heroic military events.* [general consequence]

In 1836, this country wasn't a confident player on the world stage [specific reason] *so the story of the Alamo spread quickly.* [specific claim]

This ***Specific Condition*** *exists,* [reason] *so this* ***Specific Consequence*** *follows.* [claim]

To accept that claim, readers must accept the following:

- The warrant is true.
- The specific reason is true.
- The specific reason is a valid instance of the general condition side of the warrant.
- The specific claim is a valid instance of the general consequence side of the warrant.
- No limiting conditions keep the warrant from applying.

If the writer thought that readers might deny the truth of that warrant or reason, she would have to make an argument supporting it.

If she thought they might think the reason or claim wasn't a valid instance of the warrant, she'd have to make yet another argument that it was.

As you gain experience, you'll learn to check arguments in your head, but until then you might try to sketch out warrants for your most debatable reasons.

After you test a warrant, add it to your storyboard where you think readers will need it.

If you need to support a warrant with an argument, outline it there.

5.4.4.4 Why warrants are especially difficult for researchers new to a field

If you're new in a field, you may find warrants difficult for these reasons:

- Advanced researchers rarely spell out their principles of reasoning, because they know their colleagues take them for granted.

  New researchers must figure them out on their own.

  (It's like hearing someone say, "Wear a long-sleeved shirt because it's above 80° tonight.")

  Prefer Arguments Based on Evidence

- Warrants typically have exceptions that experts also take for granted and therefore rarely state, forcing new researchers to figure them out as well.
- Experts also know when *not* to state an obvious warrant or its limitations, 1 more thing new researchers must learn on their own.

  E.g., if an expert wrote *It's early June, so we can expect that we'll soon pay more for gasoline*, he wouldn't state the obvious warrant: *When summer approaches, gas prices rise.*

If you offer a well-known but rarely stated warrant, you'll seem condescending or naive.

But if you fail to state one that readers need, you'll seem illogical.

You just need to know when readers need one and when they don't.

And that takes time and familiarity with the conventions of your field.

So don't be dismayed if warrants seems confusing; they're difficult even for experienced writers.

But knowing about them should encourage you to ask this crucial question: in addition to the *truth* of your reasons and evidence, will your readers see their *relevance* to your claim?

If they might not, you should consider making an argument to demonstrate it.

5.5 Prefer Arguments Based on Evidence to Arguments Based on Warrants

Finally, it's important to note that readers judge arguments based on warrants and arguments based on evidence differently.

- The 1st type of argument infers a claim from a reason and warrant.

  The claim in that kind of argument is believed to be *certainly* true.

- The 2nd type supports a claim with reasons based on evidence.

  The claim in that kind of argument is considered to be *probably* true.

Contemporary readers generally trust the 2nd kind more than the 1st.

Compare these 2 examples:

Example 5.5.1. *We should do what we can to discourage teenagers from texting and driving,* [claim] *because that behavior increases the risk of having an accident.* [reason 1] *Driving is difficult and texting a distraction,* [reason 2 supporting reason 1] *and we know that when people are distracted while performing complex tasks, their performance suffers.* [warrant linking reason 2 & reason 1]

We should do what we can to discourage teenagers from texting and driving, [claim] *because distracted driving is a leading cause of teenager deaths.* [reason] *According to the CDC, motor vehicle accidents are responsible for over a 3rd of all fatalities among people aged 12–19, and texting while driving exponentially increases the likelihood that any driver will be involved in one. Moreover, . . .* [evidence]

If you are like most contemporary readers, you probably preferred the 2nd of these arguments to the 1st.

That's because its warrant is not controversial (and therefore goes without saying) and its claims is supported by a reason based on solid evidence.

That 1st argument seems plausible because an uncontroversial (and therefore unstated) warrant connects reason 1 to the claim, and an explicit warrant connects reason 2 to reason 1 (reason 1 is the claim reason 2 supports).

In other words, reason 1 and reason 2 are good instances of that warrant's general consequence and condition, resp.

Even so, as tight as that argument is, most readers still want some hard evidence.

In particular, you can rarely support a claim of face with a warrant and reason alone.

Generally, we can't just *reason* our way to the facts' we have to *discover* them through research.

Except in a few fields - some branches of mathematics, philosophy, theology - the way to demonstrate a claim of fact is to show with evidence that what you are claiming is, *in fact*, the case.

All arguments rely on warrants, but readers of *research* arguments are likely to mistrust arguments from principle alone.

So whenever you can, rely not on elaborate lines of reasoning based on warrants but on hard evidence.

5.6 Assemble an Argument

Here is a small argument that fits together all 5 parts:

Example 5.6.1. *Video games aimed at children can aid their intellectual development, but that contribution is largely offset by a factor that could damage their emotional development - too much violence.* [claim]

Example, parents and child psychologists agree, is a major influence on children's development.

It seems plausible, then, that when children see images of degrading or disturbing behavior, they will be adversely affected by it. [warrant]

In a single day, children see countless examples of violence. [reason]

Every day the average child plays over 3 hours of video games and sees over 20 acts of violence (Duarte 2012). [evidence]

Kim has shown that children don't confuse video-game violence with real life (2015), [acknowledgment of alternative point of view] *but because of their interactivity, video-game violence may affect them nonetheless.* [response]

We cannot ignore the possibility that childhood exposure to violent video games encourages the development of violent adults. [claim restated & amplified]

Most of those elements could be expanded to fill many paragraphs.

Arguments in different fields look different, but they all consist of answers to just these 5 questions:

- What are you claiming?
- What are your reasons?
- What evidence supports your reasons?
- But what about other points of view?
- What principle makes your reasons relevant to your claim?

Your storyboard should answer those questions many times.

If it doesn't, your paper will seem incomplete and unconvincing.

Chapter 6

Planning a 1st Draft

Once you have assembled your argument, you might be ready to draft it.

But experienced writes know that the time they invest in planning a draft more than pays off when they write it.

To draft effectively, you need more than just the elements of a sound argument; you need a plan to assemble those elements into a coherent draft.

Some plans, however, are better than others.

6.1 Avoid Unhelpful Plans

When you are doing research, it is good to write early, but your early writing should not determine the plan for your final paper: the real purpose of that early writing is to help you *discover* your ideas, not to *communicate* them to your readers.

If you find yourself adopting 1 of the following plans, pause.

It means that you are not get ready to draft your paper.

1. **Narrative of discovery.** If you find yourself telling the story of your project - *1st I investigated . . . Then I compared . . .* - you are probably still too close to the activity of research to put your readers' needs before your own.

   Most readers want to know your ideas, not necessarily the steps through which you arrived at them.

2. **Patchwork of sources.** If your draft is just a series of quotations, paraphrases, and summaries of sources, you are probably overwhelmed by your sources and have not yet developed a controlling claim of your own.

   Readers want to know that you've done your research thoroughly, but they also want to know what *you* think.

   Note: if you submit a paper that is stitched together from other sources, especially if you have cut-and-pasted from the web, you also risk being seen not just as an amateur but also a plagiarist (see **7.9**).

3. **Mirror of the assignment.** If you writing for a course and your draft simply mirrors the organization of your assignment, you have to ask why.

   Are you just playing it safe or going through the motions?

   You owe it to yourself and your teacher to do more.

   Are you still developing your own ideas?

   Let the structure of your paper reflect them.

   Are you struggling to organize your paper another way?

   (See Sect. **6.2.5** for some possibilities.)

6.2 Create a Plan That Meets Your Readers' Needs

Some fields stipulate the plan of a paper.

In the experimental sciences, e.g., readers expect papers to follow an organization like this:

Example 6.2.1. *Introduction - Methods & Materials - Results - Discussion - Conclusion*

If your paper must follow a conventional plan, find a model in a secondary source or ask your mentor, teacher, or advisor for guidance.

In most fields, however, you have to create a plan of your own, but that plan must make sense to your readers.

To create one, start with your storyboard or outline.

6.2.1 Convert Your Storyboard into an Outline

If you prefer to work from an outline, you can turn your storyboard into one:

- Start with a sentence numbered *I* that states your claim.
- Add complete sentences under it numbered *II,III,...*, each of which states a reason supporting your claim.
- Under each reason, use capital letters to list sentences summarizing your evidence; then list by numbers the evidence itself.

  E.g. (the data are invented for the illustration):

  Example 6.2.2. *I. Introduction: Educational benefits of writing on laptops are uncertain.*

  II. Different uses have different effects.

  A. All uses increase number of words produced.

  1. Study II.A.1: 950 vs. 780

  2. Study II.A.2: 1,103 vs. 922

  B. Study II.B: Using laptops encourages writer's block.

  III. Studies show limited benefit on revision.

  A. Study III.A: Writers using laptops are more wordy.

  1. Average of 2.3 more words per sentence

  2. Average of 20% more words per essay.

  B. Study III.B: Writers need hard copy to revise effectively.

  1. 22% fewer typos when done on hard copy vs. computer screen

  2. 2.26% fewer spelling errors

  IV. Conclusion: Disadvantages of laptops may outweigh advantages.

A sparer outline is just phrases, with no formal layers of I, A, 1, and so on.

Example 6.2.3. *Introduction: Benefits uncertain*
Different uses/different effects
More words
Writer's block
Revision studies
Study 1 longer sentences
Study 2 longer essays
Study 3 hardcopy better
Conclusion: Disadvantages outweigh advantages

When you start a project, a spare outline may be the best you can do, and for a short project it may be all you need, so long as you know the point of each item.

But an outline of complete sentences is usually more useful.

More useful yet is a storyboard, especially for a long project.

6.2.2 Sketch a Working Introduction

Writers are often advised to write their introduction last, but most of us need a working introduction to start us on the right track.

Expect to write your introduction twice, 1st a sketch for yourself and later a final one for your readers.

That final introduction will usually have 4 parts (see Chap. 9), so you might as well sketch your working introduction to anticipate them.

1. At the top of the 1st page of your storyboard, sketch a *brief* summary of *only* the key points in *only* those sources most relevant to your argument.

   Include only the sources that you intend to challenge, modify, or expand on.

   Then order those sources in a way that is useful to your readers: chronologically, by quality, significance, point of view, or the like.

   Don't just follow the order in which you happened to read them or record them in your notes.

2. Rephrase your research question as a statement about a flaw or gap that you see in your sources:

   Example 6.2.4. *Why is the Alamo story so important in the United States' national mythology?*

   → *Few of these historians, however, have explained why the Alamo story has become so important in the United States' national mythology.*

3. If you can, sketch an answer to *So what*?

   What larger issue will your readers not understand if you don't answer your question?

   You may be only guessing, but try to find *some* answer.

   Example 6.2.5. *If we understood how such stories become national legends, we would better understand the United States' national values, perhaps even what makes the American character distinct.*

   If you can't think of any answer, skip it; we return to it in Chap. 10.

4. State the answer to your question as your claim, or promise an answer in a launching point.

   You have 2 choices here:

   - State your claim at the end of your introduction to frame what follows and again near the beginning of your conclusion.
   - State it only it your conclusion, as a climax to your reasoning.

This is a crucial choice, because it creates your social contract with your readers (see 1.2).

If you state your main claim toward the end of your introduction, you put your readers in charge: they know what's coming, and they can decide to read on - or not.

On the other hand, if you wait until your conclusion to state your main claim, you create a more controlling relationship: you ask them to trust that they'll find your claim - when you eventually reveal it - worth the investment of their time.

Most readers prefer to see your main claim at the end of your introduction, because that lets them read what follows faster, understand it better, and remember it longer.

In your storyboard, put that claim at the bottom of your introduction page.

Then restate it in different words at the top of your storyboard's conclusion page.

If you can, make this concluding claim more specific than the one in the introduction.

Some new researchers fear that if they reveal their main point too early, readers will be "bored" and stop reading.

Others worry about repeating themselves.

Both fears are baseless.

If you ask an interesting question, readers will want to see how well you can answer it, and knowing where your argument is going will help your readers follow it.

But if you do decide to announce your claim only in your conclusion, you still need a sentence at the end of your introduction that launches your reader into the body of your paper.

That sentence should include the key terms that will run through your paper (see **6.2.3**).

You'll be better prepared to write that sentence after you draft your final introduction, so when planning, just put your main claim at the bottom of your storyboard's introduction page (you can move it later).

Finally, some writers add a "road map" at the end of their introduction:

Example 6.2.6. *In part 1, I discuss the issue of... Part 2 addresses... Part 3 examines...*

Road maps are common in the social sciences, but many in the humanities find them clumsy.

You can add a road map to your storyboard to guide your drafting, then cut it from your final draft.

If you keep it, make it short.

6.2.3 Identify Key Concepts that Will Run through Your Paper

For your paper to seem coherent, readers must see a few key concepts running through all its parts.

But readers won't recognize those repeated concepts if you prefer to them in many different words.

Readers need to see specific terms that repeatedly refer to those concepts, not every time you mention them but often enough that readers can't miss them.

You might find them among the terms you used to categorize your notes, but they must include keywords from the sentences stating your problem and main point.

On the introduction page, circle 4 or 5 words or phrases that express those concepts.

Ignore words that name your general topic; focus on those relevant to your specific question:

Example 6.2.7. *gender, education level, major, choice of profession, wage gap*

If you find few words or phrases that can serve as key terms, your topic and point might be too general (review **5.4.1**).

You can also use this procedure to identify concepts that distinguish your sections from each other.

Look at the reason you stated at the top of each reason page, and circle its important words.

Some of those words should be related to the words circled in the introduction and conclusion, but others will be specific to that section.

When you draft, you can use these lists of key terms to keep yourself on track and to recognize when you might be saying something new (see **7.3**).

6.2.4 Use Key Terms to Create Subheads that Uniquely Identify Each Section

Even if papers in your field don't use subheads (see **A.2.2.4** in the appendix), the authors recommend that you use them in your drafts.

Create them out of your key terms.

If you cannot find key terms that distinguish a section, consider its contribution to the whole: if little differentiates that section from others, readers may find it repetitive or irrelevant.

If your field avoids subheads, use them to keep yourself on track, and delete them from your last draft.

6.2.5 Order Your Paper

When you assembled your argument, you may not have put your reasons in any particular order (1 benefit of a storyboard).

But when you plan a draft, you must impose some order on them.

That is not easy, especially when you're writing on a new topic in a new field.

The best order is the one that best meets your readers' needs.

When you're not sure how to order your reasons, consider the following options.

You can organize your paper according to your subject matter:

- *Chronological.* This is the simplest: earlier-to-later or cause-to-effect.
- *Part by part.* If you can break your topic into its constituent parts, you can deal with each in turn, but you must still order those parts in some way that helps readers understand them: by their functional relationships, hierarchy, or the like.
- *Comparison & contrast.* Choose this form if you are comparing 2 or more entities, concepts, or objects.

  You have 2 options, and one is usually better than the other.

  If you were comparing Hopi masks to Inuit masks, you might decide to devote the 1st half of your paper to the former and the 2nd half to the latter.

  But this kind of organization often results in a pair of disconnected summaries.

  Instead, try to treat the objects of comparison together, aligning parallel aspects as you go.

  In our example, you might write 1st about the masks' designs, then about the stages of their evolution, then about their use of symbolism, and so on.

You can also organize your paper to accommodate your readers' prior knowledge and to facilitate their efforts to grasp your argument:

- *Short to long, simple to complex.* More readers prefer to deal with less complex issues before they work through more complex ones.
- *More familiar to less familiar.* Most readers prefer to read what they know about before they read what they don't.
- *Less contestable to more contestable.* Most readers move more easily from what they agree with to what they don't.
- *Less important to more important (or vice versa).* Readers prefer to read more important reasons 1st, but those reasons may have more impact when they come last.
- *Earlier understanding as a basis for later understanding.* Readers may have to understand some events, principles, definitions, and so on before they understand another thing.

Often these principles cooperate: what readers agree with and easily understand might also be short and familiar.

But these principles may also conflict: readers might reject most quickly reasons that are most important.

Whatever your order, it must reflect *your readers*' needs, not the order that the material seems to impose on itself (e.g. chronology), least of all the order in which those reasons occurred to you.

6.2.6 Make Your Order Clear with Transitional Words

Your readers must be able to recognize the order you choose.

In your storyboard, start each page of reasons with words that make the *principle* of order clear: *1st, 2nd, Later, Finally, More important, A more complex issue is…, As a result.*

Don't worry if these words feel awkwardly obvious.

At this point, they're more for your benefit than for your readers'.

You can revise or even delete the clumsy ones from your final draft.

6.2.7 Sketch a Brief Introduction to Each Section & Subsection

Just as your whole paper needs an introduction that frames what follows, so does each of its sections.

If a section is only a page or 2, you need just a short paragraph; for a section several pages long, you might need to sketch in 2 or more paragraphs.

This introduction should announce the key terms that are special to the section, ideally in a sentence and its end expressing the section's point, which might be a reason, a response to a different point of view, or a warrant you must explain.

6.2.8 For Each Section, Sketch Evidence, Acknowledgments, Warrants, & Summaries

In their relevant sections, sketch out the parts of your argument.

Remember that many of those parts will themselves make points that must be supported by smaller subarguments.

6.2.8.1 Evidence

Most sections consist primarily of evidence supporting reasons.

Sketch the evidence after the reason it supports.

If you have different kinds of evidence supporting the same reason, group and order them in a way that will make sense to your readers.

6.2.8.2 Explanations of evidence

You may have to explain your evidence - where it came from, why it's reliable, exactly how it supports a reason.

Usually these explanations follow the evidence, but you can sketch them before if that seems more logical.

6.2.8.3 Acknowledgments & responses

Imagine what readers might object to in your argument and sketch responses.

Responses are typically subarguments with at least a claim and reasons, often including evidence and even another response to an imagined objection to your response.

6.2.8.4 Warrants

If you think you need a warrant to justify the relevance of a reason, develop it before you state the reason.

(If you're using a warrant only for emphasis, put if after the reason.)

If you think readers will question the truth of the warrant, sketch a subargument to support it.

If readers might think that your reason or claim isn't a valid instance of the warrant, sketch a subargument showing that it is.

6.2.8.5 Summaries

If your paper is long and "fact heavy" with dates, names, events, or numbers, you might end each major section by briefly summarizing the progress of your argument.

What have you established in that section?

How does your argument shape up so far?

If in your final draft those summaries seem clumsy, cut them.

Writers in different fields may arrange these elements in slightly different ways, but the elements themselves and their principles of organization are the same in every field and profession.

What's crucial in every paper, regardless of field, is that you must order the parts of your argument not merely to reflect your own thinking but to help your readers understand it.

6.2.9 Sketch a Working Conclusion

You should have stated your concluding claim at the top of the conclusion page of your storyboard.

If you can add to the significance of that claim (another answer to *So what?*), sketch it after the claim (see **10.2** for more on conclusions).

6.3 File Away Leftovers

Once you have a plan, you may discover that you have a lot of material left over that doesn't fit into it.

Resist the impulse to shoehorn these leftovers into your paper just to show your work.

In fact, if you don't have more leftovers than what you used, you may not have done enough research.

Leftovers aren't waste: at a minimum, they helped you discover your ideas.

You might decide that you need to use them after all.

And they could even contain the seeds of another project.

Chapter 7

Drafting Your Paper

Some writers think that once they have an outline or storyboard, they can draft by just grinding out sentences.

If you've written a lot to explore your ideas, you may even think that you can plug that preliminary writing into a draft.

Experienced writers know better.

They know 2 things: exploratory writing is crucial but often not right for a draft, and thoughtful drafting can be an act of discovery that planning and storyboarding can prepare them for but never replace.

In fact, most writers don't know what they *can* think until they see it appear in words before them.

Indeed, *your experience 1 of the most exciting moments in research when you discover yourself expressing ideas that you did not know you had until that moment.*

So don't treat drafting as merely translating a storyboard or outline into words.

If you draft with an open mind, you can discover lines of thought that you couldn't have imagined before you started.

But like other steps in the process, even surprises work better with a plan.

7.1 Draft in the Way That Feels Most Comfortable

Writers draft in different ways.

Some are slow and careful: they have to get every paragraph right before they start the next one.

To do that, they need a meticulous plan.

So if you draft slowly, plan carefully.

Other writers let the words flow, skipping ahead when they get stuck, omitting quotations, statistics, and so on what they can plug in later.

If they are stopped by a stylistic issue e.g. whether to represent numbers in words or numerals, they insert a "[?]" and keep on writing until they run out of gas, then go back and fix it.

But quick drafters need lots of time to revise.

So if you draft quickly, start early.

Draft in whatever way works for you, but experienced writers usually draft quickly, then revise extensively.

7.2 Develop Effective Writing Habits

Most of us learn to write in the least efficient way - under pressure, rushing to meet a deadline, with a quick draft the night before and maybe a few minutes in the morning for proofreading.

That rarely works for a short paper, almost never for a longer one.

You need time and a plan that sets small, achievable goals but keeps your eye on the whole.

Most importantly, draft regularly and often, not in marathon sessions that dull your thinking and kill your interest.

Set a small goal and a reasonable quota of words for each session, and stick to it.

When you resume drafting, you need not start where you left off: review your storyboard to decide what you're ready to draft today.

Review how it will fit into its section and the whole: *What reason does this section support? Where does it fit in the overall logic? Which key terms state the concepts that distinguish this section?*

If you're blocked, skip to another section.

Whatever you do, don't substitute more reading for writing.

Chronic procrastinators are usually so intimidated by the size of their project that it paralyzes them, so they just keep putting off getting started.

You can overcome that destructive habit by breaking your project into small, achievable goals (see **7.111**).

7.3 Keep Yourself on Track through Headings & Key Terms

Here are 2 techniques you can use to keep yourself on track as you draft.

Use headings - ideally full sentences - to break your draft into manageable chunks and to show how your sections are related to one another.

Even if papers in your field don't ordinarily use headings and subheadings, you can still use them as you draft and delete them later.

You can also use lists of key terms to keep yourself on track.

As you draft, keep in front of you both the terms that should run through your whole paper and those specific to individual sections (see **6.2.3**–**6.2.4**).

From time to time, check how often you've used those words, both those that run through the whole paper and those that distinguish 1 section from another.

If you find yourself writing something that lacks those terms, pause and reflect: *are you just off track, or are you discovering something new?*

You need not stay yoked to your original plan: you are free to follow a new path to see where it leads, but do that as a choice - not just because you got lost along the way.

7.4 Quote, Paraphrase, & Summarize Appropriately

The authors covered this issue when they discussed note-taking (**4.2.2**).

You should build most of your paper our of your own words that reflect your own thinking.

But you'll support much of that thinking with the words of others, delivered in quotations, paraphrases, and summaries.

As they've said, different fields use these techniques differently: researchers in the humanities quote more than do social and natural scientists, who typically paraphrase and summarize.

But you must decide each case for itself, depending on how you use the information.

Here again are some principles:

- Summarize when details are irrelevant or a source isn't important enough to warrant much space.
- Paraphrase when you can state what a source says more clearly or concisely or when your argument depends on the details in a source but not on its specific words.
- Quote for these purposes:
  - The words themselves are evidence that backs up your reasons.
  - The words are from an authority who backs up your claims.
  - The words are strikingly original or express your key concepts so compellingly that the quotation can frame an extended discussion.
  - The words express a claim you disagree with, and to be fair you want to state it exactly.

Readers value research only to the degree that they trust its sources.

So for every summary, paraphrase, or quotation you use, cite its bibliography data in the appropriate style (see **15.5**).

Under no circumstances should you stitch together passages from the internet with a few sentences of your own.

Teachers grind their teeth reading such papers, dismayed by their lack of original thinking.

Readers of advanced projects reject such patchworks out of hand.

7.5 Integrate Quotations in Your Text

Signal direct quotations in 1 of 2 ways:

- For 4 or fewer quoted lines, run them into your text, surrounded by quotation marks.
- For 5 or more lines, set them off as an indented block.

You can insert run-in and block quotations in your text in 3 ways.

1. Drop in the quotation with a few identifying words (*Author says, According to Author, As Author puts it*, etc.).

   Example 7.5.1. *Diamond says, "The histories of the Fertile Crescent and China... hold a salutary lesson for the modern world: circumstances change, and past primacy is no guarantee of future primacy" (417).*

2. Introduce the quotation with a sentence that interprets or characterizes it.

   Example 7.5.2. *Diamond suggests what we can learn from the past: "The histories of the Fertile Crescent and China... hold a salutary lesson for the modern world" (417).*

3. Weave the grammar of the quotation into the grammar of your own sentence.

   Example 7.5.3. *Diamond suggests that the chief "lesson for the modern world" in the history of the Fertile Crescent and China is that "circumstances change, and past primacy is no guarantee of future primacy" (417).*

You can modify a quotation, so long as you don't change its meaning and you signal deletions with 3 dots (called *ellipses*) and changes with square brackets.

This sentence quotes the original intact:

Example 7.5.4. *As Hariman argues, "The realist style radically separates power and textuality, constructing the political realm as a state of nature and the political actor as someone either rationally calculating vectors of interest and power or foolishly believing in such verbal illusions as laws or ethical ideals" (4).*

This version modifies the quotation to emphasize 1 part of it and to fit the grammar of the writer's sentence:

Example 7.5.5. *Hariman argues that "the realist style radically separates power and textuality, constructing... the political actor as someone either rationally calculating vectors of interest and power or foolishly believing in such verbal illusions as laws or ethical ideals" (4).*

See Chap. 25 for more on integrating quotations with your text.

When you refer to a source the 1st time, use the author's full name.

Do not precede it with *Mr., Mrs., Ms.,* or *Prof.* (see **24.2.2** for the use of *Dr., Reverend, Senator*, and so on).

When you mention a source thereafter, use just the author's last name:

Example 7.5.6. *According to Steven Pinker, "claims about a language instinct... have virtually nothing to do with possible genetic differences between people." Pinker goes on to explain that...*

Except when referring to royalty, never refer to an author only bu his or her 1st name.

You might write this:

Example 7.5.7. *In a recent speech, Prince Charles described his efforts to preserve the village pubs that are such a part of British culture.*

But never this:

Example 7.5.8. *According to Steven Pinker, "claims about a language instinct..." Steven goes on to explain that...*

7.6 Use Footnotes & Endnotes Judiciously

If you are using notes-style citations (see **3.5.1**), you will have to decide as you draft how to use footnotes and endnotes (for their formal requirements, see Chap. 16).

You must cite every source in a note, of course, but you may also decide to use footnotes and endnotes for substantive material that you don't want to include in the body of your text but also don't want to omit.

(You might also use such substantive notes in combination with parenthetical citations in author-date style; see **18.3.3**.)

- If you cite sources in endnotes, put substantive material in footnotes.

  Otherwise you force readers to keep flipping to the back of your paper to check every endnote to see whether it is substantive or bibliographical.

- Use substantive footnotes sparingly.

  If you create too many, you risk damaging the flow of your writing with asides and digressions.

In any event, keep in mind that many readers ignore substantive footnotes on the principle that information not important enough for you to include in the main text is not important enough for them to read in a footnote.

7.7 Show How Complex or Detailed Evidence Is Relevant

By this point you may be so sure that your evidence supports your reasons that you'll think readers can't miss its relevance.

But evidence never speaks for itself, especially not long quotations or complex sets of numbers.

You must therefore speak for it: introduce it by stating what you want your readers to get out of it.

E.g., this passage bases a claim about Hamlet on the evidence of the following quotation:

Example 7.7.1. *When Hamlet comes upon his stepfather Claudius at prayer, he demonstrates cool rationality:*[claim]
Now might I do it [kill him] pat, now 'a is a-praying,
And now I'll do't. And so 'a goes to heaven,
And so am I reveng'd... [Hamlet pauses to think]
[But this] villain kills my father, and for that,
I, his sole son, do this same villain send
To heaven.
Why, this is hire and salary, not revenge. (3.3) [report of evidence]

Since that quotation doesn't explicitly refer to Hamlet's rationality, readers might not see how it supports the claim.

So show them what is means by introducing it with a reason:

Example 7.7.2. *When Hamlet comes upon his stepfather Claudius at prayer, he demonstrates cool rationality.*[claim] *He impulsively wants to kill Claudius but pauses to reflect: if he kills the praying Claudius, he will send his soul to heaven, but he wants Claudius damned to hell, so he coolly decides to kill him later:*[reason]
Now might I do it [kill him] pat,... [report of evidence]

Now we see the connection.

This kind of explanatory introduction is even more important when you present data in a table or figure (see **8.3.1**).

7.8 Be Open to Surprises

If you write as you go and plan your best case before you draft, you're unlikely to be utterly surprised by how your draft develops.

Even so, be open to new directions from beginning to end:

- When your drafting starts to head off on a tangent, go with it for a bit to see whether you're onto something better than you planned.
- When reporting your evidence leads you to doubt a reason, don't ignore that feeling.

  Follow it up.
- When the order of your reasons starts to feel awkward, experiment with new ones, even if you thought you were almost done.
- Even when you reach your final conclusion, you may think of a way to restate your claim more clearly and pointedly.

If you get helpful new ideas early enough before your deadline, invest the time to make the changes.

It is a small price to pay for a big improvement.

7.9 Guard against Inadvertent Plagiarism

It will be as you draft that you risk making 1 of the worst mistakes a researcher can make: leading readers to think that you're trying to pass off the work of another writer as your own.

Do that and you risk being accused of plagiarism, a charge that, if sustained, could mean for a professional writer an irreparably damaged reputation, or for a student writer a failing grade or even expulsion.

Students know they cheat when, say, they submit as their own work papers bought online.

Most also know they cheat when they pass off as their own long passages copied directly from their sources.

For those cases, there's nothing to say beyond *Don't*.

But many inexperienced researchers fail to realize that they risk being charged with plagiarism even if they were not intentionally dishonest but only misinformed or careless.

You run that risk when you do any of the following:

- You quote, paraphrase, or summarize a source but fail to cite it.

- You use ideas or methods from a source but fail to cite it.
- You use the exact words of a source and you do cite it, but you fail to put those words in quotation marks or in a block quotation.
- You paraphrase a source and cite it, but paraphrase too closely (see **7.9.2**).

7.9.1 Signal Every Quotation, Even When You Cite Its Source

Even if you cite the source, readers must know exactly which words are yours and which you quote.

If, however, you borrow only a few words, you enter a gray area.

Read this:

Example 7.9.1. *Because technology begets more technology, the importance of an invention's diffusion potentially exceeds the importance of the original invention.*

Technology's history exemplifies what is termed an autocatalytic process: i.e., one that speeds up at a rate that increases with time, because the process catalyzes itself (Diamond 1998, 301).

If you were writing about Jared Diamond's ideas, you would probably have to use some of his words, e.g. *the importance of an invention.*

But you might not put that phrase in quotation marks, because it shows no originality of thought or expression.

2 of his phrases, however, are so striking that they do require quotation marks: *technology begets more technology* and *autocatalytic process.* E.g.:

Example 7.9.2. *The power of technology goes beyond individual inventions because "technology begets more technology."*

It is, as Diamond puts it, an "autocatalytic process" (301).

Once you cite those words, you can use them again without quotation marks or citation:

Example 7.9.3. *As 1 invention begets another one and that one still another, the process becomes a self-sustaining catalysis that spreads across national boundaries.*

This is a gray area: words that seem striking to some are not to others.

If you put quotation marks around too many ordinary phrases, readers might think you're naive, but if you fail to use them when readers think you should, they may suspect you of plagiarism.

Since it's better to seem naive than dishonest, especially early in your career, use quotation marks freely.

(You must, however, follow the standard practices of your field.

Lawyers, e.g., often use the exact language of a statute or judicial option with no quotation marks.)

7.9.2 Don't Paraphrase Too Closely

You paraphrase appropriately when you represent an idea in your own words more clearly or pointedly than the source does.

But readers will think that you plagiarize if they can match your words and phrasing with those of your source.

E.g., here is a passage from p. 38 of Malcolm Gladwell's *Outliers: The Story of Success*:

Example 7.9.4. *Achievement is talent plus preparation.*

The problem with this view is that the closer psychologists look at the careers of the gifted, the smaller the role innate talent seems to play and the bigger the role preparation seems to play.

This too-close paraphrase is plagiarism:

Example 7.9.5. *Success seems to depend on a combination of talent and preparation.*

However, when psychologists closely examine the gifted and their careers, they discover that innate talent plays a much smaller role than preparation (Gladwell 2008, 38).

This paraphrase does not plagiarize:

Example 7.9.6. *As Gladwell (2008, 38) observes, summarizing studies on the highly successful, we tend to overestimate the role of talent and underestimate that of preparation.*

This phrasing is not a close match to the original.

And notice that we chose not to put *talent* or *preparation* in quotes.

We decided that those words are common enough to use as our own.

To avoid seeming to plagiarize, read the passage, look away, think about it for a moment; *then, still looking away*, paraphrase it in your own words.

Then check whether you can run your finger along your sentence and find synonyms for the same ideas in the same order in your source.

If you can, try again.

7.9.3 Usually Cite a Source for Ideas Not Your Own

This rule is more complicated than it seems, because few ideas are entirely new.

Readers don't expect you to cite a source for the idea that the earth is round.

But they do expect you to cite a source for an idea when

1. it is associated with a specific person *and*
2. it is new enough *not* to be part of a field's common knowledge.

E.g., psychologists claim that we think and feel in different parts of our brains.

No knowledgeable reader would expect you to cite a source for that idea, because it's so familiar that no one would think you are implying it is yours.

On the other hand, some psychologists argue that emotions are crucial to rational decision-making.

That idea is so new and tied to particular researchers that you'd have to cite them.

7.9.4 Don't Plead Ignorance, Misunderstanding, or Innocent Intentions

To be sure, what looks like plagiarism is often just honest ignorance of how to use and cite sources.

Some students sincerely believe that they don't have to cite material downloaded from the Web because it's free and publicly available.

They are wrong.

Others defend themselves by claiming they didn't *intend* to mislead.

The problem is that we read words, not minds.

So think of plagiarism not as an *intended* act but as a *perceived* one.

Here is the best way to think about this: If the author of the source you borrowed from were to read your paper, would she recognize any of it as hers, including paraphrases and summaries, or even general ideas or methods from her original work?

If so, you must cite those borrowings.

7.10 Guard against Inappropriate Assistance

Experienced writers regularly show their drafts to others for criticism and suggestions, and you should too (see Chap. 12).

But check how much assistance is appropriate and how you should acknowledge the assistance you receive.

1. How much help is appropriate?

   - For a class paper, most instructors encourage students to get general criticism and minor editing, but not detailed rewriting or substantive suggestions.
   - For work submitted for publication, writers are free to get all the help they can from colleagues, reviewers, and others so long as these people don't become virtual ghostwriters.

   Theses and dissertations lie between these extremes.

   If you are in doubt, ask someone with authority - your teacher or your advisor - where the line is drawn.

   Then get all the help you can on the RHS of it.

2. What help must you acknowledge?

   - For a class paper, you usually aren't required to acknowledge general criticism, minor editing, or help from a school writing tutor, but you must acknowledge help that's special or extensive.

     Your instructor sets the rules, so ask.
   - For a thesis, dissertation, or published work, you're not required to acknowledge routine help, though it's courteous and often politic to do so in a preface (see **A.2.1.9**–**A.2.1.10**).

     But you must acknowledge special or extensive editing and cite in a note major ideas or phrases provided by others.

7.11 Work Through Chronic Procrastination & Writer's Block

If you can't seem to get started on a 1st draft or if you struggle to draft more than a few words, you may have writer's block.

Writer's block is tremendously frustrating.

Some cases of it arise from mental health issues related to school pressures, and for those you should consider seeking professional help.

But most cases have causes you can address: you may be stuck because you have no goals or have goals that are too high; you may feel so intimidated by the size of the task that you don't know where to begin; you may feel that you have to make every sentence or paragraph perfect before you move on to the next one; you may just allow yourself to be easily distracted.

Here are some good practices that can help with all of these causes:

- Create a routine and set small, achievable goals.
- Use devices to keep yourself moving, e.g. a progress chart or a timer for writing sessions.
- Write routinely as you research, not just after.
- Lower the bar by telling yourself that you're not writing a draft but only sketching out some ideas.
- Write without looking at the page or with your screen turned off.
- Disconnect your computer from your network, or turn off wifi.
- Find a writing partner or join a writing group.
- Visit your school's learning center or writing center, which will be staffed with consultants who can help.

And remember that if you take perfection as you goal, you will never finish or, worse, never start.

We all have to compromise to get the job done.

On the other hand, sometimes writer's block is a sign that you need to let your ideas simmer in the back of your mind, where they might combine and recombine into something new and surprising.

If you're stuck but have time (another reason to start early), do something else for a few hours: go on a run, read a book, take a nap.

If you have even more time, put your draft aside for a day or 2.

Then return to the task to see if you can get back on track.

Chapter 8

Presenting Evidence in Tables & Figures

If your data are in the form of numbers, most readers grasp them more easily if you present them graphically.

But you face many choices of graphic forms, and some forms will suit your data and message better than others.

In this chapter we show you how to choose the right graphic form and design it so that readers can see both what your data are and how they support your argument.

(See pp. 426–28 in the bibliography for guides to creating and using graphics; see Chap. 26 for details on formatting graphics.)[1]

8.1 Choose Verbal or Visual Representations of Your Data

Ordinarily, present quantitative data verbally when they include only a few numbers.

(See Chap. 23 for presenting numbers in text.)

Present them graphically when most of your evidence is quantitative or you must communicate a large set of data.

But when the data are few and simple, readers can grasp them as easily in a sentence as in a table like Table 8.1.

Example 8.1.1. *In 2013, on average, men earned §50,033 a year, women $39,157, a difference of $10,876.*

Table 8.1. Male-female salaries ($), 2013.

But if you present more than 4 or 5 numbers in a passage, readers will struggle to keep them straight, particularly if they must compare them, like this:

Example 8.1.2. *Between 1970 and 2010, the structure of families changed in 2 ways.*

In 1970, 85% of families had 2 parents but by 1980 that number had declined to 77%, then to 73% by 1990, to 68% by 2000, and to 64% by 2010.

The number of 1-parent families rose, particularly families headed by a mother.

In 1970, 11% of families were headed by a single mother.

By 1980 that number rose to 18%, by 1990 to 22%, to 23% by 2000, and to 27% by 2010.

There were some marginal changes among single fathers (headed 1% of families in 1970, 2% in 1980, 3% in 1990, and 4% in 2000 and 2010).

Families with no adult in the home have remained stable at 3–4%.

Those data can be presented more effectively in graphic form, as in Table 8.2.

Table 8.2. Changes in family structure, 1970–2010.

8.2 Choose the Most Effective Graphic

When you graphically present data as complex as in that paragraph, you have many choices.

The simplest and most common are tables, bar charts, and line graphs, each of which has a distinctive rhetorical effect.

[1] A noet on terminology: In this chapter we use the term *graphics* to refer to all visual representations of evidence.

Another term sometimes used for such representations is *illustrations*.

Traditionally, graphics are divided into *tables* and *figures*.

A table is a grid with columns and rows that present data in numbers or words organized by categories.

Figures are all other graphic forms, including graphs, charts, photographs, drawings, and diagrams.

Figures that present quantitative data are divided into *charts*, typically consisting of bars, circles, points, or other shapes, and *graphs*, typically consisting of continuous lines.

For a survey of common figures, see Table 8.7.

- To emphasize specific values, use a table like Table 8.2.
- To emphasize comparisons that can be seen at a glance, use a bar chart like Fig. 8.1.

  Fig. 8.1. Changes in family structure, 1970–2010.
- To emphasize trends, use a line graph like Fig. 8.2.

  Fig. 8.2. Changes in family structure, 1970–2010.

While each of these forms communicates the same data, readers respond to them in different ways:

- A table seems precise and objective.

  It emphasizes individual numbers and forces readers to infer relationships or trends (unless you state them).
- Both charts and graphs emphasize an image that communicates values less precisely but more quickly than do the exact numbers of a table.

  But they differ:

  - A bar chart emphasizes comparisons among discrete items.
  - A line graph emphasizes trends, usually over time.

Choose the graphic form that best achieves the effect you intend.

Your choices also depend on your experience.

If you're new to quantitative research, limit your choices to basic tables, bar charts, and line graphs.
Your computer software may offer more choices, but ignore those that you aren't familiar with.
If you're doing advanced research, readers will expect you to draw from a larger range of graphics favored in your field.
In that case, consult Table 8.7, which describes the rhetorical uses of other common forms.
You may have to consider more creative ways of representing data if you are writing a dissertation or article in a field that routinely display complex relationships in larger data sets.

8.3 Design Tables & Figures

Computer programs create graphics so dazzling that you might be tempted to let your software determine their design.
But readers don't care how fancy a graphic looks if it doesn't communicate your point clearly.
Here are some principles for designing effective graphics.
To follow them, you may have to change default settings in your graphics software.
(See **A.3.1.3**–**A.3.1.4** on creating and inserting tables and figures in your paper.)

8.3.1 Frame Each Graphic to Help Your Readers Understand it

A graphic representing complex numbers rarely speaks for itself.
You must frame it to show readers know what to see in it and how to understand its relevance to your argument.

1. Introduce tables and figures with a sentence in your text that states how the data support your point.

   Include in that sentence any specific number that you want readers to focus on.

   That number must also appear in the table or figure.
2. Label every table and figure in a way that describe its data and, if possible, their important relationships.

   For a table, the label is called a *title* and is set flush left above the table; for a figure, the label is called a *caption* and is set flush left below the figure.

   (For the forms of titles and captions, see Chap. 26.)

   Keep titles and captions short but descriptive enough to distinguish every graphic from every other one.

   - Avoid making the title or caption a general topic:

     Example 8.3.1. *Not: Heads of households*
     But: Changes in 1- and 2-parent heads of households, 1970–2010
   - Use noun phrases; avoid relative clauses in favor of participles:

Example 8.3.2. *Not: Number of families that subscribe to online streaming services*
But: Number of families subscribing to online streaming services

- Do not give background information or characterize what the data imply:

  Example 8.3.3. *Not: Weaker effects of counseling on depressed children before professionalization of staff, 1995–2014*
  But: Effect of counseling on depressed children, 1885–2014

- Be sure labels distinguish graphics presenting similar data:

  Example 8.3.4. *Not: Risk factors for high blood pressure*
  But: Risk factors for high blood pressure among men in Maywood, Illinois
  Or: Risk factors for high blood pressure among men in Kingston, Jamaica

3. Insert into the table or figure information that helps readers see how the data support your point.

   E.g., if numbers in a table show a trend and the size of the trend matters, indicate the change in a final column.

   If a line on a graph changes in response to an influence not mentioned on the graph, as in Fig. 8.3, add text to the image to explain it:

   Example 8.3.5. *Although reading and math scores initially declined by almost 100 points following redistricting, that trend was substantially reversed by the introduction of supplemental math and reading programs.*

   Fig. 8.3. SAT scores for Mid-City High, 1990–2005.

4. Introduce the table or figure with a sentence that explains how to interpret it.

   Then highlight the part of the table or figure that you want readers to focus on, particularly any number or relationship mentioned in that introductory sentence.

   E.g., we have to study Table 8.3 to see how it supports the sentence that introduces it:

   Example 8.3.6. *Most predictions about gasoline consumption have proved wrong.*

   Table 8.3. Gasoline consumption.

   We need another sentence explaining how the numbers support or explain the claim.

   We also need a more informative title and visual help that focuses us on what we should see in the table (Table 8.4):

   Example 8.3.7. *Gasoline consumption has not grown as predicted.*
   Though Americans drove 28% more miles in 2010 than in 1970, they used 32% less fuel.

   Table 8.4. Per capita mileage & gasoline consumption, 1970–2010.

   The added sentence tells us how to interpret the key data in Table 8.4, and the highlight tells us where to find it.

8.3.2 Keep All Graphics as Simple as Their Content Allows

Some guides encourage you to put as much data as you can into a graphic.
But readers want to see only the data relevant to your point, free of distractions.

- For All Graphics

  1. Include only relevant data.
     If you include data only for the record, label it accordingly and put it in an appendix (see **A.2.3.2**).
  2. Keep the visual impact simple.
     - Box a graphic only if you group 2 or more figures.
     - Use caution in employing shading or color to convey meaning.
       Even if you print your paper on a color printer or submit it as a PD, it may be printed or copied later in black and white.
     - Do not use a 3D background for a 2D graphic.
       The added depth contributes nothing and can distort how readers judge values.
  3. Use clear labels.

- Label rows and columns in tables and both axes in charts and graphs.

  (See Chap. 26 for punctuation and spelling in labels.)

- Use tick marks and labels to indicate intervals on the vertical axis of a graph (see Fig. 8.4).

- If possible, label lines, bar segments, and the like on the image rather than in a caption set to the side.

  Do so in the caption only if labels would make the image too complex to read.

- When specific numbers matter, add them to bars, segments or segments in charts or to dots on lines in graphs.

- For Tables

  - Never use both horizontal and vertical lines to divide columns and rows.

    Use light gray lines if you want to direct your reader's eyes in 1 direction to compare data or if the table is unusually complex.
  - For tables with many rows, lightly shading every 5th row will improve legibility.
  - To ensure legibility, do not use a font size smaller than 8 points.

- For Charts & Graphs

  - Use grid lines only if the graphic is complex or readers need to see precise numbers.

    Make all grid lines light gray.
  - Color or shade lines or bars only to show a contrast.

    If you do use shading, make sure it does not obscure any text, and do not use multiple shades, which might not reproduce distinctly.
  - Plot data on 3D only when you cannot display the data in any other way and your readers are familiar with such graphs.
  - Never use iconic bars (e.g., images of cars to represent automobile production).

    They can distort how readers judge values, and they look amateurish.

8.3.3 Follow Guidelines for Tables, Bar Charts, & Line Graphs

8.3.3.1 Tables

Tables with lots of data can seem dense, so organize them to help readers.

- Order the rows and columns by a principle that lets readers quickly find what you want them to see.

  Do not automatically choose alphabetic order.

- Round numbers to relevant values.

  If differences of less than 1,000 don't matter, then 2,123,499 and 2,124,886 are irrelevantly precise.

- Sum totals at the bottom of a column or at the end of a row, not at the top or left.

  Cf. Tables 8.5 and 8.6.

  Table 8.5. Unemployment in major industrial nations, 2010–2015.

  Table 8.6. Changes in unemployment rates of industrial nations, 2010–2015.

  Table 8.5 has a vague title and its items aren't helpfully sorted.

  Table 8.6 is clearer because it has an informative title and its items are organized to let us see patterns more easily.

8.3.3.2 Bar charts

Bar charts communicate as much by visual impact as by specific numbers.
But bars arranged in no pattern imply no point.
If possible, group and arrange bars to create an image that matches your point.
E.g., look at Fig. 8.4 in the context of the explanatory sentence before it.
Fig. 8.4. World's 10 largest deserts.
The items are listed alphabetically, an order that doesn't help readers see the point.
In contrast, Fig. 8.5 supports the claim with a coherent image.

Fig. 8.5. World distribution of large desserts.

In standard bar charts, each bar represents 100% of a whole.

But sometimes readers need to see specific values for parts of the whole.

You can do that in 2 ways:

- Divide the bars into proportional parts, creating a "stacked bar," as in the chart on the left in Fig. 8.6.

  Fig. 8.6. Stacked bar chart compared to grouped bar chart.

- Give each part of the whole its own bar, then group the bar into clusters, as in the chart on the right in Fig. 8.6.

Use stacked bars only when it's more important to compare whole values than it is to compare their segments.

Readers, however, can't easily gauge proportions by eye alone, so if you do use stacked bars, do this:

- Arrange segments in a logical order.

  If possible, put the largest segment at the bottom in the darkest shade.

- Label segments with specific numbers and connect corresponding segments with gray lines to help clarify proportions.

Fig. 8.7 shows how a stacked bar chart is more readable when irrelevant segments are eliminated and those kept are logically ordered and fully labeled.

Fig. 8.7. Stacked bar charts showing generators of nuclear energy, 1980–1999.

A grouped bar chart makes it easy for readers to compare parts of a whole, but difficult for them to compare different wholes because they must do mental arithmetic.

If you group bars because the segments are more important than the wholes, do this:

- Arrange groups of bars in a logical order; if possible, put bars of similar size next to one another (order bars within groups in the same way).

- Label groups with the number for the whole, either above each group or below the labels on the bottom.

Most data that fit a bar chart can also be represented in a pie chart.

Pie charts are popular in magazines, tabloids, and annual reports, but they're harder to read than bar charts and invite misinterpretation because readers must compare proportions of segments whose sizes are often hard to judge.

Most researchers avoid pie charts, especially to convey quantitative data.

They use bar charts.

8.3.3.3 Line graphs

Because a lien graph emphasized trends, readers must see a clear image to interpret it correctly.

Do the following:

- Choose the variable that makes the line go in the direction, up or down, that supports your point.

  If the good news is a reduction (down) in high school dropouts, you can more effectively represent the same data as an increase in retention (up).

  If you want to emphasized bad news, find a way to represent your data as a falling line.

- Plot more than 6 lines on 1 graph only if you cannot make your point in any other way.

- Do not depend on different shades of gray to distinguish lines, as in Fig. 8.8.

- If you plot fewer than 10 or so values (called *data points*), indicate each with a dot, as in Fig. 8.9.

  If those values are relevant, you can add numbers above the dots.

  Do not add dots to lines plotted from 10 or more data points.

Cf. Fig. 8.8 and Fig. 8.9.

Beyond its general story, Fig. 8.8 is harder to read because the shades of gray do not distinguish the lines well and because our eyes have to flick back and forth to connect lines with variables and their numbers.

Fig. 8.8. Foreign-born residents in the United States, 1870–1990.

Fig. 8.9 makes those connections clearer.

Fig. 8.9. Foreign-born residents in the United States, 1870–1990.

These different ways of showing the same data can be confusing.

To cut through that confusion, test different ways of representing the same data.

(Your software program will usually let you do that quickly.)

Then ask someone not familiar with your data to judge the representations for their impact and clarity.

Be sure to introduce your graphics with a sentence that states the claim you want the figures to support.

8.4 Communicate Data Ethically

Your graphics must be not only clear and accurate but also honest.

Do not distort the image of your data to make a point.

E.g., the 2 bar charts in Fig. 8.10 display identical data yet send different messages.

Fig. 8.10. Capitol City pollution index, 2002–2014.

The 0–100 scale in the figure on the left a creates fairly flat slope, which makes the drop in pollution seem small.

The vertical scale in the figure on the right, however, begins not at 0 but at 80.

When a scale is truncated, its sharper slope exaggerates small contrasts.

Graphs can also mislead by implying false correlations.

Someone claiming that unemployment goes down when union membership goes down might offer Fig. 8.11 as evidence.

Fig. 8.11. Union membership & unemployment rate, 2004–2010.

And indeed, in that graph, union membership and the unemployment rate do seem to move together so closely that a reader might infer one causes the other.

But the scale for the left axis in Fig. 8.11 (union membership) differs from the scale for the right axis (unemployment rate).

The 2 scales have been deliberately skewed to make it seem the 2 downward trends are related.

They may be, but the distorted image doesn't prove it.

Graphs can also mislead when the image encourages readers to misjudge values.

The 2 charts in Fig. 8.12 represents exactly the same data but seem to communicate different messages.

Fig. 8.12. Representation of collar counties among State U. undergraduates (percentage of total).

These "stacked area" charts represent differences in values not by the *angles* of the lines but by the areas *between* them.

In both charts, the bands for south, east, and west are roughly the same width throughout, indicating little change in the values they represent.

The band for the north, however, widens sharply, representing a large increase in the value it represents.

In the chart on the left, readers are likely to misjudge the top 3 bands because they are on top of the rising north band, making those bands seem to rise as well.

In the chart on the right, on the other hand, those 3 bands do not rise because they are on the bottom.

Here only the band for the north rises.

Here are 4 guidelines for avoiding visual misrepresentations:

- Do not manipulate a scale to magnify or reduce a contrast.
- Do not use a figure whose image distorts values.
- Do not make a table or figure unnecessarily complex or misleadingly simple.
- If the table or figure supports a point, state it.

Table 8.7. Common graphic forms & their uses.

Graphic forms: **Data**: **Rhetorical uses**

1. **Bar chart**: Compares the value of 1 variable across a series of items called *cases* (e.g., average salaries for service worker [variable] in 6 companies [cases]).:

   Creates strong visual contrasts among individual cases, emphasizing comparisons.

   For specific values, add numbers to bars.

   Can show ranks or trends.

   Vertical bars (called *columns*) are most common, but bars can be horizontal if cases are numerous or have complex labels. See Sect. 8.3.3.2.

2. **Bar chart, grouped or split**: Compares the value of 1 variable, divided into subsets, across a series of cases (e.g., average salaries [variable] for men and women service workers [subsets] in 6 companies [cases]).:

   Contrasts subsets within and across individual cases; not useful for comparing total values for cases.

   For specific values, add numbers to bars.

   Grouped bars show ranking or trends poorly; useful for time series only if trends are unimportant. See Sect. 8.3.3.2.

3. **Bar chart, stacked**: Compares the value of 1 variable, divided into 2 or more subsets, across a series of cases (e.g., harassment complaints [variable] segmented by region [subsets] in 6 industries [cases]).:

   Best for comparing totals across cases and subsets *within* cases; difficult to compare subsets *across* cases (use grouped bars).

   For specific values, add numbers to bars and segments.

   Useful for time series.

   Can show ranks or trends for total values only. See Sect. 8.3.3.2.

4. **Histogram**: Compares 2 variables, with 1 segmented into ranges that function like the cases in a bar graph (e.g., service workers [continuous] whose salary is $0–5,000, $5–10,000, §10–15,000, etc. [segmented variable]).:

   Best for comparing segments within continuous data sets.

   Show trends but emphasizes segments (e.g., a sudden spike at $5–10,000 representing part-time workers).

   For specific values, add numbers to bars.

5. **Image chart**: Show value of 1 or more variables for cases displayed on a map, diagram, or other image (e.g., state [cases] colored red or blue to show voting patterns [variable]).:

   Shows the distribution of the data in relation to preexisting categories; deemphasizes specific values.

   Best when the image is familiar, as in a map or diagram of a process.

6. **Pie chart**: Shows the proportion of a single variable for a series of cases (e.g., the budget share [variable] of US cabinet departments [cases]).:

   Best for comparing 1 segment to the whole.

   Useful only with few segments or segments that are very different in size; otherwise comparisons among segments are difficult.

   For specific values, add numbers to segments.

   Common in popular venues, frowned on by professionals. See Sect. 8.3.3.2.

7. **Line graph**: Compares continuous variables for 1 or more cases (e.g., temperatures [variable] and viscosity [variable] in 2 fluids [cases]).:

   Best for showing trends; deemphasizes specific values.

   Useful for time series.

   To show specific values, add numbers to data points.

   To show the significance of a trend, segment the grid (e.g., below or above average performance). See Sect. 8.3.3.3.

8. **Area chart**: Compares 2 continuous variables for 1 or more cases (e.g., reading test scores [variable] over time [variable] in a school district [case]).:

   Show trends; deemphasizes specific values.

   Can be used for time series.

   To show specific values, add numbers to data points.

   Areas below the lines add no information and will lead some readers to misjudge values.

   Confusing with multiple lines/areas.

9. **Area chart, stacked**: Compares 2 continuous variables for 2 or more cases (e.g., profit [variable] over time [variable] for several products [cases]).:

   Shows the trend for the total of all cases, plus how much each case contributes to that total.

   Likely to mislead readers on the value or the trend for any individual case, as explained in Sect. 8.4.

10. **Scatterplot**: Compares 2 variables at multiple data points for a single case (e.g., housing sales [variable] and distance from downtown [variable] in 1 city [case]) or at 1 data point for multiple cases (e.g., brand loyalty [variable] and repair frequency [variable] for 10 manufacturers [cases]).:

    Best for showing the distribution of data, especially when there is no clear trend or when the focus in on outlying data points.

    If only a few data points are plotted, it allows a focus on individual values.

11. **Bubble chart**: Compares 3 variables at multiple data points for a single case (e.g., housing sales, [variable] distance from downtown, [variable] and prices [variable] in 1 city [case]) or at 1 data point for multiple cases (e.g., image advertising, [variable] repair frequency, [variable] and brand loyalty [variable] for 10 manufacturers [cases]).:

    Emphasizes the relationship between the 3rd variable (bubbles) and the 1st 2; most useful when the question is whether the 3rd variable is a product of the others.

    Readers easily misjudge relative values shown by bubbles; adding numbers mitigates that problem.

Chapter 9

Revising Your Draft

Some new researchers think that once they have a draft, they're practically done.

Experienced researchers know better.

They write that 1st draft not for their readers but for themselves, to see whether they can make the case they hoped to (or even a better one).

Then they revise until they think their draft meets the needs and expectations of their readers.

That's hard, because we all know our own work too well to read it as others will.

To revise effectively, you must know what readers look for and whether your draft helps them find it.

To that end, the authors' advice may seem mechanical.

But only when you can analyze your draft objectively can you avoid reading into it what you want your readers to get out of it.

The authors suggest 1st revising overall organization, especially the "outer frame" of the introduction and conclusion; then sections, paragraphs, sentences; and finally stylistic issues e.g. spelling and punctuation (for guidance on these issues, see Part 3).

Of course no one revises so neatly.

All of us fiddle with words as we move paragraphs around and reorganize as we revise a sentence.

But you're likely to work best if you revise from whole to part.

Many experienced researchers find that they can edit hard copy more reliably than they can edit text on a screen.

Even if you prefer to edit on a screen, consider reading at least 1 later draft on paper.

You may catch more errors, and you'll get a better sense of your paper's overall structure than you can from the screen alone.

9.1 Check for Blind Spots in Your Argument

Completing a draft is an accomplishment, but don't move immediately to fine-tuning sentences.

After the 1st draft, parts of your argument will likely still not stand up to a robust challenge.

If you spend a lot of time polishing sentences, it can be hard to later accept that whole sections need to be rearranged or even cut.

Instead, check your argument's reasoning.

Have you considered the strongest relevant counterarguments?

Have you looked for evidence that challenges or complicates your reasons?

Have you considered alternative interpretations of your evidence?

If not, now is the time.

If you find it difficult to think of significant alternatives to your argument now that you have completed a draft, talk with your instructor or advisor about where your argument overlook likely objections.

9.2 Check Your Introduction, Conclusion, & Claim

Your readers must recognize 3 things quickly and unambiguously:

- where your introduction ends
- where your conclusion begins

- what sentences in 1 or both state your claim

To make the 1st 2 clearly visible, you might insert a subhead or extra space between your introduction and body and another between the body and conclusion.

(Chap. 10 discusses revising your introduction and conclusion in detail.)

9.3 Make Sure the Body of Your Report Is Coherent

Once you frame your paper clearly, check its body.

Readers will find your paper coherent when they can see the following:

- what key terms run through all of its sections
- where each section and subsection ends and the next begins
- how each section relates to the one before it
- what role each section plays in the whole
- what sentence in each section and subsection states its point
- what distinctive key terms run through each section

To ensure that your readers will see those features, check for the following:

1. Do key terms run through your whole paper?

   - Circle key terms in the claim as stated in your introduction and conclusion (review **7.3**).
   - Circle those same terms in the body of your paper.
   - Underline other words related to concepts named by those circled terms.

   If readers don't see at least 1 of your key terms in most paragraphs, they may think your paper wanders.

   Revise by working those terms into parts that lack them.

   If you underlined many more words than you circled, be sure that readers will recognize how the underlined words relate to the concepts named in your circled key terms.

   If readers might miss the connections, change some of those related words to the key terms.

   If that's difficult, you may have gotten off track and will need to rewrite or discard some passages.

2. Is the beginning of each section and subsection clearly signaled?

   You can use headings to mark transitions from 1 section or subsection to the next (review **6.2.4**).

   In a relatively short paper, rather than use headings, you might add an extra space at the major joints.

   If you can't decide what words to use in headings or even where to put them, your readers will likely have a problem with your paper's organization.

   (For styles of different levels of heads, see **A.2.2.4**.)

3. Does each major section begin with words that signal how that section relates to the one before it?

   Readers must not only recognize where sections begin and end but also understand why they are ordered as they are (see **6.2.5**–**6.2.6**).

   Signal the logic of your order with words e.g. *Consequently…, In contrast…, Some have objected that…*, or even just *1st,… 2nd,…*

4. Is it clear how each section relates to the whole?

   For each section, ask, *What question does this section answer?*

   If it doesn't help to answer 1 of the 5 questions whose answers constitute an argument (see **5.2**), think about its relevance: does it create a context, explain a background concept or issue, or help readers in some other way?

   If you can't explain how a section relates to your claim, consider cutting it.

5. Is the point of each section stated in a sentence at the end of a brief introduction to that section (or at its end)?

   If you have a choice, state the point of a section at the end of its introduction.

   Under no circumstances should you bury the point of a section in its middle.

   If a section is longer than 4 or 5 pages, you might conclude by restating your point and summarizing your argument.

6. Do the specific terms that distinguish a section run through it?

   Each section and subsection needs its own key terms to unify and distinguish it from others.

   Repeat step 1 for each section: find the sentence that expresses its point and circle in it the key terms that distinguish that section from the others.

   Then check whether those terms run through that section.

   If you find no terms that differ from those running through the whole, then you readers might no see what distinct ideas that section contributes to the whole.

9.4 Check Your Paragraphs

Each paragraph should be relevant to the point of its section.

And like sections, each paragraph should have a sentence or 2 introducing it, usually stating its point and including the key concepts that the rest of the paragraph develops.

If the opening sentences of a paragraph don't state its point, then its last one should.

Order your sentences by some principle and make them relevant to the point of the paragraph (for principles of order, see **6.2.5**).

Paragraphs vary in length depending on the type of writing in which they appear.

E.g., they tend to be shorter in brief researcher reports and longer in critical essays or book chapters.

If you find yourself stringing together several short paragraphs, it may mean your points are not well developed.

If your paragraphs run more than a page, it may mean that you are losing focus.

Reserve the use of 2- or 3-sentence paragraphs for lists, transitions, introductions and conclusions to sections, and statements that you want to emphasize.

(The authors use short paragraphs here so that readers can more easily skim - rarely a consideration in scholarly writing.)

9.5 Let Your Draft Cool, Then Paraphrase It

If you start your project early, you'll have time to let your revised draft cool.

What seems good 1 day often looks different the next.

When you return to your draft, don't read it straight through; skim its top-level parts: its introduction, the 1st paragraph of each major section, and the conclusion.

Then, based only on what you have read, paraphrase it for someone who hasn't read it.

Does the paraphrase hang together?

Does it fairly sum up your argument?

Even better, ask someone else to skim your paper by reading just its introduction and the introduction to each major section: how well that person summarizes your paper will predict how well your readers will understand it.

Finally, be receptive to feedback, especially from more experienced researchers and your teacher or advisor.

You don't have to follow every suggestion, but you should consider each carefully.

In Chap. 12 the authors tell you how to get the most out of comments on writing.

Chapter 10

Writing Your Final Introduction & Conclusion

Once you have a final draft and can see what you have actually written, you can write your final introduction and conclusion.

These 2 framing parts of your paper crucially influence how readers will understand and remember the rest of it, so it's worth your time to make them as clear and compelling as you can.

In Chap. 2 the authors showed you how to develop a project around a research problem.

Here they show you how to use that problem to craft an introduction likely to engage your readers and a conclusion likely to solidify their understanding and prompt new questions.

What seizes readers' attention in a research paper is not a snappy hook but a problem they think needs a solution, and what holds their attention is a promise that you've found it.

As we've said, you can always work with readers who say *I don't agree.*

What you can't survive are those who shrug and say *I don't care.*

Your introduction has 3 aims.

It should do the following:

- put your research in the context of other research
- make readers understand how your paper addresses a problem they care about
- give them a framework for understanding it (which sometimes, but not always, includes announcing your claim)

Most introductions run about 10% of the whole (in the sciences they are often shorter).

Your conclusion also has 3 aims.

It should do the following:

- leave readers with a clear idea of your claim
- make readers understand its importance
- suggest further research

Conclusions are usually shorter than introductions.

In article-length papers, they are usually sections; in theses and dissertations, they are usually separate chapters.

10.1 Draft Your Final Introduction

Introduction to papers in different fields can seem different, but behind most of them is a pattern with the 4 parts described in **6.2.2**:

1. Opening context or background.

   When this summarizes relevant research, it's called a *literature review* that puts your project in the context of other research and sets up the next step.

   Keep it short.

2. A statement of your research question, or your research problem's condition.

   This is typically a statement of what isn't known or understood or of what is flawed about the research you cited in step 1.

   It often begins with *but, however*, or another word signaling a qualification.

3. A statement of the significance of your question, or your research problem's consequence.

   This answers *So what?*

   It is key to motivating your readers.

4. Your claim or a promise of one.

   This addresses the research problem expressed in Step 2.

   Here is an abbreviated example (each sentence could be expanded to a paragraph or more):

   Example 10.1.1. *For centuries, risk analysts have studied risk as a problem in statistics and the rational uses of probability theory.* [context]

   But risk communicators have discovered that ordinary people think about risk in ways that seem unrelated to statistically based probabilities. [question/problem-condition]

   Until we understand how nonexperts think about risk, an important aspect of human cognition will remain a puzzle. [significance/problem-consequence]

   It appears that nonexperts judge risk by visualizing worst-case scenarios, then assessing how frightening the image is. [claim]

10.1.1 Establish a Context of Prior Research

Not every paper opens with a survey of research.

Some begin directly with a research question stated as something not known or understood, followed by a review of the relevant literature.

This is a common strategy when the gap in knowledge or understanding is well known:

Example 10.1.2. *The relationship between secondhand smoke and heart disease is still contested.*

But if that gap isn't well known, such an opening can feel abrupt, like this one:

Example 10.1.3. *Researchers do not understand how ordinary people think about risk.*

As a rule, a research paper prepares readers by describing the prior research that it will extend, modify, or correct.

If the paper is intended for general readers, the context it provides can be brief:

Example 10.1.4. *We all take risks every day - when we cross the street or eat high-fat food, and even when we take a bath.*

The study of risk began with games of chance, so it has long been treated mathematically.

By the 20th century, researchers used mathematical tools to study risk in many areas: investments, commercial products, even war.

As a result, most researchers think that risk is a statistically quantifiable problem and that decisions about it should be rationally based.

In a paper intended for other researchers, this opening context typically describes the specific research the paper will extend or modify.

It is important to represent this prior research fairly, so describe it as you think the researchers who conducted it would.

Example 10.1.5. *Ever since Girolamo Cardano thought about games of chance in quantitative terms in the 16th century (Cardano 1545), risk has been treated as a purely mathematical problem.*

Analyses of risk significantly improved in the 17th century when Pascal, Leibniz, and others developed the calculus (Bernstein 1996).

In the 20th century, researchers widened their focus to study risk in all areas of life: investments, consumer products, the environment, even war (Stimson 1990; 1998).

These problems, too, have been addressed almost exclusively from a mathematical perspective.

[Detailed discussion of contemporary research follows.]

Some papers, especially theses and dissertations, go on like that for pages, citing scores of books and articles only marginally relevant to the topic, usually to show how widely the researcher has read.

That kind of survey can provide helpful bibliography to other researchers, especially new ones, but busy readers want to know about only the *specific* research that the researcher intends to extend, modify, or correct.

Early in your career you might not be able to write this review of prior research with much confidence, because you're unlikely to know much of it.

If so, imagine your reader as someone like yourself *before* you started your research.

What would you have wanted to know?

What did you get wrong that your research has corrected?

How has it improved your own flawed understanding?

This is where you can use a working hypothesis that you rejected: *It might seem that X is so, but...* (see also **4.1.2**).

10.1.2 Restate Your Question as Something Not Known or Fully Understood

After establishing the context, state what that prior research hasn't done or how it's incomplete, even wrong.

Introduce that statement with *but, however*, or some other term indicating that you're about to modify the received knowledge and understanding that you just surveyed:

Example 10.1.6. *Ever since Girolamo Cardano... mathematical perspective.*[context]

But risk communicators have discovered that ordinary people think about risk in ways that are irrational and unrelated to statistically realistic probabilities.

What is not understood is whether such nonexpert risk assessment is based on random guesses or whether it has systematic properties. [question restarted]

10.1.3 State the Significance of Your Question

Now you must show your readers the *significance* of answering your research question.

Imagine a reader asking *So what?*, then answer it.

Frame your response as the consequence of not knowing the answer to your research question:

Example 10.1.7. *Ever since Girolamo Cardano... mathematical perspective.*[context]

But risk communicators have discovered that...

What is not understood is whether such nonexpert risk assessment is based on random guesses or whether it has systematic properties.[question related]

[So what?] Until we understand how risk is understood by nonexperts, and important aspect of human reasoning will remain a puzzle: the kind of cognitive processing that seems systematic but lies outside the range of what is called "rational thinking." [significance]

Alternatively, you can phrase the consequence as a benefit:

Example 10.1.8. *Ever since Girolamo Cardano... mathematical perspective.*[context]

But risk communicators have discovered that...

What is not understood is whether such nonexpert risk assessment is based on random guesses or whether it has systematic properties.[question related]

[So what?] If we could understand how ordinary people make decisions about risks in their daily lives, we could better understand a kind of cognitive processing that seems systematic but lies outside the range of what is called "rational thinking."[significance]

You may struggle to answer that *So what?*

It is a problem that only experience can solve, but the fact is, even experienced researchers can be vexed by it.

10.1.4 State Your Claim

Once you state that something isn't known or understood and why it should be, readers want to see your claim, the answer to your research question (we abbreviate a good deal in what follows):

Example 10.1.9. *Ever since Girolamo Cardano... mathematical perspective.*[context]

But risk communicators have discovered that ordinary people think about risk in ways that are systematic but irrational and unrelated to statistically realistic probabilities.[question]

[So what?]

Until we understand how risk is understood by nonexperts, an important kind of human reasoning will remain a puzzle: the kind of cognitive processing that seems systematic but lies outside the range of what is called "rational thinking."[significance]

It appears that nonexperts assess risk not by assigning quantitative probabilities to events that might occur but by visualzing worst-case scenarios, then assigning degrees of risk according to how vivid and frightening the image is.[claim]

If you have reason to withhold your claim until the end of your paper, write a sentence to conclude your introduction that uses the key terms from that claim and that frames what follows:

Example 10.1.10. *It appears that nonexperts assess risk not by assigning quantitative probabilities but by systematically using properties of their visual imagination.*[promise of claim]

Those 4 steps may seem mechanical, but they constitute the introductions to most research papers in every field, both inside the academic world and out.

As you read your sources, especially journal articles, watch for that 4-part framework.

You will not only learn a range of strategies for writing your own introductions but better understand the ones you read.

10.1.5 Draft a New 1st Sentence

Some writers find it so difficult to write their 1st sentence that they fall into clichés.

Avoid these:

- Do not repeat the language of your assignment.
- Do not quote a dictionary definition: *Webster defines risk as...*
- Do not begin too generally: *For centuries, philosophers have debated the question of...*

  (Remember that you are not writing to everyone, only to your research community.)

If you want to begin with something livelier than prior research, try 1 or more of these openers (but note the warning that follows):

1. A pithy quotation:

   Example 10.1.11. *As Dale Carnegie once said, "All life is a chance."*

2. A striking fact:

   Example 10.1.12. *Many people drive rather than fly because the vivid image of an airplane crash terrifies them, even though they are many times more likely to die in a car crash than a plane wreck.*

3. A relevant anecdote:

   Example 10.1.13. *George Miller always drove long distances to meet clients because he believed that the risk of an airplane crash was too great.*

   Even when he broke his back in an automobile accident, he still thought he had made the right calculation.

   "At least I survived.

   The odds of surviving an airplane crash are zero!"

You can combine all 3:

Example 10.1.14. *As Dale Carnegie once said, "All life is a chance."*

E.g., many people drive rather than fly because the vivid image of an airplane crash terrifies them, even though they are more likely to die in a car crash than a plane wreck.

George Miller always drove long distances to meet clients because he believed that the risk of an airplane crash was too great.

Even after he broke his back in an automobile accident, he still thought he had made the right calculation.

"At least I survived.

The odds of surviving an airplane crash are zero!"

Be suer to include in these openers terms that refer to the key concepts you'll use when you write the rest of the introduction (and the rest of the paper).

In this case, they include *calculating, risk, vivid image, more likely.*

Now the warning: before you write a snappy opening, be sure that others in your field use them.

In some fields they're considered too journalistic for serious scholarship.

10.2 Draft Your Final Conclusion

Your conclusion sums up your argument, but just as important, it offers an opportunity to raise new questions suggested by your research.

You can build your conclusion around the same elements as your introduction, only in reverse order.

10.2.1 Restate Your Claim

Restate your claim early in your conclusion, more fully than in your introduction:

Example 10.2.1. *Ordinary people make decisions about risk not on a rational or quantifiable basis but on the basis of at least 6 psychological factors that not only involve emotion but systematically draw on the power of visual imagination.*

At this point you're probably sure what your claim is, but even so, take this last chance to rephrase it to make it as specific and complete as you can.

10.2.2 Point Out a New Significance, a Practical Application, or Opportunities for Further Research (or All 3)

After stating your claim, remind readers of its significance, or better, state a new significance or a practical application:

Example 10.2.2. *These findings suggest a hitherto unsuspected aspect of human cognition, a quantitative logic independent of statistical probabilities involving degrees of precision or realism in visualization.*

Once we understand this imaginative but systematic assessment of risk, it should be possible for risk communicators to better explain risk in everyday life.

Finally, suggest further research.

This gesture suggests how the community of researchers can continue the conversation.

It mirrors the opening context:

Example 10.2.3. *Although these factors improve our understanding of risk, they do not exhaust the "human" factors in judgments of it.*

We must also investigate the relevance of age, gender, education, and intelligence. E.g, . . .

When you state what remains to do, you keep the conversation alive.

So before you write your final words, imagine other researchers who are intrigued by your work and want to follow it up.

What more would *you* like to know, as *their* reader?

What research would you suggest they do?

10.3 Write Your Title Last

Your title is the 1st thing your readers read, but it should be the last thing you write.

It should both announce your topic *and* communicate its conceptual framework, so build it out of the key terms that you earlier identified (review **9.3**).

Compare these 3 titles:

Example 10.3.1. *Risk*

Thinking about Risk

Irrational but Systematic Risk Assessment: The Role of Visual Imagination in Calculating Relative Risk

The 1st title is accurate but too general to give us much guidance about what is to come.

The 2nd is a bit more specific, but the 3rd uses both a title and a subtitle to give us advane notice of the paper's key terms.

When readers see the key terms from the introduction turn up throughout a paper, they are more likely to think it coheres, or holds together.

2-part titles - a main title followed by a subtitle - give your more room for key terms.

Chapter 11

Revising Sentences

Your final task is to make your writing as understandable as you can for your readers, for it is their judgment of your ideas - as you have expressed them in your writing - that matters most.

Readable writing, of course, depends on more than clear sentences, but clear sentences will go a long way toward making your writing readable.

In this chapter, therefore, the authors offer some advice on how to revise your sentences so that readers will find them clear.

Sometimes you will know your writing is awkward, especially if you're writing about an unfamiliar and complex topic for intimidating readers.

Other times, though, you may overestimate your writing, thinking it clear when your readers won't.

You need a way to revise sentences in both of these situations: to revise those you know need help and, even more, to identify and improve those that you think are fine but that readers might not.

The authors can't tell you how to fix every problem in every sentence, but they can tell you how to deal with those that most often afflict writers who, in struggling to sound like "serious scholar," end up sounding merely pretentious.

Here is a short example:

Example 11.0.1. *1a. A better understanding of student learning could achieve improvement in teaching effectiveness.*

However impressive that sounds, the student who wrote it meant only this:

Example 11.0.2. *1b. If we better understood how students learn, we could teach them more effectively.*

To diagnose 1a and revise it into 1b, you must know a few grammatical terms: *noun, verb, active verb, passive verb, whole subject, simple subject, main clause, subordinate clause.*

If they're only a dim memory, skim a grammar guide before you go on.

11.1 Focus on the 1st 7 or 8 Words of a Sentence

Just as the key to a clearly written paper, section, or paragraph is in its 1st few sentences, so the key to a clearly written sentence is in its 1st few words.

When readers grasp those 1st 7 or 8 words easily, they read what follows faster, understand it better, and remember it longer.

It is the difference between these 2 sentences:

Example 11.1.1. *2a. The United Nations' insistence on acceptance by all nations of the principles of equal rights and self-determination of peoples in a product of its recognition that maintenance of stability in the world order requires that nations be guided by values beyond narrow self-interest.*

2b. The United Nations insists that all nations accept the principles of equal rights and self-determination of peoples, because it recognizes that maintaining a stable world order requires that nations be guided by values beyond narrow self-interest.

To write sentence like 2b, or to revise one like 2a into 2b, follow these 7 principles:

- Get to the subject of your sentence quickly; begin sentences with long phrases and clauses only occasionally.
- Make subjects short and concrete, ideally naming the character that performs the action expressed by the verb that follows.
- Avoid separating the subject and verb with more than a word or 2.

- Put key actions in verbs, not in nouns.
- Put information familiar to readers at the beginning of a sentence, new information at the end.
- Choose an active or passive verb based on which lets you best apply the preceding principles.
- Use 1st-person pronouns appropriately.

Those principles are not inviolable rules but guidelines you can rely on to communicate clearly with your readers.

They add up to this: readers want to get past a short, concrete, familiar subject quickly and to a verb expressing a specific action.

When a sentence lets them do that, the rest of the sentence will usually take care of itself.

To diagnose your own writing, skim the 1st 7 or 8 words of every sentence.

Look for sentences in which you don't follow the authors' 7 principles, and then revise them as follows.

11.1.1 Get to the Subject of Your Sentence Quickly

Compare these 2 sentences (introductory phrases are boldfaced, whole subjects italicized):

Example 11.1.2. *3a.* ***In view of claims by researchers on higher education indicating at least 1 change by most undergraduate students of their major field of study,*** 1st-year students *may need better guidance when they choose a major.*

3b. Researchers on higher education *claim that* most students *change their major field of study at least once during their undergraduate career.*

If that is so*, the* 1st-year students *may need better guidance when they choose a major.*

Most readers find 3a harder to read than 3b, because it makes them work through a 24-word phrase before they reach its subject (*1st-year students*).

In the 2 sentences in 3b, readers immediately start with the subject (*Researchers on higher education*) or reach it after a very short clause (*If that is so*).

The principle is this: start most of your sentences directly with their whole subjects.

Begin only a few sentences with introductory phrases or clauses longer than 10 or so words.

You can usually revise long introductory phrases and subordinate clauses into separate independent sentences, as in 3b.

11.1.2 Make Subjects Short & Concrete

Readers must grasp the subject of a sentence easily, but they can't when the subject is long, complex, and abstract.

Compare these 2 sentences (the whole subject in each is italicized; the 1-word simple subject is boldfaced):

Example 11.1.3. *4a.* A school system's successful **adoption** of a new reading curriculum for its elementary schools *depends on the demonstration in each school of the commitment of its principal and the cooperation of teachers in setting reasonable goals.*

4b. A school **system** *will adopt a new reading curriculum for elementary schools successfully only when* each **principal** *demonstrates that* she *is committed to it and* teachers *cooperate to set reasonable goals.*

In 4a, the whole subject is 14 words long, and its simple subject is an abstraction - *adoption.*

In 4b, the clearer version, the whole subject of every verb is short, and each simple subject is relatively concrete: *school system, each principal, she, teachers.*

Moreover, each of those subjects performs the action in its verb: ***system*** *will adopt,* ***principal*** *demonstrates,* ***she*** *is committed,* ***teachers*** *cooperate.*

The principle is this: readers tend to judge a sentence readable when the subject of its verb names the main character in a few concrete words, ideally a character that is also the "doer" of the action expressed by the verb that follows.

But there's a complication.

The authors are not saying that to be clear you must write only about people and concrete things.

In fact, writers often tell clear stories about abstract characters:

Example 11.1.4. *5.* No skill *is more valued in the professional world than problem solving.*

The ability to solve problems quickly *requires us to frame situations in different ways and to find more than 1 solution.*

In fact, effective problem solving may define general intelligence.

Few readers have trouble with those abstract subjects, because they're short and familiar: *no skill, the ability to solve problems quickly*, and *effective problem solving.*

What gives readers trouble is an abstract subject that is long and unfamiliar.

To fix sentences with long, abstract subjects, revise in 3 steps:

- Identify the main character in the sentence.
- Find its key action, and if that is buried in an abstract noun, make it a verb.
- Make the main character the subject of that new verb.

E.g., compare 6a and 6b (actions are boldfaced; verbs are capitalized):

Example 11.1.5. *6a. Without a means for* ***analyzing interactions*** *between social class and eduation in regard to the* ***creation*** *of job opportunities, success in* ***understanding*** *economic mobility will REMAIN limited.*

6b. Economists do not entirely ***UNDERSTAND*** *economic mobility, because they cannot* ***ANALYZE*** *how social class and education* ***INTERACT*** *to* ***CREATE*** *job opportunities.*

In both sentences the main character is *economists*, but in 6a that character isn't the subject of any verb; in fact, it's not in the sentence at all: we must infer it from actions buried in nouns - *analyzing* and *understanding* (what economists do).

We revise 6a into 6b by making the main characters, *economists, social class*, and *education*, subjects of the explicit verbs *understand, analyze, interact*, and *create*.

Readers want subjects to name the main characters in your story, ideally flesh-and-blood characters, and specific verbs to name their key actions.

11.1.3 Avoid Separating Subjects & Verbs with More than a Word or 2

One past a short subject, readers want to get to a verb quickly, so avoid splitting a verb from its subject with long phrases and clauses:

Example 11.1.6. *7a. Some economists, because they write in a style that is impersonal and objective, do not communicate with laypeople easily.*

In 7a, the *because* clause separates the subject *some economists* from the verb *do not communicate*, forcing us to mentally suspend our breath.

To revise, move the interrupting clause to the beginning or end of its sentence, depending on whether it connects more closely to the sentence before or the one after.

When in doubt, put it at the end (for more on this, see **11.1.5**).

Example 11.1.7. *7b. Because some economists write in a style that is impersonal and objective, they do not communicate with laypeople easily. This inability to communicate...*

7c. Some economists do not communicate with laypeople easily because the write in a style that is impersonal and objective. They use passive verbs and...

Readers manage short interruptions more easily:

Example 11.1.8. *8. Few economists* deliberately *write in a style that is impersonal and objective.*

11.1.4 Put Key Actions in Verbs, Not in Nouns

Readers want to get to a verb quickly, but they also want that verb to express a key action.

So avoid using an empty verb e.g. *have, do, make*, or *be* to introduce an action buried in an abstract noun.

Make the noun a verb.

Compare these 2 sentences (nouns naming actions are boldfaced; verbs naming actions are capitalized; verbs expressing little action are italicized):

Example 11.1.9. 9a. During the early years o the 1st World War, the Great Powers' **attempt** at **enlisting** the United States on their side *was met* with **failure**.

9b. During the early years of the 1st World War, the Great Powers ATTEMPTED to ENLIST the United States on their side but FAILED.

In 9a, 3 important actions aren't verbs but nouns: *attempt, enlisting, failure.*

Sentence 9b seems more direct because it expresses those actions in verbs: *attempted, enlist, failed.*

11.1.5 Put Information Familiar to Readers at the Beginning of a Sentence, New Information at the End

Readers understand a sentence most readily when they grasp its subject easily, and the easiest subject o grasp is not just and concrete but also *familiar*.

Compare how the 2nd sentence in each of the following passages does or doesn't contribute to a sense of "flow":

Example 11.1.10. *10a. New insights into global weather patterns are emerging from recent research on the large low-pressure zones rotating above the Earth's poles, known as the polar vertices. Environmental changes that are leading temperatures at the poles to rise, this research suggests, are affecting the vortices. These temperature increases cause the vortices to deviate toward the equator, bringing with them the frigid air responsible for our recent colder winters.*

10b. New insights into global weather patterns are emerging from recent research on the large low-pressure zones rotating above the Earth's poles, known as the polar vortices. The vortices, this research suggests, are being affected by environmental changes that are leading temperatures at the poles to rise. These temperature increases cause the vortices to deviate toward the equator, bringing with them the frigid air responsible for our recent colder winters.

Most readers think 10b flows better than 10a, partly because the subject of the 2nd sentence, *The vortices*, is shorter than the longer subject of 10a: *Environmental changes that are leading temperatures at the poles to rise.*

But 10b also flows better because the order of its ideas is different.

In 10a, the 1st words of the 2nd sentence express new information:

Example 11.1.11. *10a... the polar vortices. Environmental changes that are leading temperatures at the poles to rise...*

Those words about rising temperatures seem to come out of nowhere.

But in 10b, the 1st words echo the end of the previous sentence:

Example 11.1.12. *10b... the polar vortices. The vortices...*

Moreover, once we make that change, the end of that 2nd sentence introduces the 3rd more cohesively:

Example 11.1.13. *10b... leading temperatures at the poles to rise. These temperature increases...*

Contrast 10a; the end of its 2nd sentence doesn't flow into the beginning of the 3rd as smoothly:

Example 11.1.14. *10a... the vortices. These temperature increases...*

That is why readers think that passage 10a feels choppier than 10b: the end of 1 sentence does not flow smoothly into the beginning of the next.

The corollary of the old-information-1st principle is to put new information last, especially new technical terms.

So when you introduce a new term, put it at the end of its sentence.

Compare these:

Example 11.1.15. *11a. Calcium blockers can control muscle spasms. Sarcomers are the small units of muscle fibers in which these drugs work. 2 filaments, 1 thick and 1 thin, are in each sarcomere. The proteins actin and myosin are contained in the thin filament. When actin and myosin interact, your heart contracts.*

11b. Muscle spasms can be controlled with drugs known as calcium blockers*. They work in small units of muscle fibers called* sarcomeres*. Each sarcomere has* 2 filaments, 1 think and 1 thin*. The thin filament contains* 2 proteins, actin and myosin*. When actin and myosin interact, your heart contracts.*

In 11a, the new technical terms are *calcium blockers, sarcomeres, filaments, the proteins actin and myosin*, but they 1st appear early in their sentences.

In contrast, in 11b those new terms 1st appear toward the ends of their sentences.

After that, they're old information and so can appear at the beginning of the next sentences.

No principle of writing is more than important than this: *old before new; familiar information introduces unfamiliar information.*

11.1.6 Choose Active or Passive Verbs to Reflect the Preceding Principles

You may recall advice to avoid passive verbs - good advice when a passive verb allows you to write a sentence that contradicts the principles we have discussed, as in the 2nd sentence of this passage;

Example 11.1.16. *12a. Climate change may have many catastrophic effects. Tropical diseases and destructive insect life even north of the Canadian border could be increased* [passive verb] *by it.*

That 2nd sentence opens with an 11-word subject conveying new information: *Tropical diseases... Canadian border.*

It is the subject of a passive verb, *be increased*, and that verb is followed by a short, familiar bit of information from the sentence before: the concept of climate change represented by the pronoun *it*.

That sentence would be clearer if its verb were active:

Example 11.1.17. *12b. Climate change may have many catastrophic effects. It could increase* [active verb] *tropical diseases and destructive insect life even north of the Canadian border.*

Now the subject is familiar, and the new information in the longer phrase is at the end.

In this case, the active verb is the right choice.

But if you always use active verbs, avoiding passive verbs entirely, you'll write sentences that contradict the old-new principle.

We saw an example in 10a:

Example 11.1.18. *10.a. New insights into global weather patterns are emerging from recent research on the large low-pressure zones rotating above the Earth's poles, known as the polar vortices.*

Environmental changes that are leading temperatuers at the poles to rise, this research suggests, are affecting [active verb] *the vortices.*

The verb in the 2nd sentence is active, but the passage flows better when it's passive:

Example 11.1.19. *10b. New insights into global weather patterns are emerging from recent research on the large low-pressure zones rotating above the Earth's poles, known as the polar vortices.*

The vortices, this research suggests, are being affected [passive verb] *by environmental changes that are leading temperatures at the poles to rise.*

A sentence is more readable when its subject is short, concrete, and familiar, regardless of whether its verb is active or passive.

So choose active or passive voice by considering which gives you the right kind of subject: short, concrete, and familiar.

11.1.7 Use 1st-Person Pronouns Appropriately

Almost everyone has heard the advice to avoid using *I* or *we* in academic writing.

In fact, opinions differ on this point.

Some teachers tell students never to use *I*, because it makes their writing "subjective."

Other encourage using *I* as a way to make writing more lively and personal.

Most instructors and editors do agree that 2 uses of *I* should be avoided:

- Insecure writers begin too many sentences with *I think* or *I believe* (or their equivalent, *In my opinion*).

  Readers assume that you think and believe what you write, so you don't have to say you do.

- Inexperienced writers too often narrate their research: *1st I consulted..., then I examined...*, and so on.

  Readers care less about the story of your research than about its results (see **6.1**).

But the authors believe, and most editors of scholarly journals agree, that the 1st person is appropriate on 2 occasions.

That last sentence illustrates 1 of them: *we believe... that the 1st person...*

- An occasional introductory *I* (or *we*) *believe* can soften the dogmatic edge of a statement.

  Compare this blunter, less qualified version:

  Example 11.1.20. *But ~~we believe, and most editors of scholarly journals agree, that~~ the 1st person is appropriate on 2 occasions.*

  The trick is not to hedge so often that you sound uncertain or so rarely that you sound smug.

- A 1st-person *I* or *we* is also appropriate when it's the subject of a verb naming an action unique to you as the writer of your argument.

  Verbs referring to such actions typically appear in introductions (*I will show/argue/prove/claim that X*) and in conclusions (*I have demonstrated/concluded that Y*).

  Since only you can show, prove, or claim what's in your argument, only you can say so with *I*:

  Example 11.1.21. *14. In this report, I will show that social distinctions at this university are...*

On the other hand, researchers rarely use the 1st person for an action that others must repeat to replicate their research.

Those words include *divide, measure, weigh, examine*, and so on.

Researchers rarely write sentences with active verbs like this:

Example 11.1.22. *15a. I* calculated *the coefficient of X.*

Instead, they're likely to write in the passive, because anyone can do that:

Example 11.1.23. *15b. The coefficient of X* was calculated

Those same principles apply to *we*, if you're 1 of 2 or more authors.
But many instructors and editors object to 2 other uses of *we*:

- the royal *we* used to refer reflexively to the writer
- the all-purpose *we* that refers to people in general

E.g.:

Example 11.1.24. *16. We must be careful to cite sources when we use data from them.*
When we read writers who fail to do that, we tend to distrust them.

In the end, though, those having authority decide.
If your instructor or editor flatly forbids *I* or *we*, then so be it.
When you have that authority, you can make a different choice.
Here is a final piece of advice.
If you follow the authors's 7 principles, your readers will likely find your writing clear.
But you can test your writing by having someone read it back to you.
If that person stumbles over a passage, you can bet your readers will struggle with it too.

11.2 Diagnose What You Read

Once you understand how readers judge what they read, you know not only how to write clear prose but also why so much of what you must read seems so dense.
You might struggle with some writing because its content is difficult.
But you may also struggle because the writer didn't write clearly.
This next passage, e.g., is by no means the thickest ever written:

Example 11.2.1. *15a. Recognition of the fact that grammars differ from 1 language to another can serve as the basis for serious consideration of the problems confronting translators of the great works of world literature originally written in a language other than English.*

But in half as many words, it means only this:

Example 11.2.2. *15b. Once we recognize that languages have different grammars, we can consider the problems of those who translate great works of literature into English.*

So when you struggle to understand some academic writing (and you will), don't blame yourself, at least not 1st.
Diagnose its sentences.
If they have long subjects stuffed with abstract nouns expressing new information, the fault is probably not yours.
Unfortunately, the more experience you get reading unclear academic prose, the greater your risk of imitating it.
In fact, that risk is common to all professions, academic or not.
So remain aware of it as you grow in expertise.

11.3 Choose the Right Word

Another bit of standard advice is *Choose the right word.*

1. Choose the word with the right meaning.

   Affect doesn't mean *effect*; *elicit* doesn't mean *illicit.*

   Common confused words are listed in many handbooks.

   If you're an inexperienced writer, invest in one.

2. Choose the word with the right level of usage.

   If you draft quickly, you risk choosing words that might mean roughly what you think they do but are too casual for a formal paper.

   Someone can *criticize* another writer or *knock* him; a risk can seem *frightening* or *scary*.

   Those pairs have similar meanings, but most readers would judge the 2nd in each pair to be a bit loose for academic writing.

On the other hand, if you try too hard to sould like a real "academic," you risk using words that are too formal.

You can *think* or *cogitate, drink* or *imbibe*.

Those pairs are close in meaning, but the 2nd in each is too fancy for a paper written in ordinary English.

Whenever you're tempted to use a word that you think is especially fine, look for a more familiar one.

The obvious advice is to look up words you're not sure of.

But they're not the problem; the problem is the ones you *are* sure of.

Worse, no dictionary tells you that a word like *visage* or *perambulate* is too fancy for just about any context.

The short-term solution is to ask someone to read your paper before you turn it in (but be cautious before accepting too many suggestions; see **7.10**).

The long-term solution is to read a lot, write a lot, endure a lot of criticism, and learn from it.

11.4 Polish It Up

Before you submit your paper, read it 1 last time to fix errors in grammar, spelling, and punctuation.

Many experienced writers read from the last sentence back to the 1st to keep from getting caught up in the flow of their ideas and missing the words.

Do not rely solely on your word-processing program's spelling and grammar checker.

It will sometimes miss words that are spelled correctly but used incorrectly - *their/there/they're, it's/its, too/to, accept/except, affect/effect, already/all ready, complement/compliment, principal/principle, discrete/discreet*, and so on - and it will flag constructions like the passive voice when they may be the best choice.

(See Chap. 20 for more on spelling.)

If you have used a lot of non-English words, numbers, abbreviations, and so on, check the relevant chapters in Part 3 of this manual.

Finally, if your paper has a table of contents that lists titles and numbers for chapters and sections, be certain that they *exactly* match the corresponding wording and numbering in your paper's body.

If in your text you refer back or forward to other sections or chapters, be sure the references are accurate.

Some students think they have to worry about the quality of their writing in English courses.

That's not true.

Instructors in every discipline appreciate clear and coherent writing, and every course in which you write is an opportunity for you to practice writing better.

11.5 Give It Up & Turn It In

If 1 thing is harder than starting to write, it's stopping.

We all want another day to get the organization right, another hour to tweak the opening paragraph, another minute to... (you get the idea).

Experienced researchers know that nothing they write will ever be perfect and that the benefit of getting the last 1% or even 5% exactly right is rarely worth the cost.

Less experienced researchers should take this lesson to heart.

Dissertation students in particular agonize over reaching a standard of perfection that exists largely in their own minds.

No thesis or dissertation has to be utterly perfect; what it has to be is *done*.

At some point, enough is enough.

Page through it 1 last time to be sure that it looks the way you want it to: look at page breaks, spacing in margins, positions of tables and figures, and so on.

Then give it up and turn it in.

Chapter 12

Learning from Comments on Your Paper

Whether you're a beginner or an advanced researcher, you can learn much from the feedback of careful readers.

But to get the most benefit from such comments - whether they are on a draft of a final version - you need to know how to use them.

As you develop as a writer and researcher, you will receive feedback from many different readers, some (like teachers and editors) having authority over you and others being people whose response you seek out.

Experienced writers know that nothing is more valuable than comments from a trusted reader.

None of us can accurately judge the way readers will respond to our writing, for the simple reason that we know too much about it.

We need readers to show us, through their responses, what we got right and where we went wrong.

12.1 2 Kinds of Feedback: Advice & Data

Commenting on a paper is hard (even if a reader is being paid to do it).

When giving feedback, that reader's responsibility is to be honest and to try to be helpful.

Your responsibility as a writer is to make the most of the comments you receive.

To do that, you must decide whether to treat a reader's comments as *advice* - observations about what you could do (or should have done) to make the paper better - or as *data* documenting a particular reader's response.

When you treat comments as advice, you must decide whether to accept or reject them.

Good advice, especially from a trusted reader, can be invaluable to new and experienced researchers alike.

Sometimes you might feel a strong obligation to accept a reader's advice, e.g. when it comes from teacher, advisor, or editor.

But even in those instances, you are not obligated to accept it wholesale: as a writer, you are responsible not only for your ideas but also for the choices you make in expressing them.

When you treat comments as *data*, you don't accept or reject them; you analyze them to understand why your reader responded to your paper as she did.

And when you understand that, you can revise the paper or make a different decision the next time.

You might know what you wanted your paper to *say*, but only your reader can tell you what it in fact *said* - at least to her.

Again, the writer is ultimately responsible: if a careful reader misunderstood your paper, you should not blame her for that misunderstanding but use the data of her feedback to figure out how and why that misunderstanding occurred.

In this sense, bad advice can be great data.

Even the most dedicated and careful reader is not infallible.

She might misunderstand your intentions or argument, or give you wrongheaded advice (even if she is a teacher).

In such cases, ignore the advice but ask yourself, *What about my paper created that misunderstanding?*

How did I lead my reader astray?

If you can answer those questions, you can still improve your paper, or the next one.

Here are some tips for making the most of the comments you receive, however you choose to interpret them.

12.2 Find General Principles in Specific Comments

When you review your reader's comments, focus on those that you can apply to your next project.

- Look for a pattern of errors in spelling, punctuation, and grammar.

  If you see one, you know what to work on.

- If your reader says you made factual errors, check your notes: Did you take bad notes or misreport them?

  Were you misled by an unreliable source?

  Whatever you find, you know what to do in your next project.

- If your reader reports only her judgments of your writing, look for what causes them.

  If she says your writing is choppy, dense, or awkward, check your sentences using the steps in Chap. 11.

  If she says it's disorganized or wandering, check it against Chap. 9.

  You won't always find what caused the complaints, but when you do you'll know what to work on next time.

12.3 Talk with Your Reader

If you receive comments that include words like *disorganized, illogical*, and *unsupported* but cannot find what triggered them, make an appointment with your reader to ask.

Such words are not descriptions of your paper but descriptions of the reader's impressions of it.

You need to find out what it is in the paper itself that is provoking those impressions.
Following these guidelines will help that conversation go well:

- If your reader marked up your spelling, punctuation, and grammar, correct those errors *before* your meeting to show that you took her comments seriously.

  You might also jot responses to more substantive comments, so that you can discuss them.

- If your reader is your teacher, don't complain about your grade.

  Be clear that you want to understand the comments so that you can do better next time.

- Prioritize: focus on those comments that address the most important issues, like your paper's argument and organization.

  It's tempting to zero in on local concerns that can be quickly corrected, especially if your reader has done a lot of line editing, or to quibble over minor points of disagreement.

- Rehearse your questions so that they'll seem amiable: not "You say this is disorganized but you don't say why," but rather "Can you help me see where I went wrong with my organization so I can do better next time?"

- Ask your reader to point to passages that illustrate her judgments and to explain what those passages should have looked like.

  Don't ask "What didn't you like?" but rather "Where exactly did I go wrong and what could I have done to fix it?"

A final word for students: you might think that meeting with your teacher is helpful only when a paper or yours receive a low grade.

But that would be wrong.
Even after a high grade, it's useful to know how you earned it.
Your next project will likely be more challenging, so it's good to know what successful practices you can build on.
Of course, in that new project you might again feel like a beginner.
That's the way it goes with research.

Chapter 13

Presenting Research in Alternative Forums

It may be too early in your career to think about publishing your work, but it's not too early to present.

Researchers at all stages communicate their work to others in *oral presentations* to a class or at a conference or through *research posters*, a form especially common in the sciences.

Increasingly, undergraduate researchers share their work with audiences beyond the classroom by participating in local research fairs or symposia.

Experienced researchers also present their work to colleagues before publishing it in professional journals.

Indeed, the ability to stand up and talk about your work clearly and confidently is a crucial skill for any career.

In this chapter the authors show you how to use your plan for a written text to prepare a talk.

The authors also discuss the research poster, which combines elements of speech and writing.

Finally, we discuss how to prepare a *conference proposal* as a 1st step, often, in presenting your research.

13.1 Plan Your Oral Presentation

An oral presentation may be easier to prepare than a written paper, but to benefit from the experience of delivering it, you must plan just as carefully.

In giving a talk, you get immediate feedback that can be very helpful if you are testing new ideas or new data.

You want to design your presentation to elicit responses that help you refine your arguments and your analysis.

13.1.1 Narrow Your Focus

Typically, a talk delivered at a conference runs about 20 minutes (or about 8–10 double-spaced pages for text read aloud).

This is not much time to communicate your ideas, so you must boil down your work to its essence or present just a part of it.

Here are 3 common options:

- *Problem statement with a sketch of your argument.* If your problem is new, focus on its originality.

  Start with a short introduction (review Chap. 10), then explain your reasons, summarizing your evidence for each.

- *Summary for a subargument.* If your argument is too big, focus on a key subargument.

  Mention your larger problem in your introduction and conclusion, but be clear that you're addressing only part of it.

- *Methodology or data report.* If you offer a new methodology or source of data, explain why it matters.

  Start with a brief problem statement, then focus on how your new methods or data solve it.

13.1.2 Understand the Difference between Listeners & Readers

Unless you know and respect the difference between listening and reading, your audience will find your presentation tedious or hard to follow.

When we read, we can pause to reflect and puzzle over difficult passages.

To keep on track, we can look at headings and even paragraph indentations.

If we wander, we can reread.

But as listeners in an audience, we can do none of these things.

We must be motivated to pay attention, and we need help to follow a complicated line of thought.

If we lose its thread, we may drift off into our own thoughts.

That's why it's important to not simply read your paper with little or no eye contact or, if using slides, merely introduce them and repeat their content.

You must engage your audience as in a conversation but with extra care for what listeners need.

You have to be explicit about your purpose and your organization.

Further, you have to make your sentence structure far simpler than in a written paper.

So favor shorter sentences with consistent subjects (see **11.1.2**).

Use "I," "we," and "you" a lot.

What seems clumsily repetitive to readers is usually welcomed by listeners.

13.2 Design Your Presentation to Be Listened To

To hold your listeners' attention, you must seem to be not lecturing *at* but conversing *with* them.

This is a skill that does not come easily, since few of us can write as we would speak and most of us need notes to stay on track.

If you do read your paper, read no faster than about 150 words per minute.

That's about 2 minutes per page.

This is faster than you speak ordinarily, so time yourself.

Inexperienced presenters tend to read more quickly than their listeners can comfortably hear and digest.

It's important that your audience see you and not just the top of your head, so build in moments when you look straight out at your audience, especially when saying something important.

Do so at least once or twice per page, ideally at the end of a paragraph.

Remember as well that a paper delivered orally differs in style from a written one.

For one, effective sentences tend to be much shorter and use consistent subjects (see **11.1.2**).

Overall, you need more repetition than you would use in writing to help your listeners stay on track with you.

It is far better to talk from notes than just to read aloud, but to do that well you need to prepare and practice.

In the next section the authors give advice on how to structure your talk and notes for this kind of delivery.

13.2.1 Sketch Your Introduction

For a 20-minute talk, you get 1 shot at motivating your audience before they tune out, so prepare your introduction more carefully than any other part of your talk.

Base it on the 4-part problem statement described in Sect. 10.1, plus a road map.

(The times in parentheses in the list below are rough estimates.)

Use your notes only to remind yourself of the 4 parts, not as a word-for-word script.

If you can't remember the content, you're not ready to give your talk.

Sketch enough in your notes to *remind* yourself of the following:

1. What research you extend, modify, or correct (no more than a minute).

2. What question your research addresses - the gap in knowledge or understanding (30 seconds or less).

3. Why your research matters - an answer to *So what?* (30 seconds).

Those 3 steps are crucial in motivating your listeners.

If your question is new or controversial, give it more time.

If your listeners know its significance, mention it quickly and go on.

4. Your claim, the answer to your research question (30 seconds or less).

   Listeners want to know your answer up front even more than readers do, so state at least its gist, unless you have a compelling reason to wait for the end.

   If you do choose to wait to give your answer, at least forecast it.

5. A forecast of the structure of your presentation (10–20 seconds).

   The most useful forecast is an oral table of contents: "1st I will discuss…"

   That can seem clumsy in print, but listeners need more help than readers do.

   Repeat that structure as you work through the body of your talk on the spirit of research.

Rehearse your introduction, not only to get it right but also so you can look your audience in the eye as you give it.

You can look down at notes later.

All told, spend no more than 3 minutes or so on your introduction.

13.2.2 Design Notes You Can Understand at a Glance

Do not write your notes as complete sentences (much less paragraphs) that you read aloud; notes should help you see at a glance the structure of your talk and cue what to say at crucial points.

So do not cut and paste sentences from a written text; create your notes from scratch.

Use a separate page for each main point.

On each page, write our your main point not as a topic but as claims, either in shortened form or (if you must) in complete sentences.

Above each point, you might add an explicit transition as the oral equivalent of a subhead: "The 1st issue is…"

Visually highlight those main points so that you spot them instantly.

Under them, list the evidence that supports them.

If your evidence consists of numbers or quotations, you'll probably have to write them out.

Otherwise, know your evidence well enough to be able to talk about it directly to your audience.

Organize your points so that you cover the most important ones 1st.

If you run long (most of us do), you can skip a later section or even jump to your conclusion without losing anything crucial to your argument.

Never build up to a climax that you might not reach.

If you must skip something, use the question-and-answer period to return to it.

13.2.3 Model Your Conclusion on Your Introduction

As in a written paper, your conclusion is your last opportunity to communicate your claim and its significance (see **10.2**).

Make your conclusion memorable, because listeners will repeat it when asked, *What did Jones say?*

Learn it well enough to present it looking at your audience, without reading from notes.

It should have these 3 parts:

- your claim, in more detail than in your introduction (if listeners are mostly interested in your reasons or data, summarize them as well)
- your answer to *So what?* (you can restate an answer from your introduction, but try to add a new one, even if it's speculative)
- suggestions for more research, what's still to be done

Rehearse your conclusion so that you know exactly how long it takes (no more than a minute or 2).

Then when you have that much time remaining, conclude, even if you haven't finished your last (relatively unimportant) points.

If you had to skip 1 or 2 points, work them into an answer during the question-and-answer period.

If your talk runs short, don't ad lib.

If another speaker follows you, make her a gift of your unused time.

13.2.4 Anticipate Questions

If you're lucky, you'll get questions after your talk, so prepare answers for predictable ones.

Expect questions about data or sources, especially if you didn't cover them much in your talk.

If you address matters associated with well-known researchers to schools of research, be ready to expand on how your work relates to theirs, especially if you contradict or complicate their results or approach.

Also be ready to answer questions about a source you never heard of.

The best policy is to acknowledge that you haven't seen it but that you'll check it out.

If the question seems friendly, ask why the source is relevant.

Don't prepare only defensive answers.

Use answers to questions to reemphasize your main points or cover matters that you may have left out.

Listen to every question carefully; then, to be sure you understand the question, *pause before you respond and think about it for a moment.*

If you don't understand the question, ask the questioner to rephrase it.

Don't snap back an answer reflexively and defensively.

Good questions are invaluable, even when they seem hostile.

Use them to refine your thinking.

13.2.5 Create Handouts

You can read short quotations or important data aloud for your listeners, but if you have many, create a handout.

If you use slides, pass out printed copies.

You can also hand out an outline of your main points, with white space for notes.

13.3 Plan Your Poster Presentation

A poster is a large board on which you lay out a summary of your research along with your most relevant evidence.

It is a cross between a talk delivered orally and a written paper.

In a poster presentation you can go into more detail than you can in a 20-minute talk.

But a poster is not juts a research paper transferred to a larger canvas.

It's a selection of key elements of a research project (question, claim, data, significance) designed to welcome an audience of individuals who choose to stop and visit.

Poster sessions are usually held in hallways or in a large room filled with many presenters.

People move from poster to poster, reading as their interests dictate but also asking questions of the presenters, who typically stand by their posters at designated times ready to discuss their research.

In this sense, a good poster can turn your imagined conversation with your research community (see Chap. 1) into a real one.

Posters combine the advantages of writing and speaking.

While posters, like written reports, should be self-explanatory, they also encourage presenters and their audiences (typically fellow researchers) to interact 1 on 1.

Those who read your poster have more control than a listener, and they can rely on prominent visual signals that you use to organize your material - boxes, lines, colors, and larger and smaller titles.

You can design your poster using available software and templates that produce a serviceable final product.

For the text of the poster itself, however, follow the guidelines for a paper to be read aloud, with 2 additional considerations:

1. Layer your argument.

   Present your argument visually in 3 levels of detail:

   - Highlight an abstract or a problem statement and summary at the top of the poster (box it, use larger type, etc.).
   - Under that introductory material, list your reasons as subheads in a section that summarizes your argument.
   - Under that section, restate your reasons and group your evidence under them.

2. Explain all graphs and tables.

   In addition to providing a caption for each graphic, add a sentence or 2 explaining what is important in the data and how they support your reason and claim (review **7.7** and **8.3.1**).

13.4 Plan Your Conference Proposal

Conferences are good opportunities to share your work, but to be invited to speak, you usually have to submit a proposal.

Write it not as a paragraph-by-paragraph summary of your work but as a 30-second "elevator story" - what you would tell someone who asked, as you both stepped into an elevator on the way to your talk, *What are you saying today?*

In fact, a carefully prepared and rehearsed elevator story is very useful for any conversation about your work, particularly interviews.

An elevator story has 3 parts:

- a problem statement that highlights an answer to *So what?*
- a sketch of your claim and major reasons
- a summary of your most important evidence

Conference reviewers are less interested in your exact words than in why anyone should want to listen to them.

Your aims are to pose your research question and to answer the reviewer's *So what?*

So focus on how your claim contributes to your field of research, especially on what's novel or controversial about it.

If you address a question established by previous research, mention it, then focus on your new data or your new claim, depending on which is more original.

Be aware that reviewers will often know less about your topic than you do and may need help to see the significance of your question.

Besides gaining acceptance to the conference, a conference proposal has 1 other objective.

Its title and often its abstract (or even the full proposal) are used to attract an audience to the session in which the proposed talk will be scheduled.

A good conference proposal thus has a clear and substantive title (see **10.3**), one that lets an audience know what to expect.

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On the Spirit of Research

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